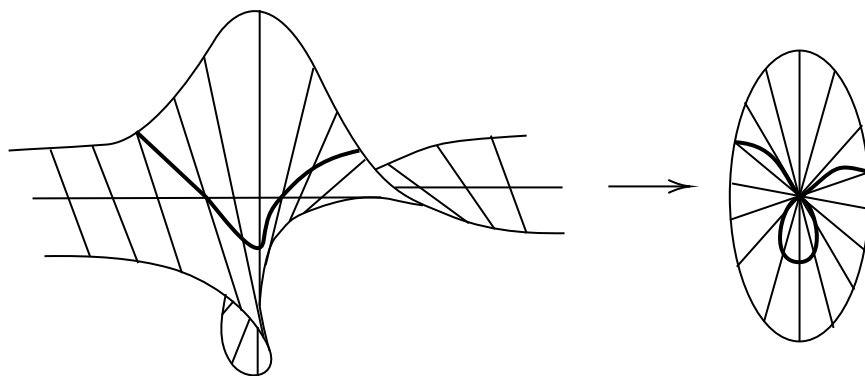


Affine and $\mathbb{P}$ rojective Geometry I

FUNG SAN GAAN



Affine and Projective Geometry I

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Week 1	3	Week 7	23
affine spaces, common zeros, radical ideals, intro to Hilbert's Nullstellensatz		results about closed sets, finite surjective/injective maps give equal dimension, the "old" dimension, dimension of hypersurfaces, principal ideal theorem	
Week 2	5	Week 8	28
Nullstellensatz with assumptions, finite/integral extensions, Cayley-Hamilton, Nakayama		additional consequences of the principal ideal theorem, Catenary property, local ring, germ, stalk, local dimension theory, prime avoidance	
Week 3	9	Week 9	31
"lying over", preparation, noether normalization, coordinate rings, all Nutz, some corollaries		spring break	
Week 4	13	Week 10	31
$K[X]$ is the set of regular functions, regular maps of varieties, facts about regular maps, isomorphism between varieties, closed immersions, dominant maps		more on the dimension theory, the dimension of fibers, intro to projective space, homogeneous Nutz	
Week 5	16	Week 11	35
regular maps on open sets, open immersions, irreducible and noetherian topological spaces, notions of dimension		12 4/5	35
Week 6	21	Appendix	36
finite maps between rings/varieties, finiteness and dimensionality, $\dim(\mathbb{A}_K^n) = n$, dimension of affine varieties, more facts about finite maps		Index	

(25/6/1) General notice: Let $A, B \in \mathbb{N}$. "according to A/B " means it dates back to the lecture on 2024/ A/B . This is my back-burner project and will receive updates once in a while.
 (25/7/31) Week 7 is done. Added new examples, yay!
 (26/1/25) Week 10 is done. Work is still in progress.

These notes are adapted from professor C. S. Dodd's lecture, SP24.

If you spot any error in these notes, please reach out to me via email at b192.168.1.91.03@gmail.com, or send me a direct message on Discord at [3fsg](#).

Week 1

Defined affine spaces, common zeros, and radical ideals. Introduced Hilbert's Nullstellensatz.

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Vocabulary 1.1 (affine space $\mathbb{A}_K^n$). $\mathbb{A}_K^n = K^n := \{(a_1, \dots, a_n) | a_i \in K\}$. We interpret K^n as a ring. We can just write $\mathbb{A}^n$ if there is an assumed field K .

Vocabulary 1.2 (set of polynomials). Let $\{f_i\}$ be a subset of $K[x_1, \dots, x_n]$. Any f is a function

$$f : \mathbb{A}_K^n \rightarrow \mathbb{A}_K^1 \quad (a_1, \dots, a_n) \mapsto f(a_1, \dots, a_n)$$

Vocabulary 1.3 (zero set in $\mathbb{A}_K^n$). $Z\{f_i\} := \{(a_1, \dots, a_n) | f_i(a_1, \dots, a_n) = 0, \forall i\} \subseteq \mathbb{A}_K^n$. In other words, $Z\{f_i\}$ is the set of common zeros.

Vocabulary 1.4 (affine algebraic variety). A subset Y of $\mathbb{A}_K^n$ is an affine algebraic variety if there exists a subset $T \subset K[x_1, \dots, x_n]$ such that $Y = Z(T)$.

Example. When $n = 1$, $X \subseteq \mathbb{A}_K^1$ is an algebraic variety if and only if there exists a collection of polynomials $\{f_i\}$ such that $X = Z\{f_i\}$. Here are all the possibilities

1. X is finite, $X = \{z_1, \dots, z_n\} = Z(f)$, $f = (x_1 - z_1) \dots (x_n - z_n)$.
2. $X = \emptyset$, $X = Z(1)$.
3. $X = \mathbb{A}_K^1$, $X = Z(0)$ (the 0 polynomial).

Vocabulary 1.5 (Zariski topology on $\mathbb{A}_K^n$). We define affine algebraic varieties to be the closed sets of this topology. The open subsets of the Zariski topology on $\mathbb{A}_K^n$ are complements of affine algebraic varieties.

Vocabulary 1.6 (closed set). They are $Z(T)$ in 1.4.

Example. The open set of the Zariski topology on $\mathbb{A}_K^1$ is the empty set. It is also the complement of finite subsets. This topology is not Hausdorff.

Proof. The space $\mathbb{A}_K^1 = K[x]$. Since K is a field, $K[x]$ is a PID. Since K is algebraically closed, any polynomial can be factorized as $f(x) = c(x_1 - a_1) \dots (x_n - a_n)$, where $c, a_i \in K$, then $Z(f) = \{a_1, \dots, a_n\}$. Hence all closed subsets are either finite or $\mathbb{A}_K^1$, which is $\mathbb{A}_K^1 \setminus \emptyset$. Suppose the space is Hausdorff. For all $x, y \in \mathbb{A}_K^1$, let a desired $U_x = \mathbb{A}_K^1 \setminus \{a_1, \dots, a_n\} \neq \emptyset$, then $U_y = \{a_1, \dots, a_n\}$, which is finite. This is a contradiction. $\square$

Now we get to the more general case of $n \geq 1$:

Lemma. Let $\{f_i\} \subseteq K[x_1, \dots, x_n]$ and $I = \langle \{f_i\} \rangle$, then $Z\{f_i\} = Z(I)$.

Proof. ($Z(I) \subseteq Z\{f_i\}$): If $(a_1, \dots, a_n) \in Z(I)$, then $g(a_1, \dots, a_n) = 0$ for all $g \in I$. This implies $f_i(a_1, \dots, a_n) = 0$ for all i . By the definition of zeros, we know that $(a_1, \dots, a_n) \in Z\{f_i\}$. ($Z\{f_i\} \subseteq Z(I)$): If $f_i(a_1, \dots, a_n) = 0$ for all i , let $g \in I$. We can write $g = \sum p_i f_i$, where $p_i \in K[x_1, \dots, x_n]$. $g(a_1, \dots, a_n) = \sum p_i(a_1, \dots, a_n) f_i(a_1, \dots, a_n) = 0$ implies $(a_1, \dots, a_n) \in Z(I)$. $\square$

Vocabulary 1.7 (radical ideal). Let R be a commutative ring with 1 and let $I \subseteq R$ be an ideal. The radical ideal of I is

$$\sqrt{I} = \{f \in R | \exists N \geq 1, f^N \in I\}$$

Example. $R = K[x]$. If $I = \langle f \rangle$ and $f = (x - a_1)^{n_1} \dots (x - a_r)^{n_r}$, then $\sqrt{I} = \langle g \rangle$ where $g = (x - a_1) \dots (x - a_r)$.

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Vocabulary 1.8 (ideal of $\mathbb{A}_K^n$). Let $X \subseteq \mathbb{A}_K^n$. Define

$$I(X) = \{f \in K[x_1, \dots, x_n] | f(X) = 0\}$$

We know the existence of maps

$$\begin{aligned}\varphi : I &\rightarrow Z(I) & \{I \subseteq K[x_1, \dots, x_n]\} &\mapsto \{\text{subvarieties of } \mathbb{A}_K^n\} \\ \psi : X &\rightarrow I(X) & \{\text{subsets of } \mathbb{A}_K^n\} &\mapsto \{I \subseteq K[x_1, \dots, x_n]\}\end{aligned}$$

Now we have a good question to ask: **When is ψ bijective?**

Lemma. Here are things we can say about ideals and their zero sets

1. Let I, J be ideals in $K[x_1, \dots, x_n]$, $I \subseteq J$ implies $Z(I) \supseteq Z(J)$.
2. $Z(I \cap J) = Z(I) \cup Z(J)$.
3. Let $\{I_i\}$ be a collection of ideals, then $Z(\sum I_i) = \bigcap Z(I_i)$.

Proof. They should be easy to prove...

1. If $(a_1, \dots, a_n) \in Z(J)$, then $f(a_1, \dots, a_n) = 0$ for all $f \in J$. This implies $f(a_1, \dots, a_n) = 0$ for all $f \in I$. It follows that $(a_1, \dots, a_n) \in Z(I)$. Moreover, if $X \subseteq Y$ are subsets of $\mathbb{A}_K^n$, then $I(X) \supseteq I(Y)$.
2. We obtain the following results from (1)

$$\begin{aligned}I \cap J \subseteq I &\Rightarrow Z(I \cap J) \supseteq Z(I) \\ I \cap J \subseteq J &\Rightarrow Z(I \cap J) \supseteq Z(J)\end{aligned}$$

Combining them together we have $Z(I) \cup Z(J) \subseteq Z(I \cap J)$. Now we do the other inclusion. Suppose the other inclusion does not hold, then there exists $(a_1, \dots, a_n) \in Z(I \cap J)$ such that $(a_1, \dots, a_n) \notin Z(I)$ and $(a_1, \dots, a_n) \notin Z(J)$. In other words, there exists some $f_1 \in I$ such that $f_1(a_1, \dots, a_n) \neq 0$ and some $f_2 \in J$ such that $f_2(a_1, \dots, a_n) \neq 0$. These together imply $f_1 f_2(a_1, \dots, a_n) \neq 0$. But $f_1 f_2 \in I \cap J$ leads to a contradiction: $Z(I \cap J) = Z(I) \cup Z(J)$.

3. If $(a_1, \dots, a_n) \in Z(\sum I_i)$, then $f \in I_i$ implies $f(a_1, \dots, a_n) = 0$. If $(a_1, \dots, a_n) \in \bigcap Z(I_i)$, then $f(a_1, \dots, a_n) = 0$ for any $f \in I_i$. If $g \in \sum I_i$, then $g = \sum p_i f_i$, where $f_i \in I_i$ and all but finitely many $p_i = 0$. So that we have $g(a_1, \dots, a_n) = \sum p_i f_i(a_1, \dots, a_n) = 0$. Conclude that $(a_1, \dots, a_n) \in Z(\sum I_i)$. □

Hilbert's Nutz I. Let $I \subseteq K[x_1, \dots, x_n]$ be an ideal, then $I(Z(I)) = \sqrt{I}$. If $X \subseteq \mathbb{A}_K^n$ is closed, $Z(I(X)) = X$. If $Y \subseteq \mathbb{A}_K^n$ is any subset, $Z(I(Y)) = \overline{Y}$.

Remark. It turned out we do have a bijection

$$I \rightarrow Z(I) \quad \{I \mid I = \sqrt{I}\} \mapsto \{\text{closed subsets of } \mathbb{A}_K^n\}$$

This map respects the inclusion-reversing property.

Lemma. Let $Y_1, Y_2 \subseteq \mathbb{A}_K^n$. Then the following properties hold.

1. if $Y_1 \subseteq Y_2$, then $I(Y_2) \subseteq I(Y_1)$;
2. $I(Y_1 \cup Y_2) = I(Y_1) \cap I(Y_2)$.

Proof. Let us quickly prove these properties.

1. For all $f \in I(Y_2)$ and $P \in Y_1$, since $Y_1 \subset Y_2$, $P \in Y_2$, so $f(P) = 0$, hence $I(Y_2) \subset I(Y_1)$.
2. Since $Y_1, Y_2 \subset Y_1 \cup Y_2$, $I(Y_1 \cup Y_2) \subset I(Y_1) \cap I(Y_2)$. For all $f \in I(Y_1) \cap I(Y_2)$ and $P \in Y_1 \cup Y_2$, $f(P) = 0$, hence $I(Y_1) \cap I(Y_2) \subset I(Y_1 \cup Y_2)$. □

Lemma. If $\mathfrak{m} \subseteq K[x_1, \dots, x_n]$ is maximal, then $\mathfrak{m} = \sqrt{\mathfrak{m}}$.

Proof. Recall a fact from algebra that if $\mathfrak{m} \subseteq R$ is maximal if and only if $R/\mathfrak{m}$ is a field. Now suppose $\mathfrak{m} \subseteq K[x_1, \dots, x_n]$ is maximal. If $f^N \in \mathfrak{m}$, then inside $K[x_1, \dots, x_n]/\mathfrak{m}$, $\overline{f}^N = 0$ would imply $\overline{f} = 0$. Thus $f \in \mathfrak{m}$. □

Lemma. If $\mathfrak{p} \subseteq K[x_1, \dots, x_n]$ is prime, then $\mathfrak{p} = \sqrt{\mathfrak{p}}$.

Proof. Let $x^n \in \mathfrak{p}$. Assume n is the minimal such natural number. If n is 1, we are done. If $n > 1$, because $\mathfrak{p}$ is prime, either $x \in \mathfrak{p}$ or $x^{n-1} \in \mathfrak{p}$. In the first case, we are done. The latter cannot happen since n is minimal. □

Hilbert's Nutz II. For each maximal ideal $\mathfrak{m} \subseteq K[x_1, \dots, x_n]$, there exists a unique $(a_1, \dots, a_n) \in \mathbb{A}_K^n$ such that $\mathfrak{m} = \langle x_1 - a_1, \dots, x_n - a_n \rangle = \ker(\chi : K[x_1, \dots, x_n] \rightarrow K)$, where $\chi(f(x_1, \dots, x_n)) = f(a_1, \dots, a_n)$.

proof using Hilbert's Nutz I. We first show the ideal $\langle x_1 - a_1, \dots, x_n - a_n \rangle$ is maximal. When $n = 1$, $K[x_1]/\langle x_1 \rangle = K$. When $n > 1$,

$$\begin{aligned} K[x_1, \dots, x_n]/\langle x_1 - a_1, \dots, x_n - a_n \rangle &\simeq (K[x_1, \dots, x_{n-1}][x_n]/\langle x_n - a_n \rangle)/(\langle x_1 - a_1, \dots, x_n - a_n \rangle/\langle x_n - a_n \rangle) \\ &\simeq K[x_1, \dots, x_{n-1}]/\langle x_1 - a_1, \dots, x_{n-1} - a_{n-1} \rangle \end{aligned}$$

By induction, $K[x_1, \dots, x_n]/\langle x_1 - a_1, \dots, x_n - a_n \rangle \simeq K$. Clearly, $x_i - a_i \in \ker(\chi)$, hence the maximal ideal

$$\langle x_1 - a_1, \dots, x_n - a_n \rangle \subseteq \ker(\chi)$$

implies $\langle x_1 - a_1, \dots, x_n - a_n \rangle = \ker(\chi)$. Now we have to show all maximal ideals look like this. Let $\mathfrak{m}$ be maximal, $I(Z(\mathfrak{m})) = \mathfrak{m}$ implies $Z(\mathfrak{m}) \neq \emptyset$. Then there is $(a_1, \dots, a_n) \in Z(\mathfrak{m})$ for all $f \in \mathfrak{m}$. If $f(a_1, \dots, a_n) = 0$, then $\mathfrak{m} \subseteq \ker(\chi)$ implies $\mathfrak{m} = \ker(\chi)$. $\square$

Week 2

Proved the Nullstellensatz with some facts assumed. Defined finite and integral ring extensions. Proved the generalized Cayley-Hamilton and Nakayama lemma.

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Remark. Hilbert's Nutz II tells us all maximal ideals of $K[x_1, \dots, x_n]$ are of the form $\langle x_1 - a_1, \dots, x_n - a_n \rangle$. Moreover, each maximal ideal corresponds to a point $(a_1, \dots, a_n) \in \mathbb{A}_K^n$. Equivalently, every maximal ideal of $K[x_1, \dots, x_n]$ is of the form $\ker(\chi)$, where $\chi : K[x_1, \dots, x_n] \rightarrow K$ is a ring homomorphism such that $\chi(\lambda) = \lambda$ for all $\lambda \in K$.

To $(a_1, \dots, a_n)$, one identifies $\chi(f(x_1, \dots, x_n)) = f(a_1, \dots, a_n) \dots$

Lemma. If $\chi : K[x_1, \dots, x_n] \rightarrow K$ is the map above, then there exists a unique $(a_1, \dots, a_n)$ such that $\chi(f(x_1, \dots, x_n)) = f(a_1, \dots, a_n)$.

Proof. Let $a_i = \chi(x_i)$, then $\chi(x_i - a_i) = \chi(x_i) - \chi(a_i) = a_i - a_i = 0$. So $\langle x_1 - a_1, \dots, x_n - a_n \rangle \subseteq \ker(\chi)$ implies $\ker(\chi) = \langle x_1 - a_1, \dots, x_n - a_n \rangle$. Define $\mathbf{I} = (i_1, \dots, i_n) \in (\mathbb{Z}^{\geq 0})^n$, $b_{\mathbf{I}} \in K$, and $x^{\mathbf{I}} = x_1^{i_1} x_2^{i_2} \dots x_n^{i_n}$. For $f(x_1, \dots, x_n) = \sum_{\mathbf{I}} b_{\mathbf{I}} x^{\mathbf{I}}$,

$$\chi(f(x_1, \dots, x_n)) = \sum_{\mathbf{I}} b_{\mathbf{I}} \chi(x^{\mathbf{I}}) = \sum_{\mathbf{I}} b_{\mathbf{I}} (\chi(x_1))^{i_1} \dots \chi(x_n)^{i_n} = \sum_{\mathbf{I}} b_{\mathbf{I}} (a_1^{i_1} \dots a_n^{i_n}) = \sum_{\mathbf{I}} b_{\mathbf{I}} a^{\mathbf{I}}$$

Note that the above is just $f(a_1, \dots, a_n)$. $\square$

Theorem 1 (Hilbert's Nutz III). Let K be a field, let $\mathfrak{m}$ be a maximal ideal in $K[x_1, \dots, x_n]$, then there is a map $K \rightarrow K[x_1, \dots, x_n]/\mathfrak{m} = F$ making F into a field extension of K . F/K is a finite field extension.

Example. When $n = 1$, $F = K[x_1]/\mathfrak{p}$. $\mathfrak{p}$ is a prime ideal.

Remark. If $K = \overline{K}$ is algebraically closed, then the theorem will imply $F = K$, i.e. the map $i : K \rightarrow K[x_1, \dots, x_n]/\mathfrak{m}$ is an isomorphism. Then the map $K[x_1, \dots, x_n] \rightarrow \overline{K}[x_1, \dots, x_n] \rightarrow K$ given by $f \mapsto \bar{f} \mapsto i^{-1}(\bar{f}) \in K$ is a ring homomorphism $\chi(\lambda) = \lambda$ for all $\lambda \in K$.

We will only prove this for K infinite since infinite fields have the property that if $p(x) \in K[x]$, then $p(a) = 0$ for all $a \in K$ implies $p = 0$.

We start by proving the following 2 facts:

Fact 1. Let $A \subset B$ be rings and both A and B be domains. Let B be a finitely generated module over A , then A is a field if and only if B is a field. Note that $(\Rightarrow)$ can be proven using just basic algebra. Suppose A is a field and $b \in B$. Then the set $P = \{p(t) \in A[t] \mid p(b) = 0\}$ is a prime ideal. So the subalgebra of B generated by A and b is isomorphic to $A[t]/P$. This is a field because P is prime in a PID.

Fact 2 (noether normalization). Let $A = K[x_1, \dots, x_n]/I$, then there exist elements $\{t_1, \dots, t_d\} \subset A$ finite, possibly empty, such that

1. $K\langle t_1, \dots, t_d \rangle = K[t_1, \dots, t_d]$. The left hand side is a subring of A generated by K and $t_1, \dots, t_d$. In other words, the map $K[y_1, \dots, y_d] \rightarrow A$ by sending $y_i \rightarrow t_i$ is an isomorphism onto $K\langle t_1, \dots, t_d \rangle$.
2. Further, the inclusion $K[t_1, \dots, t_d] \subset A$ makes A a finite $K[t_1, \dots, t_d]$ -module.

proof of Hilbert's Nutz III. Let $I = \mathfrak{m} \in K[x_1, \dots, x_n]$ be maximal, then $F = K[x_1, \dots, x_n]/\mathfrak{m}$ is a field. Find $\{t_1, \dots, t_d\}$ by Fact 2 such that $K[t_1, \dots, t_d] \subset F$ and F are finitely generated over $K[t_1, \dots, t_d]$ (a $K[t_1, \dots, t_d]$ -module). By Fact 1, F is a field implies $K[t_1, \dots, t_d]$ is a field. It follows that $\{t_1, \dots, t_d\} = \emptyset$ since variables are not invertible. Therefore, $K \subset F$ is finitely generated. $\square$

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We are trying to prove Fact 1, but we should first recall some basic definitions from algebra.

Vocabulary 2.1 (finite). Let $A \subset B$ be rings, then

1. B is finite over A if B is finitely generated as an A -module.
2. B is finitely generated if there exists $\{x_1, \dots, x_n\} \subset B$ such that B is the smallest ring containing A and $\{x_1, \dots, x_n\}$. This is equivalent to the existence of a surjective map

$$\pi : A[t_1, \dots, t_d] \twoheadrightarrow B$$

such that $\pi(a) = a$ for all $a \in A$.

Remark. For the definition above, (1.) implies (2.) because (1.) holds if and only if there exists $\{t_1, \dots, t_d\} \subset B$ such that $B = At_1 + At_2 + \dots + At_d$. It follows that B is the ring generated by A and $\{t_1, \dots, t_d\}$. However, (2.) does *not* imply (1.). To see this, just take $B = A[t]$. B is finitely generated as a ring but not a module.

Example. (Facts about algebraic elements) Suppose A, B are fields, B is finite over A if and only if (1) for all $\alpha \in B$, there exists $p(x) \in A[t] \setminus \{0\}$ such that $p(\alpha) = 0$. (2) There exists $\{\alpha_1, \dots, \alpha_n\}$ such that $B = A[\alpha_1, \dots, \alpha_n] = A(\alpha_1, \dots, \alpha_n)$. Note that $A(\alpha_1, \dots, \alpha_n)$ is a *subring* of B generated by A and $\{\alpha_1, \dots, \alpha_n\}$. $A[\alpha_1, \dots, \alpha_n]$ is a *subfield* of B generated by those two things (if and only if B is algebraic over A and B is a finitely generated A -algebra).

Example. Let K be a field, $A = K[t] \subseteq B = K[t, t^{-1}] = K[t, y]/\langle yt - 1 \rangle = A[y]/\langle yt - 1 \rangle$.

1. B is finitely generated over A .
2. B is not finite over A : If

$$b = \sum_{i=-N}^N a_i t^i \in B \Rightarrow K[t] \cdot b \in \left\{ \sum_{i=-N}^M c_i t^i \right\}$$

So $K[t] \cdot b \neq B$ since $t^{-N-1} \notin K[t] \cdot b$.

On the other hand, there exists $p(y) \in A[y]$ such that $B = A\langle b \rangle$ and $p(b) = 0$ when we take $b = t^{-1}$, $p(y) = yt - 1$. It turned out $yt - 1$ not being monic will cause problems.

Solution: Only consider monic polynomials.

Lemma. Let $p(t) \in A[t]$ be monic of degree n , then

$$B = A[t]/\langle p(t) \rangle = A \oplus At \oplus At^2 \oplus \dots \oplus At^{n-1}$$

as an A -module. In particular, B is finite over A .

Proof. Let $g(t) \in A[t]$. Then there exist $q(t)$ and $r(t)$ such that $g(t) = q(t)p(t) + r(t)$ with $\deg(r(t)) < n$. Thus $\bar{g}(t) = \bar{r}(t) \in A[t]/\langle p(t) \rangle$. Further, if $\bar{r}(t) = 0$ and $\deg(r(t)) < n$, then $r(t) = 0$, as

$$\bar{r} = 0 \iff r(t) = \sum_{\text{monic of deg } n} p(t) f(t) \Rightarrow \deg(r(t)) \geq n$$

this is a contradiction. Then we conclude that all $g(t)$ should be nonzero after reduction. $\bar{g}(t)$ is a polynomial in A with degree less than n . Then it is in $A \oplus At \oplus At^2 \oplus \dots \oplus At^{n-1}$. This tells us B is finite over A because $\bar{g}$ can be any element of B . $\square$

But what about the converse? In the lemma above, we showed for any $p(t) \in A[t]$ monic of degree n , we can construct a corresponding A -module B such that it is finite over A . Recall that

1. If A is a field and B is finite over A , $b \in B$ implies the existence of $p(t) \in A[t] \setminus \{0\}$ such that $p(b) = 0$.
2. then, if $A \subseteq B$ are general rings and B is finite over A , is there a monic $p(t) \in A[t]$ such that $p(b) = 0$ for $b \in B$?

Maybe now it is a good time to recall the Cayley-Hamilton Theorem from linear algebra without proof (and maybe *recall* is not a good word to use here because I did not know this at all).

Theorem 2 (Cayley-Hamilton). Every matrix A satisfies its own characteristic equation: $\det(A - I\lambda) = 0$.

Proof of 1. The set $\{1, b, b^2, \dots\}$ is linearly dependent because $\dim_A(B) < \infty$. □

Proof of 2. Consider $L_b : B \rightarrow B$ by $L_b(x) = b \cdot x$. L_b is A -linear. There exists a polynomial $p(t)$ such that $p(L_b) = 0$ is the characteristic polynomial of L_b . The existence of $p(t)$ is guaranteed by Cayley-Hamilton. Note that $p(L_b) = 0$ takes place in $\text{End}_A(B)$. Now we apply this to $1 \in B$ and get

$$p(L_b)(1) = 0 = p(b)$$

Here $p(t) = \det(tI_B - L_b)$ and $\deg(p(t)) = \dim(B)$. □

Remark. The *proof of 2* works for fields, but we want to generalize it to rings.

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Theorem 3 (generalized Cayley-Hamilton). Let A be a ring, M a finite A -module. Let $\varphi : M \rightarrow M$ be A -linear. Let $I \subset A$ be an ideal such that $\varphi(M) \subset IM$. Then there exists a monic polynomial $p(t) \in A[t]$ such that $p(\varphi) = 0 \in \text{End}_A(M)$. Furthermore, $p(t)$ can be taken as follows: Let $M = \langle \{m_1, \dots, m_n\} \rangle$ over A . Let $\varphi(m_i) = \sum_{j=1}^n a_{ij}m_j$. Let $X = (a_{ij})$. Then

$$p(t) = \det(tI_n - X) \in A[t]$$

Moreover, if $p(t) = t^n + a_{n-1}t^{n-1} + \dots + a_1t + a_0$, then all $a_i \in I$.

Remark. Before the proof of this theorem, let us recall that

$$IM = \left\{ \sum_{i=1}^n a_i m_i \mid a_i \in I \right\} \subseteq M$$

If $I = A$, then $IM = M$. And also recall a lemma from Lang's "Algebra" under the section "determinants".

Lemma. Let R be a ring, $M \in M_{n \times n}(R)$,

1. $\det(M) \in R$.

2. There exists a $n \times n$ matrix C such that
$$\begin{pmatrix} \det(M) & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \det(M) \end{pmatrix} \cdot C$$
 is the matrix whose (i, j) entry is $(-1)^{i+j} \det(M_{ji})$, where M_{ji} is (j, i) -minor. C is also known as the cofactor matrix.

Proof of the theorem. Choose a set $\{m_1, \dots, m_n\}$ such that $M = \text{span}_A\{m_1, \dots, m_n\}$ and then consider the matrix N whose (i, j) entry is determined by $\varphi(m_j) = \sum_{i=1}^n a_{ij}m_i$ (If $\varphi(M) \in IM$, then we can choose $a_{ij} \in I$). Now turn M into an $A[t]$ -module by letting $t \cdot M = \varphi(M)$ for any $m \in M$. Note that

$$F = \bigoplus_{i=1}^n (A[t])_i \xrightarrow{\pi} M \quad \pi(e_i) = \pi(0, \dots, 1, \dots, 0) = m_1$$

For any $n \times n$ matrix T over $A[t]$, consider $L_T : F \rightarrow F$ by $L_T(f) = T \cdot f$. The map $L_N : F \rightarrow F$ takes F into

$\ker(\pi)$. To see this, we should think of M as $\begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix}$, where m_i is a generator of M . And

$$L_{tI_n - N}(M) = (tI_n - N)M = tI_n \cdot M - N \cdot M = \begin{bmatrix} \varphi(m_1) \\ \vdots \\ \varphi(m_n) \end{bmatrix} - \begin{bmatrix} \sum_{i=1}^n a_{i1}m_i = \varphi(m_1) \\ \vdots \\ \sum_{i=1}^n a_{in}m_i = \varphi(m_n) \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

We conclude that $L_{tI_n - N}$ takes F into $\ker(\pi)$, so is $L_{C \cdot (tI_n - N)}$ where C is any $n \times n$ matrix. In particular,

the cofactors of $tI_n - N$. Moreover, let $H := \begin{pmatrix} \det(tI_n - N) & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \det(tI_n - N) \end{pmatrix}$, L_H takes F to $\ker(\pi)$ and

$p(t) := \det(tI_n - N)$ takes F to $\ker(\pi)$ (note that $tI_n \in A[t]$). Therefore, $p(t)$ acts as 0 on M if and only if $p(\varphi) = 0 \in \text{End}_A(M)$. Since all entries of N are in I ,

$$\det(tI_n - M) = \det \begin{pmatrix} t - a_{11} & \dots & a_{1j} \\ \vdots & \ddots & \vdots \\ a_{ji} & \dots & t - a_{nn} \end{pmatrix}$$

Then all coefficients below t^n of $p(t)$ are in I . □

Theorem 4 (integrality). Let $A \subseteq B$ be rings. Let $b \in B$. Then the following statements are equivalent

1. There exists a monic polynomial $p(t) \in A[t]$ such that $p(b) = 0$.
2. The subring generated by A and b , $A \langle b \rangle$ is finite as an A -module.
3. There exists a subring $A \subseteq B' \subseteq B$, and $b \in B'$ such that B' is finite as an A -module.

Proof. (1. $\Rightarrow$ 2.) This was shown in 1/24 as the first lemma. We have $A \langle b \rangle = A \oplus Ab \oplus Ab^2 \oplus \dots \oplus Ab^{k-1}$, $k = \deg(p)$. (2. $\Rightarrow$ 3.) This is achieved by setting $B' = A \langle b \rangle$. (3. $\Rightarrow$ 1.) Consider $L_b : B' \rightarrow B'$ by b multiplication. By Cayley-Hamilton, there exists a monic $p(t) \in A[t]$ such that $p(L_b) = 0$ in $\text{End}_A(B')$. Then for $1 \in B'$, $p(L_b)(1) = 0 = p(b)$ inside $B' \subseteq B$. Note that we need B' finite over A to use Cayley-Hamilton. □

Vocabulary 2.2 (integral). Let $A \subseteq B$ be rings, $b \in B$ is integral over A if it satisfies the theorem above. B is integral over A means all of $b \in B$ are integral over A . Note that B is finite over A implies B is integral over A by condition 3 of integrality.

Corollary. Let $A \subseteq B$ be rings, $A^{\text{int}} = \{b \in B \mid b \text{ is integral over } A\}$, then A^{int} is a subring of B .

Proof. Let b_1, b_2 be integral over A . There exists B_1 satisfying $A \subseteq B_1 \subseteq B$ and $b_1 \in B_1$ by condition 3 of integrality. Since b_2 is integral over A and $A \subseteq B_1$, by condition 1, b_2 satisfies some $p(t) \in A[t] \subseteq B_1[t]$. Thus b_2 is integral over B_1 . Then we apply condition 3 again to show the existence of B_2 , where B_2 satisfies $B_1 \subseteq B_2 \subseteq B$ and B_2 is finite over B_1 for $b_2 \in B_2$. So $b_1 + b_2, b_1b_2 \in B_2$.

Now we check that B_2 is finite over A . If

$$B_1 = \text{span}_A\{d_1, \dots, d_n\} \quad B_2 = \text{span}_{B_1}\{e_1, \dots, e_s\}$$

Then $B_2 = \text{span}_A\{d_i e_j\}$. B_2 is clearly finite over A , which implies both $b_1 + b_2$ and b_1b_2 are integral over A . □

Remark (very quick). $A \subseteq A^{\text{int}}$ as $a \in A$ satisfies $t - a$.

Example. Let $A = \mathbb{Z}$ and B be a finite field extension of $\mathbb{Q}$, then $A^{\text{int}} = \{b \in B \mid b \text{ is integral over } \mathbb{Z}\}$. This is the set of algebraic integers.

Example. Let K be an algebraically closed field. Let $A = K[t_1, \dots, t_n]$ and B be a finite field extension of $K(t_1, \dots, t_n)$. A^{int} is finitely generated K -algebra, which is finite over $K[t_1, \dots, t_n]$.

Proposition 1 (Nakayama lemma). Let A be a ring, M a finitely generated A -module, $I \subseteq A$ an ideal, then for some $I \subseteq A$, $M = IM$ implies the existence of $a \in I$ such that $(1 - a)M = 0$, i.e. $M = \langle a \rangle M$.

Proof. Apply Cayley-Hamilton for $\varphi = \text{id}$. Let $N = (a_{ij})$, where $\text{id}(m_j) = \sum_{i=1}^n a_{ij}m_i = m_j$. All a_{ij} are chosen from I . Then we have $p(t) = t^n + a_{n-1}t^{n-1} + \dots + a_1t + a_0 \in A[t]$, $a_i \in I$ such that $p(\text{id}) = 0$ is the 0 transformation with $a_i \in I$. Now apply this to m and get

$$p(\text{id})(m) = m + a_{n-1}m + \dots + a_0m = (1 + a_{n-1} + \dots + a_0)m = 0$$

Taking $a = -a_{n-1} - a_{n-2} - \dots - a_0$, we have recovered $(1 - a)M = 0$. $\square$

Week 3

Proved "lying over", preparation, and noether normalization. Defined coordinate rings. Finished all versions of Hilbert's Nullstellensatz. Proved some corollaries that followed from all this.

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Theorem 5 ("lying over"). Let $A \subseteq B$ be integral. Suppose $\mathfrak{m} \subsetneq A$ is a maximal ideal, then there exists $\mathfrak{m}' \subsetneq B$ maximal such that $\mathfrak{m}' \cap A = \mathfrak{m}$.

Proof. To prove this theorem, we need to use the following claims:

Claim. Recall that $\mathfrak{m}B = \{\sum m_i b_i | m_i \in M, b_i \in B\}$. If $\mathfrak{m}B = B$, then there exists B' that satisfies $A \subseteq B' \subseteq B$, B' is finite over A such that $\mathfrak{m}B' = B'$.

Proof: $\mathfrak{m}B = B$ if and only if $1 \in \mathfrak{m}B$ if and only if there exists $\{m_1, \dots, m_r\}$ and $\{b_1, \dots, b_r\}$ such that $\sum_{i=1}^r m_i b_i = 1$. This implies $\mathfrak{m}A \langle b_1, \dots, b_r \rangle = \langle b_1, \dots, b_r \rangle$ because both sides have 1. Setting $B' = A \langle b_1, \dots, b_r \rangle$, we have $\mathfrak{m}B' = B'$. Since B is integral over A , each b_i is integral over A . In particular, B' is integral over A . $\square$

Claim. If $\mathfrak{m}B' = B'$, then there exists $a \in \mathfrak{m}$ such that $(1 - a)B' = 0$.

Proof: Recall that in commutative algebra, both ideals and quotient rings are modules. Applying Nakayama to B' yields $(1 - a)B' = 0$ for some $a \in \mathfrak{m}$. Then $1 - a = 0 \in B'$ and $a = 1$ in B' hence also 1 in A . This is a contradiction because $\mathfrak{m} \subsetneq A$. $\square$

From the claims above, we conclude that $\mathfrak{m}B \subsetneq B$. Note that $\mathfrak{m}B$ is both an A -submodule and the ideal generated by M . Then there exists $\mathfrak{m}' \subseteq B$ maximal such that $\mathfrak{m}B \subseteq \mathfrak{m}'$. This implies $\mathfrak{m}B \cap A \subseteq \mathfrak{m}' \cap A$. In particular, $\mathfrak{m} \subseteq \mathfrak{m}B \cap A$ implies $\mathfrak{m}' \cap A = \mathfrak{m}$. $\square$

Remark. $A \subseteq B$ being integral is important. Otherwise, we can set $A = K[t]$, $B = K[t, t^{-1}]$, and $\mathfrak{m} = \langle t \rangle$. There is no $\mathfrak{m}'$ as in the theorem since t is a *unit* in B .

Corollary (Fact 1). Let A, B be domains and $A \subseteq B$ integral. Then A is a field if and only if B is a field.

Proof. $(\Rightarrow)$ A is a field. Let $b \in B$, then $B' = A \langle b \rangle$ is a finite dimensional vector space over A . The linear map $L_b : A \langle b \rangle \rightarrow A \langle b \rangle$ is injective since B is a domain. Then L_b is surjective, which implies the existence of some $b' \in A \langle b \rangle$ such that $bb' = 1$. $(\Leftarrow)$ Let B be a field, let $\mathfrak{m} \subseteq A$ be maximal, then there exists $\mathfrak{m}' \subseteq B$ maximal $\mathfrak{m}' \cap A = \mathfrak{m}$. Since B is a field, its maximal ideal is 0. It is also the maximal ideal of A . Thus A is a field. $\square$

Before getting into the preparation lemma and noether normalization, we should do the following lemma to help us a little bit.

Lemma. Let K be an infinite field and let $f \in K[x_1, \dots, x_n] \setminus \{0\}$. There exists $(\lambda_1, \dots, \lambda_n) \in \mathbb{A}_K^n$ such that $f(\lambda_1, \dots, \lambda_n) \neq 0$.

Proof. We use induction on n . When $n = 1$, let $f \in K[x] \setminus \{0\}$, then

$$\{\# \text{ of zeros of } f\} \leq \deg(f) \leq \infty$$

This implies the existence of $\lambda \in K$ such that $f(\lambda) \neq 0$ since K is an infinite field. Now consider the case when $n > 1$. Without loss of generality, we can assume $x_1 \in f$. Then

$$f = x_1^{d_1} g_{d_1}(x_2, \dots, x_n) + x_1^{d_1-1} g_{d_1-1}(x_2, \dots, x_n) + \dots + x_1 g_1(x_2, \dots, x_n) + g_0(x_2, \dots, x_n)$$

By assumption, $d_1 \geq 1$. Thus, can choose $\lambda_2, \dots, \lambda_n$ such that $g_{d_1}(\lambda_2, \dots, \lambda_n) \in K[x_1]$. Note that f here is nontrivial of degree $d_1 \geq 1$. Then we can use the $n = 1$ case and claim the existence of λ_1 such that $f(\lambda_1, \dots, \lambda_n) \neq 0$. $\square$

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Proposition 2 (preparation). Let K be a field, let $f \in K[x_1, \dots, x_n] \setminus \{0\}$, there exists $\{t_1, \dots, t_n\} \subseteq K[x_1, \dots, x_n]$ such that

1. $K\langle t_1, \dots, t_n \rangle = K[t_1, \dots, t_n] = K[x_1, \dots, x_n]$.
2. $f(t_1, \dots, t_n)$ is of the form

$$\alpha (t_n^d + t_n^{d-1} g_{d-1}(t_1, \dots, t_{n-1}) + t_n^{d-2} g_{d-2}(t_1, \dots, t_{n-1}) + \dots + t_n g_1(t_1, \dots, t_{n-1}) + g_0(t_1, \dots, t_{n-1}))$$

where $\alpha \in K^\times$. In other words, up to α , f is monic in $K[t_1, \dots, t_{n-1}][t_n]$. $\deg(f)$ is to the total degree of f .

Example. Let $f(x, y) = xy$, $t_1 = x$, $t_2 = x + y$, and $t_2 - t_1 = y$. $f(t_1, t_2) = t_1(t_2 - t_1) = -(t_1^2 - t_1 t_2)$. This is monic in t_1 .

Fact 3. Let K be an infinite field and let $f(x_1, \dots, x_n) \neq 0$ in $K[x_1, \dots, x_n]$, then there exists $(\lambda_1, \dots, \lambda_n) \in \mathbb{A}_K^n$ such that $f(\lambda_1, \dots, \lambda_n) \neq 0$. This was proven as a Lemma after the proof of Fact 1.

Proof of Preparation. (1) When $n = 1$, this problem is trivial. So without loss of generality, we can assume $n > 1$.

Set $\begin{cases} t_i = x_i - \lambda_i x_n = x_i - \lambda_i t_n & \text{for } i < n \\ t_n = x_n & \text{for } i = n \end{cases}$ for $\lambda_i \in K$ to be chosen later. The map $\varphi : K[x_1, \dots, x_n] \rightarrow K[t_1, \dots, t_n]$ given by $\varphi(x_i) = t_i$ and $\varphi(\lambda) = \lambda$. This is obviously a well-defined ring homomorphism. $\varphi^{-1} : K[t_1, \dots, t_n] \rightarrow K[x_1, \dots, x_n]$ given by $\varphi^{-1}(t_i) = x_i + \lambda_i x_n$ and $\varphi^{-1}(x_n) = x_n$ makes φ a K -algebra automorphism. Hence $K[x_1, \dots, x_n] = K[t_1, \dots, t_n]$. (2) Let $d = \deg(f(x_1, \dots, x_n))$. Recall $\mathbf{I} = (i_1, \dots, i_n) \in (\mathbb{Z}^{\geq 0})^n$ and $|\mathbf{I}| = i_1 + \dots + i_n$. Then rewrite

$$f = \sum_{|\mathbf{I}|=d} a_{\mathbf{I}} x^{\mathbf{I}} + g, \quad \deg(g) < d$$

$a_{\mathbf{I}} x^{\mathbf{I}} = a_{\mathbf{I}} x_1^{i_1} \dots x_n^{i_n} = a_{\mathbf{I}} (t_1 + \lambda_1 t_n)^{i_1} (t_2 + \lambda_2 t_n)^{i_2} \dots t_n^{i_n}$. Expanding

$$\begin{aligned} a_{\mathbf{I}} x^{\mathbf{I}} &= a_{\mathbf{I}} (t_1 + \lambda_1 t_n)^{i_1} = a_{\mathbf{I}} \left(\sum_{j=0}^{i_1} \binom{i_1}{j} t_1^{i_1-j} (\lambda_1 t_n)^j \right) = a_{\mathbf{I}} ((\lambda_1 t_n)^{i_1} + t_n \text{ terms with degree } < i_1) \\ &= a_{\mathbf{I}} \lambda_1^{i_1} \lambda_2^{i_2} \dots \lambda_{n-1}^{i_{n-1}} t^{i_1+\dots+i_n=d} + p_{\mathbf{I}}(t_{i_1}, \dots, t_{i_n}) \end{aligned}$$

The degree of $p_{\mathbf{I}}$ in t_n is less than d . Let g be some polynomial whose degree in t_n is less than d . We rewrite

$$f = \left(\sum_{|\mathbf{I}|=d} a_{\mathbf{I}} \lambda_1^{i_1} \dots \lambda_{n-1}^{i_{n-1}} \right) t_n^d + \sum_{|\mathbf{I}|=d} p_{\mathbf{I}}(t_1, \dots, t_n) + g$$

By the above Lemma, $|K| = \infty$ means that we can choose $\lambda_1, \dots, \lambda_{n-1}$ such that $\sum_{|\mathbf{I}|=d} a_{\mathbf{I}} \lambda_1^{i_1} \dots \lambda_{n-1}^{i_{n-1}} \neq 0 \in K$. With this choice, f is monic of degree d . $\square$

Theorem 6 (noether normalization). Let K be an infinite field and A be a finitely generated K -algebra. Then there exist algebraically independent $\{t_1, \dots, t_d\} \subseteq A$, possibly empty, such that

1. $K\langle t_1, \dots, t_d \rangle = K[t_1, \dots, t_d]$.
2. A is finite over $K[t_1, \dots, t_d]$, i.e. finitely generated as a $K[t_1, \dots, t_d]$ -module.

Proof. Let n represents the number of generators. Induct on n

- (i) If $n = 0$, set $A = K$, $\{t_1, \dots, t_d\} = 0$.
- (ii) If $n = 1$, set $A = K[t]/I$
 - (a) If $I = 0$, take $t_1 = t$, $A = K[t] = K[t_1]$ is a finitely generated module over itself.
 - (b) If $I \neq 0$, for $A = K[t]/I \neq 0$, we must have t involved in I . But this will limit the power of t in the quotient to some finite number. Therefore, $\dim_K(A) < \infty$. This shows A is a finite dimensional vector space over K . We take $\{t_1, \dots, t_d\} = 0$.
- (iii) If $n > 1$, set $A = K[x_1, \dots, x_n]/I$.
 - (a) If $I = 0$, take $t_i = x_i$ for all i . Then $K[x_1, \dots, x_n]$ is a finitely generated $K[t_1, \dots, t_n]$ -module.
 - (b) If $I \neq 0$, choose $0 \neq f \in I$. We get a surjection

$$K[x_1, \dots, x_n]/\langle f \rangle \twoheadrightarrow K[x_1, \dots, x_n]/I$$

Now choose $t_1, \dots, t_n$ as in the Proposition 2 for f making it monic of degree d in t_n . By change of variables, $K[x_1, \dots, x_n]/\langle f \rangle = K[t_1, \dots, t_n]/\langle f \rangle$. In other words, for any $g \in K[t_1, \dots, t_n]/\langle f \rangle$, $g = g_1 + g_2 t_n + \dots + g_d t_n^{d-1}$, where $g_i \in K[t_1, \dots, t_{n-1}]$. Define

$$K[t_1, \dots, t_{n-1}] \xrightarrow{\quad} K[x_1, \dots, x_n]/\langle f \rangle \twoheadrightarrow K[x_1, \dots, x_n]/I = A$$

$\searrow \varphi$

Set $B = \text{im}(\varphi)$. B is a sub- K -algebra generated by $n - 1$ elements. A is a finite B -module as the images of $\{1, t_n, \dots, t_n^{d-1}\}$ generate A as a B -module.

Suppose I is nonzero. By our inductive hypothesis, we know the results for algebras generated by $n - 1$ or fewer elements. For B , there exists $\{t_1, \dots, t_d\} \subseteq B$, B is finite over $K[t_1, \dots, t_d]$ and $K\langle t_1, \dots, t_d \rangle = K[t_1, \dots, t_d]$. This implies A is finite over $K[t_1, \dots, t_d]$. □

Remark. Note that $K\langle t_1, \dots, t_d \rangle$ is the image of the evaluation map $\text{ev} : K[t_1, \dots, t_d] \rightarrow A$. Therefore, $K\langle t_1, \dots, t_d \rangle = K[t_1, \dots, t_d]$ if t_i are algebraically independent. Now we should be able to prove the weak Nutz.

Corollary. Let A be a finitely generated K -algebra where A is a field. Then A is a finite field extension of K .

Proof. Assuming $|K| = \infty$, let $K[t_1, \dots, t_n] \subseteq A$ as in Theorem 6 (noether normalization). Since inclusion implies integral, it follows that A is a field implies $K[t_1, \dots, t_n]$ is a field. $\{t_1, \dots, t_n\} = 0$ implies A is finite over K . □

Corollary (weak Nutz). Let $K = \overline{K}$. Let A be a finitely generated K -algebra. Each maximal ideal of A is the kernel of a unique K -algebra homomorphism $\chi : A \rightarrow K$. In other words, $\chi(\lambda) = \lambda$, for all $\lambda \in K$.

Proof. Let $\mathfrak{m}$ be a maximal ideal, $A/\mathfrak{m}$ is a field. By the previous Corollary, $A/\mathfrak{m}$ is a finite field extension of K . But algebraically closed fields have no proper finite field extensions. It follows that $A/\mathfrak{m} = K$. So

$$\mathfrak{m} = \ker(A \rightarrow A/\mathfrak{m} = K) = \ker(\chi)$$

To show χ is unique, observe that $\chi : A \rightarrow K$ is surjective and χ is K -linear. So $\chi(\lambda a) = \chi(\lambda)\chi(a) = \lambda \cdot \chi(a)$ for all $\lambda \in K$. We write $A = K \cdot 1 \oplus \ker(\chi)$ because $a \in A$ implies $a - \chi(a) \in \ker(\chi)$ and $a = \chi(a) + a - \chi(a)$. Then $\chi = \text{proj}(A \rightarrow K \cdot 1)$. We see that $\ker(\chi)$ determines χ and hence χ is unique (Aside: We have seen this proof for $A = \text{polynomial rings}$). □

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Now we can prove Hilbert's Nutz I. In 1/22 we simply assumed two important facts without proofs. Recall that $I(Z(I)) = \{f \in K[x_1, \dots, x_n] \mid f|_{Z(I)} = 0\}$.

Theorem 7 (Hilbert's Nutz I). Let $I \subseteq K[x_1, \dots, x_n]$ be an ideal, then $I(Z(I)) = \sqrt{I}$.

Proof. Rewriting this in terms of bijections, we start from

$$\mathbb{A}_K^n \leftrightarrow \text{maximal ideal of } K[x_1, \dots, x_n]$$

then note that for all $f \in I$: $(a_1, \dots, a_n) \in Z(I)$ if and only if $f(a_1, \dots, a_n) = 0$ if and only if $f \in \langle x_1 - a_1, \dots, x_n - a_n \rangle$ if and only if $I \subseteq \langle x_1 - a_1, \dots, x_n - a_n \rangle$. From this, we can conclude the following correspondence

$$Z(I) \leftrightarrow \{\mathfrak{m} \in K[x_1, \dots, x_n] \mid I \subseteq \mathfrak{m}\}$$

Let $X \subseteq \mathbb{A}_K^n$, $I(X) = \{f \in K[x_1, \dots, x_n] \mid f|_X = 0\}$. We have the correspondence $X \leftrightarrow$ maximal ideals $\{\mathfrak{m}_x\}$. This $\mathfrak{m}_x$ simply means the maximal ideal associated with the point $x \in X \subseteq \mathbb{A}_K^n$. Rewrite $I(X) = \{f \mid f \in \mathfrak{m}_x, \forall x \in X\} = \bigcap_{x \in X} \mathfrak{m}_x$. Let $I \subseteq K[x_1, \dots, x_n]$ be an ideal. We want to show $\bigcap_{I \subseteq \mathfrak{m}} \mathfrak{m} = \sqrt{I}$ ($\iff I(Z(I)) = \sqrt{I}$). Aside: this is not true for general rings¹. So we

Claim. Let A be a finitely generated K -algebra (in our case $K[x_1, \dots, x_n]$ satisfies these requirements), then for all ideals $I \subseteq A$, $\sqrt{I} = \bigcap_{I \subseteq \mathfrak{m}} \mathfrak{m}$.

Proof: ($\subseteq$) In any ring, $\sqrt{I} \subseteq \bigcap_{I \subseteq \mathfrak{m}} \mathfrak{m}$. To see this, note that $f \in \sqrt{I}$ implies $f^N \in I$ for some $N \geq 1$. If $I \subseteq \mathfrak{m}$, then $f^N \in \mathfrak{m}$ implies $f \in \mathfrak{m}$. Thus $f \in \bigcap_{I \subseteq \mathfrak{m}} \mathfrak{m}$. ($\supseteq$) Assume $f \notin \sqrt{I}$, we want to show there exists $\mathfrak{m} \subseteq A$ maximal such that $f \notin \mathfrak{m}$. With this assumption in mind, $\bar{A} := A/I$ if and only if $\bar{f} := f + I$ is non-nilpotent. Then the localization at $S = \{\bar{f}\}$ gives $S^{-1}\bar{A} \simeq \bar{A}[t]/\langle t\bar{f} - 1 \rangle$ by Theorem A1(5). This is a nonzero finitely generated K -algebra. Then apply the Weak Nutz: there exist a maximal ideal $\bar{\mathfrak{m}}$ of $S^{-1}\bar{A}$ and a map $\chi : S^{-1}\bar{A} \rightarrow K$ such that $\bar{\mathfrak{m}} = \ker(\chi)$.

$$\begin{array}{ccc} \bar{A} & \xrightarrow{\varphi} & S^{-1}\bar{A} \xrightarrow{\chi} K \\ & \searrow \psi & \nearrow \\ & & K \end{array}$$

ψ is a K -algebra homomorphism. Since the preimage of maximal ideal of finitely generated algebra is maximal, $\ker(\psi) = \varphi^{-1}(\bar{\mathfrak{m}}) \cap \bar{A}$ is maximal in $\bar{A}$. $\bar{f} \notin \ker(\psi)$ otherwise $\psi(\bar{f}) = 0$ would imply $\chi(\bar{f}) = 0$. This is a contradiction since $\bar{f}$ is invertible in $S^{-1}\bar{A}$. Recall that suppose J is an ideal of R containing I and let $\bar{J} := J/I$ be the corresponding ideal in the quotient ring. Then J is radical/prime/maximal if and only if $\bar{J}$ is radical/prime/maximal. Now that $\ker(\psi)$ is maximal in $\bar{A}$, then $\ker(\psi) = \mathfrak{n}/I =: \bar{\mathfrak{n}}$ for some $\mathfrak{n}$ maximal in A containing I . It follows that $\bar{f} \notin \ker(\psi) = \bar{\mathfrak{n}}$. Thus $f \notin \mathfrak{n}$. $\square$

The theorem follows from this claim: $I(Z(I)) = I(X) = \bigcap_{I \subseteq \mathfrak{m}} \mathfrak{m} = \sqrt{I}$. $\square$

Corollary. The maps $\{\text{closed subsets of } \mathbb{A}_K^n\} \leftrightarrow \{\text{radical ideals of } K[x_1, \dots, x_n]\}$ given by

$$X \rightarrow I(X) \quad Z(I) \leftarrow I$$

are inclusion-reversing inverses.

Proof. The inclusion-reversing part is done. The maps are defined. We need to show (1) $X = Z(I(X))$ and (2) $I = I(Z(I))$. But as shown above, $I = I(Z(I))$ when $I = \sqrt{I}$. For (1), let $X = Z(I)$ for some I since X is closed. Then $Z(I(X)) = Z(I(Z(I))) = Z(\sqrt{I}) = Z(I)$ for all ideals I . Then

$$I \subset \sqrt{I} \Rightarrow Z(I) \supseteq Z(\sqrt{I})$$

Let $(a_1, \dots, a_n) \in Z(I)$, $f(a_1, \dots, a_n) = 0$ for all $f \in I$. Let $g \in \sqrt{I}$ and $g^N \in I$ for some N . Then $g^N(a_1, \dots, a_n) = 0$, which implies $g(a_1, \dots, a_n) = 0$. So $(a_1, \dots, a_n) \in Z(\sqrt{I})$ implies $Z(I) \subseteq Z(\sqrt{I})$. $\square$

Vocabulary 3.1 (coordinate ring in $\mathbb{A}_K^n$). Let $X \subseteq \mathbb{A}_K^n$ be closed. The coordinate ring is $K[X] = K[x_1, \dots, x_n]/I(X)$. We get the map $\sigma : K[X] \hookrightarrow \{\text{functions } |X \rightarrow \mathbb{A}_K^1\}$. If $U \subseteq X$ is a closed subset, then $K[U] = K[X]/I(U)$.

Remark. If $f \in K[x_1, \dots, x_n]$, then f yields a function and its restriction to X

$$\text{ev}(f) : \mathbb{A}_K^n \rightarrow \mathbb{A}_K^1 \quad \text{ev}(f)|_X : X \rightarrow \mathbb{A}_K^1$$

The map $K[x_1, \dots, x_n] \mapsto \{\text{functions } |X \rightarrow \mathbb{A}_K^1\}$ is given by $f \mapsto \text{ev}(f)|_X$ with kernel $I(X)$. So we get the map σ in the definition.

¹For example, take $A = K[[t]]$, where K is a field. A has a unique maximal ideal $\mathfrak{m} = \{f \in K[[t]] \mid f = a_1 t + a_2 t^2 + \dots\}$. Let $I = 0$, $\sqrt{I} = 0$, then $\sqrt{I} \neq \bigcap_{I \subseteq \mathfrak{m}} \mathfrak{m} = \mathfrak{m}$. Rings that satisfy this are called *Jacobson rings*.

Corollary (Nutz for $K[X]$). Let $I \subseteq K[X]$ be an ideal, then $I(Z(I)) = I\{x \in X | f(x) = 0, \forall f \in I\} = \sqrt{I}$. For $Y \subseteq X$, $I(Y) = \{f \in K[X] | f|_Y = 0\}$. As a result, there exists an inclusion-reversing bijection

$$\{Y \subseteq X | Y \text{ is closed}\} \leftrightarrow \{I \subseteq K[X] | I = \sqrt{I}\}$$

Two sides of the map are given by $Y \mapsto I(Y)$ and $I \mapsto Z(I)$.

Proof. The proof of this corollary is similar to what we have done

1. $X \leftrightarrow \{\text{maximal ideals of } K[X]\} = \{\text{maximal ideals of } K[x_1, \dots, x_n] \text{ containing } I(X)\}$.
2. $J \subseteq K[X]$ an ideal. Then $Z(J) \leftrightarrow \{\mathfrak{m} \subseteq K[X] | \mathfrak{m} \text{ maximal and } \mathfrak{m} \supseteq J\}$. Write $J = \overline{J'}$, $J' \subseteq K[x_1, \dots, x_n]$ ideal containing $I(X)$ (remember that $K[X] = K[x_1, \dots, x_n]/I(X)$).
3. Observe

$$I(Z(J)) = \bigcap_{\substack{\mathfrak{m} \subseteq K[X] \\ J \subseteq \mathfrak{m}}} \mathfrak{m} = \sqrt{J}$$

□

Week 4

Showed $K[X]$ is the set of regular functions. Defined regular maps of varieties. Facts about regular maps. Notion of isomorphism between varieties. Defined closed immersions and dominant maps.

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Another way to see the last corollary is that for each $X \subseteq \mathbb{A}_K^n$ closed, we attached $K[X]$ a finitely generated K -algebra with no nonzero nilpotent elements.

Q: If $X \subseteq \mathbb{A}_K^n$ and $Y \subseteq \mathbb{A}_K^m$ are closed, what is the correct notion of map $X \rightarrow Y$ in algebraic geometry?

A: Given the above assumption, a map $\varphi : X \rightarrow Y$ should be of the form

1. $\varphi = (\varphi_1, \dots, \varphi_m)$, $\varphi_i \in K[\mathbb{A}_K^n] = K[x_1, \dots, x_n]$.
2. Desired functions that $X \rightarrow \mathbb{A}_K^1$ should form a ring.
3. If $h : X \rightarrow \mathbb{A}_K^1$ is desired and has no zeros, then $1/h$ is also desired.
4. From (1), deduce that $K[X] \subseteq \{\text{desired functions } X \rightarrow \mathbb{A}_K^1\}$. $K[X]$ consists of polynomials in n variables and they satisfy the requirements. Now for some $p : (X \subseteq \mathbb{A}_K^n) \rightarrow \mathbb{A}_K^1$, p is desired implies $p|_X$ is desired.

Vocabulary 4.1 (locally regular function). Let $X \subseteq \mathbb{A}_K^n$ be closed, $\varphi : X \rightarrow \mathbb{A}_K^1$ be continuous. φ is regular if for all $d \in X$, there exists some open U_d that contains d and $f, g \in K[x_1, \dots, x_n]$ such that $g|_{U_d} \neq 0$, $f/g = \varphi$ on U_d . We can also view this as "locally rational".

Remark. If $V \subseteq X$ is any open subset, $\varphi : V \rightarrow \mathbb{A}_K^1$ can be defined to be regular using the same definition.

Theorem 8. $K[X] \subseteq \{\text{set of regular functions on } X\}$.

Proof. Any function φ in $K[X]$ is of the required form f/g with $g = 1$. We need to show φ is continuous with respect to Zariski. In other words, we are showing the preimage of a closed set under φ is closed for $\varphi : X \rightarrow \mathbb{A}_K^1$. Let $Z \subseteq \mathbb{A}_K^1$ be closed; it is the zero set of a polynomial $f \in K[y]$. If $f = 0$, then $Z = \mathbb{A}_K^1$ as everything vanishes at 0. If $f \neq 0$, it has finitely many roots; hence Z is a finite set. Write $Z = \bigcup_{i=1}^r \{z_i\}$. Then $\varphi^{-1}(Z)$ is closed if and only if $\varphi^{-1}(z_i)$ for each i is closed. Without loss of generality, assume $Z = \{z_i\}$, $z_i \in K$. $\varphi^{-1}(z_i) = \{x \in X | \varphi(x) = z_i\} = Z(\varphi(y) - z_i)$. Since $\varphi(y) \in K[X] \subseteq K[y]/I(X) \subseteq K[y]$, $\varphi(y) - z_i$ is also a polynomial. By definition, $Z(\varphi(y) - z_i)$ is closed. □

Vocabulary 4.2 (basic open). Let $X \subseteq \mathbb{A}_K^n$ be closed. Let $f \in K[X]$. $D(f) = \{x \in X | f(x) \neq 0\} = X \setminus Z(f)$ is a basic open subset of X .

Lemma. Let $U \subseteq X$ be open, $p \in U$. There exists a basic open $D(f)$ such that $p \in D(f) \subseteq U$. In other words, basic open sets generate the Zariski topology.

Proof. Let $Z = X \setminus U$, $Z \subsetneq X$ be closed. So there exists an ideal $0 \neq I \subsetneq K[X]$ such that $Z(I) = Z$ (correspondence between $I \subseteq K[\dots]$ and subvarieties). Define the maximal ideal that vanishes at p : $\mathfrak{m}_p = \{f \mid f(p) = 0\}$. $p \notin Z$ implies $I \not\subseteq \mathfrak{m}_p$. Then there exists $g \in I$ such that $g \notin \mathfrak{m}_p$. Consider $D(g) \subseteq X$, ① $p \in D(g)$ because $g(p) \neq 0$ ② $Z(I) = Z \subseteq Z(g)$ because $f \in I$. Deduce that $D(g) \subseteq U$. $\square$

Theorem 9. $\{\text{set of regular functions on } X\} \subseteq K[X]$.

Proof. By definition and the lemma above, $X = \bigcup_{\alpha} D(h_{\alpha})$ such that $\varphi|_{D(h_{\alpha})} = g_{\alpha}/k_{\alpha}$, where k_{α} does not vanish on $D(h_{\alpha})$ and h_{α} does not vanish on $D(h_{\alpha})$. Then $g_{\alpha}/k_{\alpha} = \frac{g_{\alpha}h_{\alpha}}{k_{\alpha}h_{\alpha}}$. Since k_{α} does not vanish on $D(h_{\alpha})$, $Z(h_{\alpha}) = Z(k_{\alpha}h_{\alpha})$. Then replace k_{α} with $k_{\alpha}h_{\alpha}$ we get (this is true when limited to $D(h_{\alpha})$)

$$\varphi|_{D(h_{\alpha})} = \frac{g_{\alpha}}{h_{\alpha}}$$

Consider $h_{\alpha}^2\varphi : X \rightarrow \mathbb{A}_K^1$ ① On $D(h_{\alpha})$, $h_{\alpha}^2\varphi = h_{\alpha}^2(g_{\alpha}/h_{\alpha}) = h_{\alpha}g_{\alpha} \in K[X]$. ② On $Z(h_{\alpha})$, $h_{\alpha}^2\varphi = 0 = h_{\alpha}g_{\alpha}$. So $h_{\alpha}^2\varphi = g_{\alpha}h_{\alpha}$ on X , $h_{\alpha}^2\varphi \in K[X]$.

Claim. $I = \{h_{\alpha}^2\}_{\alpha}$.

Proof: We want to show I is not contained in any maximal ideal (See Theorem A6). By the Nutz, this is the same as showing $Z(I) = \emptyset$ (recall that if I is the whole thing, its corresponding subvariety is $\emptyset$). Suppose $p \in Z(I)$, $h_{\alpha}^2(p) = 0$ for all α implies $h_{\alpha}(p) = 0$ for all α . But $X = \bigcup_{\alpha} D(h_{\alpha})$, there exists α' such that $h_{\alpha'} \neq 0$. This is a contradiction. $\square$

This means the elements of $\{h_{\alpha}^2\}_{\alpha}$ can make up $1 \in K[X]$. Then there exists a finite set of functions $\{t_{\alpha}\}$ in $K[X]$ so that for a finite set F

$$\sum_{\alpha \in F} t_{\alpha} h_{\alpha}^2 = 1 \Rightarrow \sum_{\alpha \in F} t_{\alpha} (h_{\alpha}^2 \varphi) = \varphi$$

on X . $t_{\alpha} \in K[X]$ and $h_{\alpha}^2\varphi \in K[X]$ imply $\sum t_{\alpha} h_{\alpha}^2\varphi \in K[X]$. Every component is in the ring implies the whole thing is in the ring. $\square$

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Vocabulary 4.3 (regular map between varieties). A continuous function between two varieties $\varphi : X \rightarrow Y$ is regular if for all $f : X \rightarrow \mathbb{A}_K^1$ regular, $f \circ \varphi : X \rightarrow Y \xrightarrow{f} \mathbb{A}_K^1$ is also regular.

Remark. Any reasonable definition of regular maps $X \rightarrow Y$ will imply this condition.

Lemma. If A is a finitely generated reduced K -algebra, then $A = K[Y]$ for some Y .

Proof. $A = K[x_1, \dots, x_n]/I$, $I = \sqrt{I}$. So $I = I(Z(I))$, set $Y = Z(I)$. Y is unique up to isomorphisms of varieties. $\square$

Proposition 3. Let $X \subseteq \mathbb{A}_K^n$ closed and $\varphi : X \rightarrow \mathbb{A}_K^1$ continuous. $\varphi \in K[X]$ if and only if φ is a regular map.

Proof. Recall that φ is regular means for all $f : \mathbb{A}_K^1 \rightarrow \mathbb{A}_K^1$ regular, $f \circ \varphi$ is regular. Assume φ is regular, $f \circ \varphi$ is regular whenever $f : \mathbb{A}_K^1 \rightarrow \mathbb{A}_K^1$ is regular. Choose $f = \text{id}_{\mathbb{A}_K^1}$ so that $f(t) = t \in K[t]$. It follows that $\text{id} \circ \varphi = \varphi$ is regular. Conversely, let $\varphi \in K[X]$, $f : \mathbb{A}_K^1 \rightarrow \mathbb{A}_K^1$ regular such that $f(t) = \sum_{i=0}^{\infty} a_i t^i$. $f \circ \varphi = \sum_{i=0}^{\infty} a_i \varphi^i$ is in $K[X]$ as $K[X]$ is a K -algebra. $\square$

Proposition 4. Let $\varphi : X \rightarrow \mathbb{A}_K^m$ be regular and $\pi_i : \mathbb{A}_K^m \rightarrow \mathbb{A}_K^1$ be the i^{th} coordinate projection, then $\pi_i \circ \varphi$ is regular for all i . Conversely, if $\pi_i \circ \varphi$ is regular for all i , $\varphi : X \rightarrow \mathbb{A}_K^m$ is regular.

Proof. For each i , the map $\pi_i : \mathbb{A}_K^m \rightarrow \mathbb{A}_K^1$ is regular as $\pi_i = t_i \in K[t_1, \dots, t_m]$. Note that t_i is the i^{th} coordinate function. So this is the same thing as π_i picking out the i^{th} coordinate. If φ is regular, $\pi_i \circ \varphi$ is regular for all i . Conversely, suppose $\pi_i \circ \varphi$ is regular for all i . Let $f : \mathbb{A}_K^m \rightarrow \mathbb{A}_K^1$ be regular. By Proposition 3, $f \in K[\mathbb{A}_K^m] = K[x_1, \dots, x_m]/0$. So $f = \sum_{\mathbf{I}} a_{\mathbf{I}} t^{\mathbf{I}}$. When composed with φ , we get

$$f \circ \varphi = \sum_{\mathbf{I}} a_{\mathbf{I}} (t_1 \circ \varphi)^{i_1} \dots (t_m \circ \varphi)^{i_m} = \sum_{\mathbf{I}} a_{\mathbf{I}} (\pi_1 \circ \varphi)^{i_1} \dots (\pi_m \circ \varphi)^{i_m}$$

It follows that $f \circ \varphi \in K[X]$. $\square$

Proposition 5. Additional facts about regular maps

1. Let X, Y, Z be varieties. Suppose $\varphi : X \rightarrow Y$ and $\psi : Y \rightarrow Z$ are regular maps. $\psi \circ \varphi : X \rightarrow Z$ is regular.
2. Let $X \subseteq \mathbb{A}_K^n$ be closed. Then the inclusion map $i : X \rightarrow \mathbb{A}_K^n$ is regular.

Proof. (1.) Let $f : Z \rightarrow \mathbb{A}_K^1$ be regular. We want to show $f \circ (\psi \circ \varphi)$ is regular. Since ψ is regular, $f \circ \psi : Y \rightarrow \mathbb{A}_K^1$ is regular by definition. Similarly, $(f \circ \psi) \circ \varphi$ is regular. But $(f \circ \psi) \circ \varphi = f \circ (\psi \circ \varphi)$. (2.) Let $\pi_i : \mathbb{A}_K^n \rightarrow \mathbb{A}_K^1$ be a projection. Observe that $\pi_i \circ i = t_i|_X \in K[t_1, \dots, t_n]$. By Proposition 4, i is regular. $\square$

Theorem 10. The association $\varphi \rightarrow \varphi^\#$ gives a bijection

$$\underbrace{\text{hom}(X, Y)}_{\text{regular maps } X \rightarrow Y} \xrightarrow{\sim} \text{hom}_{K\text{-alg}}(K[Y], K[X])$$

Specifically, if $\varphi : X \rightarrow Y$ is a regular map, then its associated map $\varphi^\# : K[Y] \rightarrow K[X]$ is given by $\varphi^\#(f) = f \circ \varphi$.

Proof. Let $\psi : K[Y] \rightarrow K[X]$ be a K -algebra homomorphism. Let $K[Y] = K[t_1, \dots, t_m]/I$ for some $I = \sqrt{I}$ and $Y \subseteq \mathbb{A}_K^m$. Define $\varphi_i := \psi(t_i) = \psi \circ t_i \in K[X]$ and

$$\varphi : X \rightarrow \mathbb{A}_K^m \quad \varphi(x) := (\varphi_1(x), \dots, \varphi_m(x)) = (\psi \circ t_1(x), \dots, \psi \circ t_m(x))$$

Note that each $\psi \circ t_i \in K[X]$ means it is a polynomial with coefficients in K . So $\varphi_i(x)$ corresponds to a polynomial evaluation. As a result, $\varphi_i(x) \in \mathbb{A}_K^1 = K$. We want to show $\varphi(X) \subseteq Y \subseteq \mathbb{A}_K^m$ so that $\varphi \in \text{hom}(X, Y)$. Recall that I is the set of all polynomials that vanish on Y . So equivalently, for all $f \in I = I(Y)$, $f \circ \varphi = 0$. This would imply $\text{im}(\varphi) \subseteq \overline{Y} = Y$. Since ψ is a K -algebra homomorphism between $K[Y]$ and $K[X]$, it sends 0 to 0. In other words, $\psi(I) = 0$. Let $f = \sum_{\mathbf{I}} a_{\mathbf{I}} t^{\mathbf{I}}$,

$$f \circ \varphi = \sum_{\mathbf{I}} a_{\mathbf{I}} (\varphi_1)^{i_1} \dots (\varphi_m)^{i_m} = f(\varphi_1, \dots, \varphi_m) = \sum_{\mathbf{I}} a_{\mathbf{I}} (\psi \circ t_1)^{i_1} \dots (\psi \circ t_m)^{i_m} = \psi(f)$$

$\psi(f) = 0$ as $f \in I$. So $f \circ \varphi = 0$. This shows $\text{im}(\varphi) \subseteq Y$. Now it remains to show that doing " φ " first then " $\#$ " is the same as doing " $\#$ " first then " φ ". Formally, we want to show $\varphi \rightarrow \varphi^\# \rightarrow \varphi$ and $\psi \rightarrow \varphi \rightarrow \varphi^\#$ are identity maps. Let $\{t_1, \dots, t_m\}$ be coordinate functions on $\mathbb{A}_K^m$. Define $\varphi^\#(t_i|_Y) = t_i \circ \varphi$. Then

$$\varphi(x) = (\varphi^\#(t_1)(x), \dots, \varphi^\#(t_m)(x)) = (t_1 \circ \varphi(x), \dots, t_m \circ \varphi(x)) = (t_1, \dots, t_m) \circ \varphi(x) = \varphi(x) \in \mathbb{A}_K^m$$

This shows $\varphi = \varphi$. Now let $\psi : K[Y] \rightarrow K[X]$. Since $\varphi(x) = (\varphi_1(x), \dots, \varphi_m(x)) = (\psi \circ t_1(x), \dots, \psi \circ t_m(x))$, the map $\varphi^\# : K[Y] \rightarrow K[X]$ is given by $t_i \mapsto t_i \circ \varphi = \psi(t_i) \in K[X]$. $\varphi^\# = \psi$ on $K[Y]$ because $\varphi^\#$ and ψ agree on $\{t_1, \dots, t_m\}$. This proves the bijection. $\square$

Remark. [There should be some pictures here...]

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We want to know the relationship between maximal ideals. Let $X, Y \subseteq \mathbb{A}_K^n$ be closed sets. Recall the correspondence $Y \leftrightarrow \{\mathfrak{m} \subseteq K[Y]\}$ and $X \leftrightarrow \{\mathfrak{m} \subseteq K[X]\}$. We have

Proposition 6. Let $\psi : K[Y] \rightarrow K[X]$ and a unique $\varphi : X \rightarrow Y$ such that $\psi = \varphi^\#$. And let $\mathfrak{m}_p$ be the associated ideal for $x \in X$. Then $\mathfrak{m}_{\varphi(p)} = \psi^{-1}(\mathfrak{m}_p)$.

Proof.

$$\begin{aligned} \psi^{-1}(\mathfrak{m}_p) &= (\varphi^\#)^{-1}(\mathfrak{m}_p) = \{g \in K[Y] \mid \varphi^\#(g) \in \mathfrak{m}_p\} \\ &= \{g \in K[Y] \mid g \circ \varphi \in \mathfrak{m}_p\} = \{g \in K[Y] \mid g(\varphi(p)) = 0 \in \mathfrak{m}_p\} \\ &= \mathfrak{m}_{\varphi(p)} \subseteq K[Y] \end{aligned}$$

$\square$

Vocabulary 4.4 (isomorphism). A regular map $\varphi : X \rightarrow Y$ is an isomorphism if there exists a regular $\psi : Y \rightarrow X$ such that $\varphi \circ \psi = \text{id}_Y$ and $\psi \circ \varphi = \text{id}_X$.

Corollary. $X \simeq Y$ if and only if $K[X] \simeq K[Y]$ as K -algebras.

Proof. The bijection $\text{hom}(X, Y) \leftrightarrow \text{hom}_{K\text{-alg}}(K[Y], K[X])$ sends isomorphisms to isomorphisms. To see this, suppose $X \simeq Y$ is realized by $f : X \rightarrow Y$ and $g : Y \rightarrow X$. Consider a regular function $h : X \rightarrow K$. We get a regular function on Y by $g^\#(h) = h \circ g$. This is a ring map and $f^\#g^\#(h) = h \circ f \circ g = h$. This shows $f^\#g^\# = \text{id}_{K[X]}$. We can show the other order of composition in the same way. Moreover, we can reverse this argument to show a K -algebra isomorphism is sent to an isomorphism between X and Y . $\square$

Example. A regular bijection is not necessarily an isomorphism. Consider $Z = Z(y^2 - x^3) \subseteq \mathbb{A}_K^2$. Let $\varphi : \mathbb{A}_K^1 \rightarrow Z$ be a map given by $t \mapsto (t^2, t^3)$. φ is surjective because for any $(x_0, y_0) \in Z$, let $\pm t$ be square roots of x_0 : $x_0 = t^2$. So $x_0^3 = t^6$ tells us $\pm t^3$ are the two square roots of x_0^3 . Conclude that either $y_0 = t^3$ or $y_0 = -t^3$. φ is injective because if $(t_1^2, t_1^3) = (t_2^2, t_2^3)$, $t_1^2 = t_2^2$ and $t_1^3 = t_2^3$. It follows that $t_1 = \pm t_2$ and $t_1 = t_2$. But φ is not an isomorphism. This is because the map

$$\varphi^\# : K[Z] = K[x, y]/\langle y^2 - x^3 \rangle \rightarrow K[t]$$

brings $x \mapsto t^2$ and $y \mapsto t^3$. But $\varphi^\#(K[Z]) = K\langle t^2, t^3 \rangle \subsetneq K[t]$. This is, indeed, not an isomorphism.

Vocabulary 4.5 (closed immersion). A map $i : X \rightarrow Y$ is a closed immersion if $i(X)$ is closed in Y and $i : X \rightarrow i(X)$ is an isomorphism.

Remark. The map in the example above is a regular injection but not isomorphic onto its image.

Proposition 7. Let $\varphi : X \rightarrow Y$ be a regular map. $\ker(\varphi^\#) = I(\overline{\varphi(X)})$.

Proof.

$$f \in \ker(\varphi^\#) \Leftrightarrow f \circ \varphi = 0 \Leftrightarrow f|_{\varphi(X)} = 0 \Leftrightarrow f|_{\overline{\varphi(X)}} = 0$$

The above holds if and only if $f \in I(\overline{\varphi(X)})$. $\square$

Proposition 8. $i : X \rightarrow Y$ is a closed immersion if and only if $i^\# : K[Y] \rightarrow K[X]$ is surjective.

Proof. Let $i : X \rightarrow Y$ be a closed immersion. So $i : X \rightarrow i(X) \subseteq Y$. Recall that if a set is larger, the corresponding ideal is smaller. This gives us the following maps

$$K[X] \xleftarrow{\sim} K[i(X)] = K[t_1, \dots, t_n]/I(i(X)) \xleftarrow{i^\#} K[Y] = K[t_1, \dots, t_n]/I(Y)$$

Let $i : X \rightarrow Y$ be such that $i^\# : K[Y] \rightarrow K[X]$ is surjective. By Proposition 7, $\ker(i^\#) = I(\overline{i(X)}) \subseteq K[Y]$. So $i^\# : K[Y]/I(\overline{i(X)}) \xrightarrow{\sim} K[X]$ is an isomorphism. Note that

$$\begin{aligned} K[Y]/I(\overline{i(X)}) &= K[t_1, \dots, t_n]/I(Y) \Big/_{I(\overline{i(X)})} = K[t_1, \dots, t_n]/(I(Y) + I(\overline{i(X)})) \\ &= K[t_1, \dots, t_n]/(I(Y \cap \overline{i(X)})) = K[t_1, \dots, t_n]/I(\overline{i(X)}) = K[\overline{i(X)}] \end{aligned}$$

$\square$

Vocabulary 4.6 (dominant). $\varphi : X \rightarrow Y$ is dominant if $\overline{\varphi(X)} = Y$.

Proposition 9. φ is dominant if and only if $\varphi^\# : K[Y] \rightarrow K[X]$ is injective.

Proof. $\ker(\varphi^\#) = 0$ if and only if $I(\overline{\varphi(X)}) = 0$ if and only if $\overline{\varphi(X)} = Y$. $\square$

Week 5

Extended regularity to open sets. Defined open immersions with some examples. Defined irreducible and noetherian topological spaces. Proved some results involving irreducible and noetherian stuff. Introduced dimension.

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So far we have shown the following facts

- (0) $\text{hom}(X, Y) \simeq \text{hom}_{K\text{-alg}}(K[Y], K[X])$.
- (1) $\{\text{closed immersion}\} \leftrightarrow \{\text{surjective algebra map}\}$.
- (2) $\{\text{dominant map}\} \leftrightarrow \{\text{injective algebra map}\}$.
- (3) $\{\text{variety isomorphism}\} \leftrightarrow \{\text{algebra isomorphism}\}$.

Vocabulary 5.1 (regular map on open sets). Let $V \subseteq X$ be open. X, Y are affine varieties. Let $\varphi : V \rightarrow Y$ be continuous. φ is regular if $f \circ \varphi : V \rightarrow \mathbb{A}_K^1$ is regular. This is well-defined because regularity is a local concept.

Vocabulary 5.2 (open immersion). Let X and Y be affine varieties. A regular map $\varphi : X \rightarrow Y$ is an open immersion if

- (1) $\varphi(X) \subseteq Y$ is open and
- (2) $\varphi : X \rightarrow \varphi(X)$ is an isomorphism

In other words, there exists a regular map $\psi : \varphi(X) \rightarrow X$ that inverts φ .

Example. Let $X = \{xy = 1\} \subseteq \mathbb{A}_K^2$. Let $Y = \mathbb{A}_K^1$. The map $\varphi : X \rightarrow Y$ is given by $\varphi(x, y) = x$. Its image

$$\varphi(X) = \{x \in \mathbb{A}_K^1 \mid \exists y, xy = 1\} = \mathbb{A}_K^1 \setminus \{0\}$$

is open in Y . Let $\psi : \mathbb{A}_K^1 \setminus \{0\} \rightarrow \mathbb{A}_K^2$ be a map given by $\psi(x) = (x, 1/x)$. Both φ and ψ are regular on each coordinate. Thus, they are regular. In particular, they are inverses of each other. So $\mathbb{A}_K^1 \setminus \{0\} \simeq X \subseteq \mathbb{A}_K^2$. This shows that φ is an open immersion. This induces an isomorphism between coordinate rings. Namely,

$$K[\mathbb{A}_K^1 \setminus \{0\}] \simeq K[X] = K[x, y] / \langle xy - 1 \rangle \simeq K[x, x^{-1}]$$

Example. Let Y be an affine algebraic variety. Let $f \in K[Y]$. $Y \times \mathbb{A}_K^1$ has a natural structure of an affine variety. Since $Y \subseteq \mathbb{A}_K^n$ is closed, $Y \times \mathbb{A}_K^1 \subseteq \mathbb{A}_K^{n+1}$ is also closed. $I(Y \times \mathbb{A}_K^1) = \{f(y, x_{n+1}) \mid f(Y, x_{n+1}) = 0\} \subseteq K[x_1, \dots, x_{n+1}]$.

$$K[x_1, \dots, x_{n+1}] / I(Y \times \mathbb{A}_K^1) = K[x_1, \dots, x_n][x_{n+1}] / \{f(y)(x_{n+1}) \mid f(Y)(x_{n+1}) = 0\} = K[Y][x_{n+1}]$$

Let $X = Z(x_{n+1}f - 1) = \{(x_{n+1}, y) \mid x_{n+1}f(y) = 1\} \subseteq Y \times \mathbb{A}_K^1$. The projection $\varphi : X \rightarrow Y$ is an open immersion onto the basic open $D(f) = \{y \in Y \mid f(y) \neq 0\}$. Indeed, φ is regular because it is the restriction of the projection $\mathbb{A}_K^{n+1} \rightarrow \mathbb{A}_K^n$. The image $\varphi(X) = \{y \in Y \mid \exists x_{n+1}, x_{n+1}f(y) = 1\} = D(f)$. Its inverse $\psi : D(f) \rightarrow X$ is given by $\psi(y) = (y, 1/f(y))$. Note that $D(f) \simeq X$ induces an isomorphism $K[X] \simeq K[D(f)]$. But

$$\begin{aligned} K[X] &= K[x_1, \dots, x_{n+1}] / I(Z(x_{n+1}f - 1)) = K[x_1, \dots, x_{n+1}] / I\left(\left\{(x_{n+1}, y) \mid x_{n+1} = \frac{1}{f}\right\}\right) \\ &= K[x_1, \dots, x_n] \left[\frac{1}{f}\right] / \left\{g(y) \left(\frac{1}{f}\right) \mid g(Y) \left(\frac{1}{f}\right) = 0\right\} = K[Y][f^{-1}] \end{aligned}$$

Proposition 10. Let Y be an affine algebraic variety. The ring of regular functions on $D(f) \subseteq Y$ is isomorphic to $K[Y][f^{-1}] = \{1, f^1, f^2, \dots\}^{-1} K[Y]$.

Proof. The ring of regular functions on $D(f)$ is $K[D(f)]$. Apply the above example. □

Vocabulary 5.3 (irreducible topological space). Let X be a topological space. X is irreducible if $X = Z \cup Y$ implies $X = Y$ or $X = Z$ where Y, Z are closed.

Example. Any nonempty open subset of an irreducible space is ① irreducible and ② dense.

Proof. Let X be an irreducible space and let Y be a nonempty open irreducible subset of X . ① If $Y = X$, then Y is irreducible. Suppose Y is not irreducible and $Y \subsetneq X$, then there exists closed subsets $X_1, X_2 \subseteq X$ such that $Y = (Y \cap X_1) \cup (Y \cap X_2)$. Since Y is open in X , $X \setminus Y$ is closed, so $X_2 \cup (X \setminus Y)$ is closed. We have $X = X_1 \cup (X_2 \cup (X \setminus Y))$, a contradiction. ② If $Y = X$, then Y is dense. Let $Y \subsetneq X$, then $X = \overline{Y} \cup (X \setminus Y)$. Since $Y \neq \emptyset$, $X \setminus Y$ is closed, so $\overline{Y}$ is open, which implies $X = \overline{Y}$. □

Example. If Y is an irreducible subset of X , then its closure $\overline{Y}$ in X is also irreducible.

Proof. Let Y be an irreducible subset of a space X . Suppose $\overline{Y}$ is not irreducible, then there exists closed subsets $Y_1, Y_2 \subseteq \overline{Y}$ such that $\overline{Y} = Y_1 \cup Y_2$. Since $\overline{Y}$ is the smallest closed set in X that contains Y , $Y \subsetneq Y_i$. Then $Y = Y \cap \overline{Y} = (Y \cap Y_1) \cup (Y \cap Y_2)$, where each $Y \cap Y_i$ is closed in the subspace topology. This is a contradiction. $\square$

Proposition 11. Let X be an affine algebraic variety. X is irreducible if and only if $K[X]$ is a domain.

Proof. ($\Rightarrow$) Suppose X is irreducible. Choose $f, g \in K[X] \setminus \{0\}$. We identify $K[X]$ with the set of regular functions $X \rightarrow K$. If $fg = 0$ in $K[X]$, then $fg(x) = f(x)g(x)$ is identically 0 on *all* of X . So the zero set is $Z(fg) = X = Z(f) \cup Z(g)$. Since X is irreducible, $X = Z(f)$ or $Z(g)$. Then it implies $f = 0$ or $g = 0$. This is a contradiction. ($\Leftarrow$) Suppose $K[X]$ is a domain. Let $X = Y \cup Z$ such that Y, Z are closed. Assume $Y \not\subseteq Z$ and $Z \not\subseteq Y$. Let $f \in I(Z) \setminus I(Y)$ and $g \in I(Y) \setminus I(Z)$. This will guarantee both f, g are nonzero. Since f is a polynomial that vanishes on Z , $Z \subseteq Z(f)$. Similarly, $Y \subseteq Z(g)$. Then $fg = 0$ on X because Y, Z cover X which is a contradiction. $\square$

Theorem 11. Let X be an affine algebraic variety. Then there exists a collection of n subspaces $\{x_i\}_{i=1}^n$ such that X_i are closed irreducible subsets and $X = \bigcup_{i=1}^n X_i$. The X_i are unique.

Example. Let $X = Z(f)$. Choose some nonzero $f \in K[x_1, \dots, x_n]$. Since it is a UFD, there is a unique factorization $f = f_1 \dots f_n$ where f_i are irreducible polynomials. $\langle f_i \rangle$ is a prime ideal. According to 1/17, $\sqrt{\langle f_i \rangle} = \langle f_i \rangle$.

$$K[Z(f_i)] = K[x_1, \dots, x_n]/I(Z(f_i)) \stackrel{\text{Nutz}}{=} K[x_1, \dots, x_n]/\sqrt{\langle f_i \rangle} = K[x_1, \dots, x_n]/\langle f_i \rangle$$

$K[Z(f_i)]$ is a domain. By Proposition 11, $Z(f_i)$ is irreducible. $Z(f) = \bigcup_{i=1}^n Z(f_i)$.

Vocabulary 5.4 (irreducible component). A subset $Z \subseteq X$ is an irreducible component if it is a maximal, closed, and irreducible subset.

Example. X_i in Theorem 11 are irreducible components: maximal with respect to being irreducible and closed.

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Vocabulary 5.5 (noetherian topological space). Let X be a topological space. X is noetherian if for any chain of irreducible closed subspaces

$$Z_0 \supseteq Z_1 \supseteq \dots \supseteq Z_n \supseteq \dots$$

there exists some $N \in \mathbb{N}$ such that $Z_i = Z_N$ for all $i \geq N$.

Proposition 12. Let X be a noetherian topological space. $X = \bigcup_{i=1}^r Z_i$ where $\{Z_i\}$ are irreducible closed sets.

Proof. Suppose X cannot be decomposed like this. Then X must be reducible. So $X = X_1 \cup X_2$. Both X_1 and X_2 are closed such that $X_1, X_2 \subsetneq X$. At least one of X_1, X_2 is not a finite union of irreducible closed subsets. Assume X_1 is not. Then X_1 must be reducible: $X_1 = X_{11} \cup X_{12}$ where X_{11} and X_{12} are closed such that $X_{11}, X_{12} \subsetneq X$. At least one of X_{11}, X_{12} is not a finite union of irreducible closed subsets. Assume X_{11} is not. It can be decomposed as $X_{11} = X_{111} \cup X_{112}$ where X_{111} and X_{112} are closed such that $X_{111}, X_{112} \subsetneq X$. This process goes on infinitely resulting an infinite chain

$$X \supsetneq X_1 \supsetneq X_{11} \supsetneq \dots \supsetneq X_{11\dots 1} \supsetneq \dots$$

It follows that X is not noetherian. $\square$

Theorem 12 (topological version). Let X be a noetherian topological space. $X = \bigcup_{i=1}^n X_i$ where $\{X_i\}$ are unique irreducible components.

Proof. Let X be a noetherian topological space. By Proposition 12, $X = \bigcup_{i=1}^r Z_i$ where $\{Z_i\}$ are irreducible closed sets. $\{Z_i\}$ is a poset under inclusion. Let $\{X_i\}_{i=1}^n$ be the set of all maximal elements of the poset. $\{X_i\}_{i=1}^n$ is finite because $\{Z_i\}$ is finite. We already know each X_i is irreducible and closed because they are chosen from $\{Z_i\}$. But to make it an irreducible component, we must show it is maximal among all other irreducible and closed sets. Let $Y \subseteq X$ be an irreducible closed set. Then $Y = \bigcup_{i=1}^r (Y \cap Z_i)$. Note that each $Y \cap Z_i$ is a closed set in X . Since it is also a subset of Y , it is also closed in Y . Y being irreducible implies $Y = Y \cap Z_i$ for some i . Then $Y \subseteq Z_i$ for some i . Since $\{X_i\}_{i=1}^n$ is maximal among $\{Z_i\}$, $Z_i \subseteq X_j$ for some j . Now we have $Y \subseteq Z_i \subseteq X_j$ for every such Y . Conclude that $\{X_i\}_{i=1}^n$ is maximal among all such Y and $\{X_i\}_{i=1}^n$ is the set of irreducible components. $\square$

Remark. The proof for Proposition 12 is not constructive. In polynomial rings, there exists an algorithm to find the exact decomposition. Theorem 12 is a generalized version of Theorem 11.

Vocabulary 5.6 (noetherian ring). Let A be a ring. A is noetherian if every ideal of A is finitely generated.

Proposition 13. Let A be a ring. Every chain of ideals in A stabilizes if and only if A is noetherian.

Proof. ($\Rightarrow$) Let $I \subseteq A$ be an ideal. If $I = 0$, we are done. So let $0 \neq f_1 \in I$. If $I = \langle f_1 \rangle$, we are done. If not, choose $f_2 \in I \setminus \langle f_1 \rangle$. If $I = \langle f_1, f_2 \rangle$, we are done. If not, choose $f_3 \in I \setminus \langle f_1, f_2 \rangle$. Continuing this process, we get $\langle f_1 \rangle \subsetneq \langle f_1, f_2 \rangle \subsetneq \langle f_1, f_2, f_3 \rangle \subsetneq \dots$. By assumption, there exists an N such that $\langle f_1, \dots, f_N \rangle = \langle f_1, \dots, f_{N+1} \rangle$. So $I = \langle f_1, \dots, f_N \rangle$. ($\Leftarrow$) Let $I_0 \subseteq I_1 \subseteq \dots \subseteq I_j \subseteq \dots$ be a chain in A . By assumption, $I = \langle g_1, \dots, g_N \rangle$. Each $g_i \in I_{j_i}$ for some natural number j_i . Let $M = \max\{j_i\}_{i=1}^N$. Then $g_i \in I_M$ for all i . It follows that $I = I_M$. In particular, $I_j = I_M$ for all $j \geq M$. $\square$

Proposition 14. Some analogous results for modules (See Proposition A1):

1. If M is an R -module, then every submodule is finitely generated if and only if every chain of submodules stabilizes. If these conditions are satisfied, we say that M is noetherian.
2. Any finitely generated module over a noetherian ring is noetherian. $\square$

Theorem 13 (Hilbert's basis theorem). Let R be noetherian. Then $R[x]$ is noetherian.

We are not going to prove Theorem 13. Instead, we will prove an other version of it:

Theorem 14. Let K be an infinite field. Then $K[x_1, \dots, x_n]$ is noetherian.

Proof. We induct on n . If $n = 0$, this is trivial. If $n = 1$, $K[x]$ is a PID. So we assume $n \geq 2$. Let $I \subseteq K[x_1, \dots, x_n]$ be an ideal. If $I = 0$, we are done. So we assume there exists some nonzero $f \in I$. By Proposition 2 (preparation), there exists a change of variable $x_i \mapsto t_i$ such that f is monic in t_n . For some polynomials a_i in $n - 1$ variables,

$$f = t_n^d + a_0(t_1, \dots, t_{n-1})t_n^{d-1} + \dots + a_{d-1}(t_1, \dots, t_{n-1})t_n^0$$

In the quotient $K[t_1, \dots, t_n]/\langle f \rangle$, $t_n^d = -(a_0t_n^{d-1} + \dots + a_{d-1}t_n^0)$. For any $g \in K[t_1, \dots, t_n]/\langle f \rangle$, it can be viewed as a polynomial in t_n : $g = b_k(t_1, \dots, t_{n-1})t_n^k + \dots + b_0(t_1, \dots, t_{n-1})t_n^0$ where b_i are polynomials in $n - 1$ variables. For any $l \geq d$ in this quotient, $t_n^l = t_n^{l \bmod d} C_l(a_0t_n^{d-1} + \dots + a_{d-1}t_n^0)$ where C_i is some constant in the field. Notice that this reduction decreases the power of t_n by at least 1. So within finite iterations,

$$g = \sum_i C_i (a_0t_n^{d-1} + \dots + a_{d-1}t_n^0) b_i(t_1, \dots, t_{n-1})t_n^{i \bmod d} = g_1 + g_2t_n + \dots + g_{d-1}t_n^{d-1}$$

where g_i are polynomials in $t_1, \dots, t_{n-1}$. Since $g \in K[t_1, \dots, t_n]/\langle f \rangle$ is arbitrary, it follows that

$$K[t_1, \dots, t_n]/\langle f \rangle = K[t_1, \dots, t_{n-1}] \oplus K[t_1, \dots, t_{n-1}]t_n \oplus \dots \oplus K[t_1, \dots, t_{n-1}]t_n^{d-1}$$

By induction, $K[t_1, \dots, t_{n-1}]$ is noetherian. $K[t_1, \dots, t_n]/\langle f \rangle$ is a finitely generated module over $K[t_1, \dots, t_{n-1}]$. By Proposition 14(2), $K[t_1, \dots, t_n]/\langle f \rangle$ is noetherian. $\bar{I} = I/\langle f \rangle \subseteq K[t_1, \dots, t_n]/\langle f \rangle$ is the corresponding ideal in the quotient. So $\bar{I}$ is a submodule over $K[t_1, \dots, t_{n-1}]$. Due to Proposition 14(1), $\bar{I}$ is a finitely generated module over $K[t_1, \dots, t_{n-1}]$. In particular, $\bar{I}$ is a finitely generated as an ideal. So $\bar{I} = \langle \bar{g}_1, \dots, \bar{g}_N \rangle$. Taking it out of the quotient, we get $I = \langle g_1, \dots, g_N, f \rangle$. $\square$

Proposition 15. If R is noetherian, R/I is also noetherian for all ideals I .

Proof. A chain of ideals $\bar{J}_0 \subseteq \bar{J}_1 \subseteq \dots$ in R/I corresponds to a chain $I \subseteq J_0 \subseteq J_1 \subseteq \dots$ in R . By this correspondence, J_i stabilizes implies $\bar{J}_i$ stabilizes. $\square$

Proposition 16. Suppose A is a finitely generated K -algebra. A is noetherian.

Proof. $A \simeq K[x_1, \dots, x_n]/\ker(\varphi_a : K[x_1, \dots, x_n] \rightarrow A)$. By Proposition 15, A is noetherian. $\square$

Theorem 15. Let A be a finitely generated K -algebra. For any chain of ideals $I_0 \subseteq I_1 \subseteq \dots$, there exists an N such that $I_j = I_N$ for all $j \geq N$.

Proof. By Proposition 16, A is noetherian. By Proposition 13, every chain of ideals in A stabilizes. $\square$

Theorem 16 (algebraic version). If X is an affine variety, X is noetherian.

Proof. If $X \subseteq Y$ are subsets of $\mathbb{A}_K^n$, then $I(X) \supseteq I(Y)$ by basic set-theoretic considerations. Nutz I tells us closed subsets of $\mathbb{A}_K^n$ corresponds to radical ideals. Therefore, a chain of closed subspaces $X_0 \supseteq X_1 \supseteq X_2 \supseteq \dots$ corresponds to a chain of radical ideals $I(X_0) \subseteq I(X_1) \subseteq I(X_2) \subseteq \dots$. Suppose X is not noetherian. Then we must have an infinite strictly increasing chain of radical ideals $I(X_0) \subsetneq I(X_1) \subsetneq I(X_2) \subsetneq \dots$ corresponding to closed $X_i \subseteq X$. This is impossible because Theorem 14 tells us $K[x_1, \dots, x_n]$ is noetherian. Conclude that X is noetherian. $\square$

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Vocabulary 5.7 (quasi-compact). Let X be a topological space. X is quasi-compact if $X = \bigcup_{i=0}^{\infty} U_i$ is an open cover implies $X = \bigcup_{i=0}^N U_i$ for some natural number N .

Proposition 17. X is a topological space. X is noetherian if and only if for all $U \subseteq X$ open, U is quasi-compact.

Proof. ($\Rightarrow$) Let X be noetherian. Let $U \subseteq X$ be open. Let U_i be open set such that $U = \bigcup_{i=0}^{\infty} U_i$. Without loss of generality, we can assume this is a nested open cover because otherwise we can form a new cover $\{V_i\}_{i=0}^{\infty}$ by setting $V_j = \bigcup_{i=0}^j U_i$. So $U_i \subseteq U_{i+1}$ for all i . Let $Z_i = X \setminus U_i$. Z_i is closed in X and $Z_i \supseteq Z_{i+1}$ for all i . X is noetherian means there exists some N such that $Z_i = Z_N$ for all $i \geq N$. Accordingly, $U_i = U_N$ for all $i \geq N$. So $U = \bigcup_{i=0}^N U_i$. ($\Leftarrow$) Let X be a topological space such that for any $U \subseteq X$ open, U is quasi-compact. Let $Z_0 \supseteq Z_1 \supseteq Z_2 \supseteq \dots$ be a chain of closed subsets in X . Let $Z = \bigcap_{i=0}^{\infty} Z_i$. Z is closed. Then $U = X \setminus Z$ is open with a nested open cover $U = \bigcup_{i=0}^{\infty} U \setminus Z_i =: \bigcup_{i=0}^{\infty} U_i$. Since X is quasi-compact, there exists some N such that $U_N = U$. This implies for the same N , $Z_N = Z$. Conclude that for all $i \geq N$, $Z_i = Z$. $\square$

Proposition 18. If X is noetherian, then every open $U \subseteq X$ is also noetherian.

Proof. Follows from Proposition 17. $\square$

Vocabulary 5.8 (dimension). Let X be a noetherian topological space, then its dimension is defined as

$$\dim(X) = \max \{r \mid \exists Z_0 \subsetneq Z_1 \subsetneq \dots \subsetneq Z_r, \text{ each } Z_i \text{ is irreducible and closed}\}$$

Additionally, $\dim(X) \in \mathbb{Z}^{\geq 0} \cup \{\infty\}$.

Vocabulary 5.9 (Krull dimension). Let A be a commutative ring, then its Krull dimension is defined as

$$\text{Krdim}(A) = \max \{r \mid \exists \mathfrak{p}_0 \supsetneq \mathfrak{p}_1 \supsetneq \dots \supsetneq \mathfrak{p}_r, \mathfrak{p}_i \in \text{spec}(A)\}$$

Additionally, $\text{Krdim}(X) \in \mathbb{Z}^{\geq 0} \cup \{\infty\}$.

Vocabulary 5.10 (height). In a ring A , the height of a prime ideal $\mathfrak{p}$, denoted $\text{ht}(\mathfrak{p})$, is the supremum of all integers n such that there exists a chain $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_n = \mathfrak{p}$ of distinct prime ideals. In other words, we defined the Krull dimension of A to be the supremum of the heights of all prime ideals.

Proposition 19. Let X be an affine variety. $\text{Krdim}(K[X]) = \dim(X)$.

Proof. Let $Z \subseteq X$ be a closed set. By Proposition 11, Z is irreducible if and only if $K[Z] = K[x_1, \dots, x_n]/I(Z)$ is a domain. Recall that $K[Z]$ is a domain if and only if $I(Z) \subseteq K[X]$ is a prime ideal. According to 1/19, prime ideals are radical. Therefore, Nutz I induces a bijection $\{I(Z) \in \text{spec}(K[X])\} \leftrightarrow \{Z \subseteq X \mid \text{closed and irreducible}\}$. $\square$

Goal. We should make sure this dimension thing matches our intuition.

1. $\dim(X) = 0$ if and only if X is a finite set.
2. $\dim(\mathbb{A}_K^n) = n$.
3. Let $f \in K[x_1, \dots, x_n]$ and $f \notin K$. Then $\dim(Z(f)) = n - 1$.
4. Let $\{f_1, \dots, f_l\} \in K[x_1, \dots, x_n]$ and $f_i \notin K$. $\dim(Z(f_1, \dots, f_l)) \geq n - l$.

Proposition 20. Let X be a noetherian topological space with irreducible components $X = \bigcup_{i=0}^r X_i$. Then

$$\dim(X) = \max_{1 \leq i \leq r} \dim(X_i)$$

Proof. Let $Z_0 \subsetneq \dots \subsetneq Z_n$ be a chain of closed irreducible subsets. Recall that irreducible components are maximal. This means $Z_n \subseteq X_i$ for some $i \in \{1, \dots, r\}$. If this chain is maximal in X , it is maximal in X_i . So $\dim(X) = \dim(X_i)$. If $\dim(X) = \infty$, for any $n \geq 0$, $Z_n \subseteq X_i$ for some $i \in \{1, \dots, r\}$. Due to the existence of an infinite chain, we must have some chain $Z_0 \subsetneq \dots \subsetneq Z_{n'} where $n' > n$. By the maximality of irreducible components, it is contained in some X_i . Eventually, $n' \rightarrow \infty$. This means some X_i contains the infinity chain. Therefore, $\dim(X_i) = \infty$. $\square$$

Example. $\dim(X) = 0$ if and only if X is a finite set. $(\Rightarrow)$ Let X be an affine variety and $\dim(X) = 0$. By Theorem 16, it is enough to show each component of X is a single point. If X_i is not a single point, $\{\text{pt}\} \subsetneq X_i$ has length 1. This is a contradiction. $(\Leftarrow)$ If X is a finite set, clearly $\dim(X) = 0$ with discrete topology.

Proposition 21. Let A be a finitely generated reduced² K -algebra. $\text{Krdim}(A) = 0$ if and only if $\dim_K(A) < \infty$. This last inequality means A is finite dimensional as a vector space.

Proof. $(\Rightarrow)$ By 2/7, $A = K[X]$ for some affine X . By Proposition 19, $\text{Krdim}(A) = \dim(X)$. By the example above, $\text{Krdim}(A) = 0$ if and only if X is finite set. So $X = \{v_1, \dots, v_r\}$. Define a homomorphism $\varphi : K[x_1, \dots, x_n] \rightarrow K^n$ given by $p \mapsto (p(v_1), \dots, p(v_r))$. For any $(k_1, \dots, k_r) \in K^r$, we can find a Lagrange interpolating polynomial in $K[x_1, \dots, x_n]$ that produces k_i when plugged in v_i . Thus φ is surjective. Its kernel is clearly $I(X)$. It follows that $K[X] = K[x_1, \dots, x_n]/I(X) \simeq K^r$ is an isomorphism. So $\dim_K(A) = \dim_K(K[X]) = r < \infty$. $(\Leftarrow)$ Let $\dim_K(A) < \infty$. Let $\mathfrak{p} \subseteq A$ be a prime ideal. Suppose $\text{Krdim}(A) > 0$. In other words, $\mathfrak{p} \subsetneq \mathfrak{m}$ is strictly contained in some maximal ideal. Since $A \twoheadrightarrow A/\mathfrak{p}$ is surjective, $\dim_K(A/\mathfrak{p}) \leq \dim_K(A) < \infty$. Then $A/\mathfrak{p}$ is a domain which is *not* a field. By Theorem 6 (noether normalization), $A/\mathfrak{p}$ is finite over some $K[t_1, \dots, t_d] \subseteq A/\mathfrak{p}$. Since A is not a field, $d > 0$. But

$$\infty = \dim_K(K[t]) \leq \dim_K(K[t_1, \dots, t_d]) \leq \dim_K(A/\mathfrak{p}) \leq \dim_K(A) < \infty$$

This is a contradiction. Therefore, $\text{Krdim}(A) = 0$. $\square$

Observation. Some quick facts about the Krdim of polynomials.

1. $\text{Krdim}(K) = 0$: 0 is the only prime ideal in a field.
2. $\text{Krdim}(K[t]) = 1$: Let p be some irreducible polynomial. $0 \subsetneq \langle p \rangle$ is the maximal chain of ideals.
3. $\text{Krdim}(K[t_1, \dots, t_n]) \geq n$: We at least have $0 \subsetneq \langle t_1 \rangle \subsetneq \langle t_1, t_2 \rangle \subsetneq \dots \subsetneq \langle t_1, \dots, t_n \rangle$.

Week 6

Definition of finite maps between rings/varieties. Results that connect finiteness and dimensionality. Dimension of $\mathbb{A}_K^n$ is n . Affine varieties are finite dimensional. Additional facts about finite maps.

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Notation. From now on, we will just use $\dim$ for Krdim .

Observation. $\dim(X) = 0$ if and only if X is a finite set. Obviously $\dim(\mathbb{A}_K^1) = 1$ since $\mathbb{A}_K^1$ itself irreducible and $\{\text{pt}\} \subsetneq \mathbb{A}_K^1$. Then $\dim(\mathbb{A}_K^n) \geq n$ by setting $p_i = (x_1, \dots, x_i) \in K[x_1, \dots, x_n]$. This implies $p_0 \subsetneq \dots \subsetneq p_n$ is a chain of length n (and we can *probably* embed $K[x_1, \dots, x_n]$ into $\mathbb{A}_K^n$ by sending coefficients coordinate-wise).

Vocabulary 6.1 (finite map between rings). Let A, B be rings, let $\psi : A \rightarrow B$, then ψ is finite if B is finite as a module over $\psi(A)$.

Vocabulary 6.2 (finite map between varieties). Let $\varphi : X \rightarrow Y$ be a map of affine varieties. φ is finite if $\varphi^\# : K[Y] \rightarrow K[X]$ is finite.

Theorem 17. We give two similar descriptions of dimensions. One in rings and one in varieties.

1. Let $A \rightarrow B$ be a finite map of rings, then $\dim(B) \leq \dim(A)$.
2. Let $X \rightarrow Y$ be a finite map of varieties, then $\dim(X) \leq \dim(Y)$.

²No nonzero nilpotent elements.

Proof. We start by proving (1). Let $\psi : A \rightarrow B$ and let $p_0 \subsetneq p_1 \subsetneq \dots \subsetneq p_r$ be a chain in $\text{spec}(B)$. Immediately have $\psi^{-1}(p_0) \subseteq \dots \subseteq \psi^{-1}(p_r)$ is a chain in $\text{spec}(A)$ ³. We still need to show these inclusions are strict. In other words, $P \subsetneq Q$ being primes in B implies $\psi^{-1}(P) \subsetneq \psi^{-1}(Q)$ in A . Due to the existence of

$$\bar{\psi} : \bar{A} := A/\psi^{-1}(P) \rightarrow \bar{B} := B/P$$

we can say $\bar{A} \subseteq \bar{B}$ as domains and $A \subseteq B$ is finite⁴ implies $\bar{A} \subseteq \bar{B}$ is finite. Since Q strictly contains P , the quotient $\bar{Q} \subseteq \bar{B}$ is a nonzero prime ideal. We already knew $\bar{A} \subseteq \bar{B}$ finite implies $\bar{B}$ is integral over $\bar{A}$. Let $b \in \bar{Q} \setminus \{0\}$. Can choose $p(t) \in \bar{A}[t]$ monic such that $p(b) = 0$. If $p(t) = \sum_{i=0}^n a_i t^i$, then $a_0 \neq 0$. To see why $a_0 \neq 0$, we suppose that $p(t) = tq(t)$, then $0 = p(b) = bq(b)$ implies $q(b) = 0$. So replacing $p(t)$ with $q(t)$ we reduced the degree of such polynomial: $\deg(q(t)) < \deg(p(t))$. If $p(t)$ is the least-degree monic polynomial such that $p(b) = 0$, $t \nmid p(t)$. Thus

$$0 = p(b) = \sum_{i=0}^n a_i b^i = a_0 + \sum_{i=1}^n a_i b^i = a_0 + b \left(\sum_{i=1}^n a_i b^{i-1} \right)$$

$a_0 = -b(\sum_{i=1}^n a_i b^{i-1})$ implies $a_0 \in \langle b \rangle \subseteq \bar{B}$ but this is a polynomial in $\bar{A}[t]$. So $a_0 \in \bar{A} \cap \bar{B} \cdot b = \bar{A} \cap \langle b \rangle$. This shows $\bar{A} \cap \langle b \rangle \neq \emptyset$. Since $b \in \bar{Q} \setminus \{0\}$, $\bar{A} \cap \langle b \rangle$ is nonempty implies $\bar{A} \cap \bar{Q}$ is nonempty. But $\bar{A} \cap \bar{Q} = \psi^{-1}(Q)/\psi^{-1}(P) \neq \emptyset$. So the chain is strict. (2) follows from the proof of (1). $\square$

Theorem 18. $\dim(\mathbb{A}_K^n) = n$.

Proof. Induction on n . It is obvious when $n = 0, 1$. Assume $n > 1$, and it is true for $n - 1$. Let $p_0 \subsetneq \dots \subsetneq p_r$ is a chain of prime ideals in $K[x_1, \dots, x_n]$. We want to show $r \leq n$. Without loss of generality, can assume $p_0 = 0$ since we can always attach a 0 ideal at the end of a chain. Let $0 \neq f \in p_1$. According to Proposition 2 (preparation), we can change $\{x_1, \dots, x_n\} \rightarrow \{t_1, \dots, t_n\}$ so that

$$f = \alpha(t_d^n + g_{d-1}(t_1, \dots, t_{n-1})t_n^{d-1} + \dots + g_0(t_1, \dots, t_{n-1}))$$

This polynomial is monic in t_n . Then by a lemma we proved in 1/24, $K[t_1, \dots, t_n]/\langle f \rangle$ is finite over $K[t_1, \dots, t_{n-1}]$. By induction, $\dim(K[t_1, \dots, t_{n-1}]) = n - 1$. The theorem above tells us $\dim(K[t_1, \dots, t_n]/\langle f \rangle) \leq n - 1$. But

$$p_1/\langle f \rangle \subsetneq p_2/\langle f \rangle \subsetneq \dots \subsetneq p_r/\langle f \rangle$$

is a chain of primes in A of length $r - 1$ (Note that the length is the number of strict inclusions, not the number of ideals). Hence $r - 1 \leq n - 1$ implies $r \leq n$. Then $\dim(\mathbb{A}_K^n) \leq n$ implies $\dim(\mathbb{A}_K^n) = n$. $\square$

Corollary. Let X be an affine variety, then $\dim(X) < \infty$.

Proof. $\dim(X) \leq n$ because $X \subseteq \mathbb{A}_K^n$. $\square$

Goal. The list of things we want to show

1. If $\varphi : X \rightarrow Y$ is finite surjective, then $\dim(X) = \dim(Y)$.
2. If d is the number appearing in Theorem 6 (noether normalization) ($K[t_1, \dots, t_n]/\langle f \rangle$ is finite over $K[t_1, \dots, t_d] \subseteq K[t_1, \dots, t_n]/\langle f \rangle$), then $d = \dim(K[t_1, \dots, t_n]/\langle f \rangle)$.

Lemma. A little more about finite maps

1. Let $\varphi : X \rightarrow Y$ and $\psi : Y \rightarrow Z$ be finite, then $\psi \circ \varphi : X \rightarrow Z$ is finite.
2. Let $i : X \rightarrow Y$ be a closed immersion, then i is finite.
3. Let $\varphi : X \rightarrow Y$ be finite, let $y \in Y$, then $\varphi^{-1}(y)$ is a finite set.

Proof. (1) For rings, this says $A \rightarrow B$; $B \rightarrow C$ is finite implies $A \rightarrow C$ is finite. To see this, we choose $\{x_i\}$ as generator of B/A , and $\{y_j\}$ as generators of C/B , then $\{y_j \psi(x_i)\}$ generate C/A . (2) $i : X \rightarrow Y$ is a closed immersion if and only if $K[Y] \rightarrow K[X]$ is surjective. But then $1 \in K[X]$ generates the image. (3) Let $\psi : K[Y] \rightarrow K[X]$ be

³The notation that we are about to use is $A \cap I := \psi^{-1}(I)$ where $I \subseteq A$ is an ideal.

⁴Technically, B is finite over the image of A under ψ .

the associated map. Let $\mathfrak{m}_y$ correspond to y , $\mathfrak{m}_y \subseteq K[Y]$

$$\begin{array}{ccc} \varphi^{-1}(y) & \longleftrightarrow & \{\mathfrak{n} \subseteq K[X] \mid \psi^{-1}(\mathfrak{n}) = \mathfrak{m}_y\} \\ & & \updownarrow \\ & & \text{maximal ideals of } K[X]/(\psi(\mathfrak{m}_y)K[X]) \end{array}$$

($\{\mathfrak{m}_y\}$ could be empty). If not empty, then

$$\psi^{-1}(\psi(\mathfrak{m}_y)K[X]) \supseteq \psi^{-1}(\psi(\mathfrak{m}_y)) \supseteq \mathfrak{m}_y$$

The last subset relation is true for general (pre)images. Since $\mathfrak{m}_y$ is maximal, $\psi^{-1}(\psi(\mathfrak{m}_y)K[X]) = \mathfrak{m}_y$. Now we briefly justify the vertical arrow in the diagram. According to 2/2, we have $\{\text{max ideals} \subseteq (K[X]/\psi(\mathfrak{m}_y)K[X])\} \leftrightarrow \{\text{max ideals} \subseteq K[X] \text{ containing } \psi(\mathfrak{m}_y)K[X]\}$. In other words, these are $\{\mathfrak{n} \subseteq K[X] \mid \psi(\mathfrak{m}_y)K[X] \subseteq \mathfrak{n}\}$. Taking the preimage of both sides, we get $\psi^{-1}(\psi(\mathfrak{m}_y)K[X]) \subseteq \psi^{-1}(\mathfrak{n})$. By what we have just shown, $\mathfrak{m}_y \subseteq \psi^{-1}(\mathfrak{n})$. Since $\mathfrak{m}_y$ is maximal, $\mathfrak{m}_y = \psi^{-1}(\mathfrak{n})$. The other direction is given by applying ψ to both sides of $\mathfrak{m}_y = \psi^{-1}(\mathfrak{n})$ and get $\psi(\mathfrak{m}_y) = \psi(\psi^{-1}(\mathfrak{n})) \subseteq \mathfrak{n}$. Since the image is not necessarily an ideal, we make it into $\psi(\mathfrak{m}_y)K[X]$. The inclusion into $\mathfrak{n}$ still holds because $\mathfrak{n}$ is maximal. This justifies the vertical arrow. Recall that $\mathfrak{m}_y$ s are maximal ideals corresponding to points $y \in Y$, where Y is a variety. The assumption is that φ is finite. By 6.2, $\psi : K[Y] \rightarrow K[X]$ is finite. Then the induced map

$$\bar{\psi} : K[Y]/\psi^{-1}(\psi(\mathfrak{m}_y)K[X]) \longrightarrow K[X]/\psi(\mathfrak{m}_y)K[X]$$

is also finite. In other words, $K = K[Y]/\mathfrak{m}_y \hookrightarrow K[X]/(\psi(\mathfrak{m}_y)K[X])$ is finite. If we let $J = \psi(\mathfrak{m}_y)K[X]$, then $K[X]/J$ is finite dimensional as a K -vector space. Thus $K[X]/\sqrt{J}$ is finite dimensional as well. By the proof of Proposition 21, this algebra is isomorphic to $K \times \dots \times K$. $K[X]/\sqrt{J}$ has finitely many maximal ideals because maximal ideals are all of the form $K \times \dots \times \mathfrak{m}_i \times \dots \times K$ where $\mathfrak{m}_i$ is a maximal ideal of the i^{th} K . This implies that $K[X]/J$ has finitely many maximal ideals. Conclude that $\varphi^{-1}(y)$ is finite. $\square$

Week 7

Showed the image of a closed set is also closed. A closed set is finite over its preimage. Finite surjective/injective maps give equal dimension. Dimension is now officially called "the number we got from noether normalization". Dimension of hypersurfaces. Proved the global principal ideal theorem.

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Proposition 22. Let $\varphi : X \rightarrow Y$ be a finite map of affine varieties. If $Z \subset X$ is closed, then $\varphi(Z) \subset Y$ is closed.

Proof. Let $\psi : A = K[Y] \rightarrow B = K[X]$ be the associated ring map. $Z(J)$ is closed in X for some ideal $J \subseteq B$. We have two correspondences

$$\begin{array}{ccc} Z(J) & \longleftrightarrow & \{\mathfrak{m} \subseteq B \mid \mathfrak{m} \supseteq J\} \\ \varphi(Z(J)) & \longleftrightarrow & \{\mathfrak{n} \subseteq A \mid \mathfrak{n} = \psi^{-1}(\mathfrak{m}) \text{ for } \mathfrak{m} \supseteq J, \mathfrak{m} \subseteq B\} \end{array}$$

The first correspondence comes from a version of the Nullstellensatz. The second one holds because once φ is applied to the left, $\varphi(Z(J)) \subseteq Y$ has to be matched with maximal ideals in A . So $\psi^{-1}(\mathfrak{m}) \subseteq \psi^{-1}(B) = A$. Clearly, $\varphi(Z(J)) \subseteq \{\mathfrak{n} \subseteq A \mid \mathfrak{n} \supseteq \varphi^{-1}(J)\}$ because $\mathfrak{m} \supseteq J$. We claim that $\varphi(Z(J)) \leftrightarrow \{\mathfrak{n} \subseteq A \mid \mathfrak{n} \supseteq \psi^{-1}(J)\}$. This would imply $Z(\psi^{-1}(J)) = \varphi(Z(J))$ according to 2/2. Since $\psi : A \rightarrow B$ is finite by assumption, $A/\psi^{-1}(J) \rightarrow B/J$ is finite as well. By Theorem 5 ("lying over"), for every $\bar{\mathfrak{n}} \subseteq A/\psi^{-1}(J)$ maximal, there exists some maximal $\bar{\mathfrak{m}} \subseteq B/J$ such that $\bar{\mathfrak{m}} \cap A/\psi^{-1}(J) = \bar{\mathfrak{n}}$. Pulling this up, we have for every $\mathfrak{n} \in A$ with $\psi^{-1}(J) \subseteq \mathfrak{n}$, there exists some maximal $\mathfrak{m} \subseteq B$ with $\mathfrak{m} \supseteq J$ such that $\psi^{-1}(\mathfrak{m}) = \mathfrak{n}$. This gives the correspondence

$$\{\mathfrak{n} \subseteq A \mid \mathfrak{n} \supseteq \psi^{-1}(J)\} \leftrightarrow \{\mathfrak{n} \subseteq A \mid \mathfrak{n} = \psi^{-1}(\mathfrak{m}) \text{ for } \mathfrak{m} \supseteq J, \mathfrak{m} \subseteq B\} \leftrightarrow \varphi(Z(J))$$

Now we have verified the claim. So $Z(\psi^{-1}(J)) = \varphi(Z(J))$. Therefore, $\varphi(Z(J))$ is closed. $\square$

Proposition 23. Let $\varphi : X \rightarrow Y$ be finite. Let $Z \subseteq Y$ be closed. Then $\varphi|_{\varphi^{-1}(Z)} : \varphi^{-1}(Z) \rightarrow Z$ is finite.

Proof. Suppose $\psi : K[Y] = A \rightarrow B = K[X]$. And $Z \subseteq Y$ is closed. So $Z = Z(I)$ for some ideal $I = \sqrt{I} \subseteq A$. Instead of having the correspondence $\varphi^{-1}(y) \leftrightarrow \{\mathfrak{m} \subseteq B | \psi(\mathfrak{m}_y)B \subseteq \mathfrak{m}\}$ like what we saw on 2/23, we have

$$\varphi^{-1}(Z(I)) \leftrightarrow \{\mathfrak{m} \subseteq B | \mathfrak{m} \supseteq \psi(I)B\}$$

This is because, according to the Nullstellensatz for coordinate rings, $Z(I)$ corresponds to I . This implies $\varphi^{-1}(Z(I)) = Z(\psi(I)B)$. So $K[\varphi^{-1}(Z(I))] = B/\sqrt{\psi(I)B}$ and $K[Z] = A/I$ are the coordinate rings. The map $\bar{\psi} : A/I \rightarrow B/\sqrt{\psi(I)B}$ is the associated map of $\varphi : \varphi^{-1}(Z) \rightarrow Z$. B is finite over A , which implies $B/\sqrt{\psi(I)B}$ is finite over A . Since $I \subseteq A \hookrightarrow B$, any $i \in I$ must have $i = \psi(i) \in \psi(I)B$. Let $b + \sqrt{\psi(I)B} \in B/\sqrt{\psi(I)B}$.

$$i(b + \sqrt{\psi(I)B}) = ib + \sqrt{\psi(I)B} = 0 + \sqrt{\psi(I)B}$$

because $(ib)^1 \in \psi(I)B \subseteq \sqrt{\psi(I)B}$. It follows that $I \subseteq \text{ann}(B/\sqrt{\psi(I)B})$. So $B/\sqrt{\psi(I)B}$ is finite over A/I . $\square$

Theorem 19. Let $\varphi : X \rightarrow Y$ be finite and surjective, then $\dim(X) = \dim(Y)$.

Proof. By Theorem 17, $\dim(X) \leq \dim(Y)$. We want to show $\dim(X) \geq \dim(Y)$. Let $Z_0 \supsetneq Z_1 \supsetneq \dots \supsetneq Z_r$ be a chain of irreducibles in Y . $\varphi^{-1}(Z_0) \subseteq X$ is closed. Let $\varphi^{-1}(Z_0) \supsetneq T_1 \supsetneq T_2 \supsetneq \dots$ be some descending chain of closed subsets of $\varphi^{-1}(Z_0)$. $T_i = X_i \cap \varphi^{-1}(Z_0)$ for some closed X_i . Notice that the sequence $X_1 \supsetneq X_1 \cap X_2 \supsetneq \dots \supsetneq X_1 \cap \dots \cap X_i \supsetneq \dots$ has the same intersection with $\varphi^{-1}(Z_0)$ as X_i . Since this sequence stabilizes, $\{T_i\}$ stabilizes as well. So $\varphi^{-1}(Z_0)$ is noetherian. By Proposition 12, $\varphi^{-1}(Z_0) = \bigcup_{i=1}^r W_{0,i}$ where each $W_{0,i}$ is closed and irreducible. Moreover,

$$W_{0,i} \xrightarrow[\text{closed}]{} \varphi^{-1}(Z_0) \xrightarrow[\text{finite}]{\text{prop 23}} Z_0$$

By Proposition 22, $\varphi(W_{0,i})$ is closed in Z_0 . Applying φ to $\varphi^{-1}(Z_0) = \bigcup_{i=1}^r W_{0,i}$ gives us $\bigcup_{i=1}^r \varphi(W_{0,i}) = \varphi(\varphi^{-1}(Z_0)) \subseteq Z_0$. Since φ is surjective, $\bigcup_{i=1}^r \varphi(W_{0,i}) = Z_0$ where each $\varphi(W_{0,i})$ is closed in Z_0 . But Z_0 is irreducible. This means that there exists some i such that $\varphi(W_{0,i}) = Z_0$. The restricted map $\varphi|_{W_{0,i}} : W_{0,i} \rightarrow Z_0$ is finite. $\varphi|_{W_{0,i}}$ is surjective and $W_{0,i}$ is irreducible. Let $W_0 := W_{0,i}$. Replace φ with $\varphi|_{W_0}$. We have $\varphi : W_0 \rightarrow Z_0 \supsetneq Z_1$. $\varphi^{-1}(Z_1)$ is closed in W_0 and $\varphi^{-1}(Z_1) \subsetneq W_0$. Repeat the argument to get W_1 as an irreducible component of $\varphi^{-1}(Z_1)$:

$$\begin{array}{ccccccc} W_0 & \supsetneq & W_1 & \supsetneq & W_2 & \supsetneq & \dots \\ \downarrow \varphi & & \downarrow \varphi & & \downarrow \varphi & & \\ Z_0 & \supsetneq & Z_1 & \supsetneq & Z_2 & \supsetneq & \dots \end{array}$$

Iterate the argument to get $W_i \supsetneq W_{i+1}$ in X that correspond to $Z_i \supsetneq Z_{i+1}$ in Y . So $\dim(X) \geq r = \dim(Y)$. $\square$

Proposition 24. Let $\psi : A \rightarrow B$ be a finite injective map of K -algebras. $\dim(A) = \dim(B)$.

Proof. Let $J = \{b \in B | b \text{ is nilpotent}\}$. $A \subseteq B$ implies $J \cap A = \{a \in A | a \text{ is nilpotent}\}$. Recall that nilpotent elements are contained in all prime ideals. So $\dim(A) = \dim(A/J \cap A)$ and $\dim(B) = \dim(B/J)$. Without loss of generality, we can assume A and B are reduced. Let $A = K[Y]$ and $B = K[X]$. $\varphi : X \rightarrow Y$ is dual to $\psi : A \rightarrow B$. By Theorem 5 ("lying over"), there is a surjection between maximal ideals of A and maximal ideals of B . But this translates to a surjection between points in X and points in Y . By Theorem 19, $\dim(X) = \dim(Y)$. By Proposition 19, $\dim(B) = \dim(X) = \dim(Y) = \dim(A)$. $\square$

Proposition 25. Let $f \in K[x_1, \dots, x_n]$ and $f \notin K$. Then $Z(f) \subseteq \mathbb{A}_K^n$ and $\dim(Z(f)) = n - 1$.

Proof. Applying Proposition 2, we can do a change of variable $x_i \mapsto t_i$ such that f is monic in t_n . Let $f = t_n^d + g_{d-1}t_n^{d-1} + \dots + g_0$ with $g_i \in K[t_1, \dots, t_{n-1}]$. The extension $K[t_1, \dots, t_{n-1}] \subseteq K[t_1, \dots, t_n]/\langle f \rangle$ is finite because it is generated by $\{1, t_n, \dots, t_n^{d-1}\}$. The variety map $\varphi : Z(f) \rightarrow \mathbb{A}_K^{n-1}$ is given by forgetting the last coordinate (i.e. $(x_1, \dots, x_{n-1}, x_n) \mapsto (x_1, \dots, x_{n-1})$). If we can show φ is finite and surjective, Theorem 19 would tell us that they are the same dimension. The corresponding ring map is

$$\begin{array}{ccc} K[t_1, \dots, t_{n-1}] & \xrightarrow{\psi} & K[t_1, \dots, t_n]/\sqrt{\langle f \rangle} \\ & \searrow \gamma & \uparrow \pi \\ & & K[t_1, \dots, t_n]/\langle f \rangle \end{array}$$

As we discussed above, γ is finite. So ψ is also finite. Thus, φ is finite. To show φ is surjective, consider the preimage of any $(x_1, \dots, x_{n-1}) \in \mathbb{A}_K^{n-1}$. Indeed, $\varphi^{-1}(x_1, \dots, x_{n-1}) = \{(x_1, \dots, x_{n-1}, t) \mid f(x_1, \dots, x_{n-1}, t) = 0\} \neq \emptyset$ because $x_1, \dots, x_{n-1}$ are predetermined and plugged in for each g_i . In other words, t is the only variable. So $f \in K[t]$. Since K is algebraically closed, we can always find a root for f . $\square$

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Proposition 26. Let X be an irreducible affine variety. Then $\dim(X) = \text{Trdeg}(\text{Frac}(K[X])/K) = \dim(K[X]) = d$, where d is the number from noether normalization (a.k.a. Theorem 6).

Proof. By Theorem 6, there exists a polynomial ring $K[t_1, \dots, t_d]$ such that $K[X]$ is a finitely generated ring extension over $K[t_1, \dots, t_d]$. The map $\gamma^\# : K[t_1, \dots, t_d] \hookrightarrow K[X]$ is a finite injective map. In particular, this is an integral extension: For any $0 \neq p \in K[X]$, there exists some monic

$$f(y) = y^m + g_{m-1}y^{m-1} + \dots + g_0 \in K[t_1, \dots, t_d][y]$$

such that $f(p) = 0$. Since $K[t_1, \dots, t_d] \subseteq K(t_1, \dots, t_d) := \text{Frac}(K[t_1, \dots, t_d])$, $f(y) \in K(t_1, \dots, t_d)[y]$. It follows that every p is also integral over $K(t_1, \dots, t_d)$. When it comes to fields, we say $K[X]$ is an algebraic extension of $K(t_1, \dots, t_d)$. Since $p \neq 0$, $1/p \in K(X) := \text{Frac}(K[X])$. Consider $h(y) = 1 + g_{m-1}y + \dots + g_1y^{m-1} + g_0y^m$. Then

$$h\left(\frac{1}{p}\right) \cdot p^m = \left(1 + g_{m-1}\left(\frac{1}{p}\right) + \dots + g_1\left(\frac{1}{p^{m-1}}\right) + g_0\left(\frac{1}{p^m}\right)\right) \cdot p^m = f(p) = 0$$

Conclude that $h(1/p) = 0$. Since $h(y) \in K(t_1, \dots, t_d)[y]$ and products of algebraic elements are algebraic, $K(X)$ is an algebraic extension of $K(t_1, \dots, t_d)$. Recall that any nonzero $p \in K[X]$ satisfies some $f(y) \in K(t_1, \dots, t_d)[y]$ due to integrality. Then $f(p) = p^m + g_{m-1}p^{m-1} + \dots + g_0 = 0$ implies

$$p^{m-1} + g_{m-1}p^{m-2} + \dots + g_1 + g_0 \frac{1}{p} = 0 \implies \frac{1}{p} = -\frac{p^{m-1} + g_{m-1}p^{m-2} + \dots + g_1}{g_0}$$

Theorem 6 gives us some generating set $U := \{u_1, \dots, u_k\}$ for $K[X]$ as a $K[t_1, \dots, t_d]$ -module. In other words, the evaluation map $\varphi_U : K[t_1, \dots, t_d][X_1, \dots, X_k] \rightarrow K[X]$ is surjective. So $p = l_k u_k + \dots + l_1 u_1 + l_0$ for some $l_i \in K[t_1, \dots, t_d]$. The induced fraction map $\Phi_U : K(t_1, \dots, t_d)[u_1, \dots, u_k] \rightarrow K(X)$ is surjective because $\Phi_U^{-1}(1/p) \neq 0$. This is exactly saying that $K(X)$ is a finite $K(t_1, \dots, t_d)$ -algebra. Combining previous results, $K(X)$ is a finitely generated algebraic extension. By 1/24, $K(t_1, \dots, t_d) \subseteq K(X)$ is a finite field extension. It follows that

$$\text{Trdeg}(K(X)/K) = \text{Trdeg}(K(X)/K(t_1, \dots, t_d)) + \text{Trdeg}(K(t_1, \dots, t_d)/K) = d$$

because an algebraic extension like $K(t_1, \dots, t_d) \subseteq K(X)$ has transcendence degree of 0. The dual map $\gamma : X \rightarrow \mathbb{A}_K^d$ is a finite map between varieties. By Proposition 9, γ is dominant. By Proposition 22, $\gamma(X)$ is closed. So $\overline{\gamma(X)} = \gamma(X) = \mathbb{A}_K^d$. Conclude that γ is surjective. By Theorem 19 and Theorem 18, $\dim(X) = \dim(\mathbb{A}_K^d) = d$. $\square$

Remark. This is the historical definition of dimension. This also shows that the d from Noether normalization is unique by the uniqueness of transcendence degree.

Let $f \in K[x_1, \dots, x_n]$ be an irreducible polynomial. Since K is a field, $K[x_1, \dots, x_n]$ is a UFD. Then the ideal generated by f is prime and the zero set $Z(f)$ is irreducible by Proposition 11. Trivially, $Z(f)$ is an affine variety.

Vocabulary 7.1 (affine curve). Let $f \in K[x, y]$ be irreducible, then the set $Z(f)$ is called the affine curve defined by $f(x, y) = 0$. The degree of f is called the degree of the affine curve.

Vocabulary 7.2 (hypersurface in $\mathbb{A}_K^n$). Let $f \in K[x_1, \dots, x_n]$ be irreducible. If $n = 3$, $Z(f)$ is called a surface; if $n > 3$, $Z(f)$ is called a hypersurface.

Theorem 20 (hypersurfaces in $\mathbb{A}_K^n$). Let $f \in K[x_1, \dots, x_n] \setminus K$. Define $f = \prod_{i=1}^l f_i^{m_i}$ where each f_i is irreducible.

1. the $\{Z(f_i)\}$ are the irreducible components of $Z(f)$. In particular, $\dim(Z(f_i)) = n - 1$.
2. $K[Z(f)] = K[x_1, \dots, x_n] / \langle f_1 \dots f_l \rangle$. This shows $\sqrt{\langle f \rangle} = \langle f_1 \dots f_l \rangle$.
3. let $X \subseteq \mathbb{A}_K^n$ be closed. $X = \bigcup_{i=1}^r X_i$ with each X_i being a irreducible component and $\dim(X_i) = n - 1$. Then there exists a f such that $X = Z(f)$.

Proof. We are proving this step by step.

1. Since $K[x_1, \dots, x_n]$ is a UFD, each $\langle f_i \rangle$ is a prime ideal. Then

$$K[Z(f_i)] = K[x_1, \dots, x_n]/I(Z(f_i)) = K[x_1, \dots, x_n]/\sqrt{f_i} = K[x_1, \dots, x_n]/\langle f_i \rangle$$

is a domain. As $f_i \neq K$, $\dim(Z(f_i)) = n - 1$ according to Proposition 25. By Proposition 11, $Z(f_i)$ is irreducible. If $f(x) = 0$, then $f_1^{m_1}(x)f_2^{m_2}(x)\dots f_l^{m_l}(x) = 0$. This implies $f_i(x) = 0$ for some i . So $Z(f) \subseteq \bigcup_{i=1}^l Z(f_i)$. On the other hand, $f_i|f$ implies $\langle f \rangle \subseteq \langle f_i \rangle$. So $Z(f_i) \subseteq Z(f)$. This gives $\bigcup_{i=1}^l Z(f_i) \subseteq Z(f)$ that leads to $\bigcup_{i=1}^l Z(f_i) = Z(f)$. Now want to show $Z(f_i) \neq Z(f_j)$ if $i \neq j$. Suppose $Z(f_i) = Z(f_j)$, then $\langle f_i \rangle = \sqrt{f_i} = \sqrt{f_j} = \langle f_j \rangle$. Thus, f_i and f_j divide each other. So $Z(f_i) = Z(\alpha f_j) = Z(f_j)$ for some unit α . This is a contradiction.

2. Let $g \in K[x_1, \dots, x_n]$. This says $g|_{Z(f)} = 0$ (i.e. $Z(f) \subseteq Z(g)$) if and only if $f_1 \dots f_l | g$. ($\Leftarrow$) By the above,

$$Z(f) = \bigcup_{i=1}^l Z(f_i) = Z\left(\bigcap_{i=1}^l \langle f_i \rangle\right) = Z(\langle f_1 \dots f_l \rangle) = Z(f_1 \dots f_l)$$

The equality in the middle holds because each f_i is irreducible. If $f_1 \dots f_l | g$, $Z(f) = Z(f_1 \dots f_l) \subseteq Z(g)$. ($\Rightarrow$) Now suppose $g|_{Z(f)} = 0$. We need to show $f_i | g$ for each i . There exists some i such that $g|_{Z(f_i)} = 0$. Thus, $g \in I(Z(f_i)) = \langle f_i \rangle$. It follows that $f_i | g$.

3. Let $X = \bigcup_{i=1}^r X_i \subseteq \mathbb{A}_K^n$. The dimension of each component of X is $n - 1$. So $\dim(X_i) = n - 1$. If $X_i = Z(f_i)$ for some f_i , then by the above argument, $X = Z(f_1 \dots f_r) = Z(f)$. Without loss of generality, we assume X is irreducible and $\dim(X) = n - 1$. Since $\dim(\mathbb{A}_K^n) = n$, $X \subsetneq \mathbb{A}_K^n$. The corresponding ring $K[X] \neq K[x_1, \dots, x_n]$. It must be that $I(X) \neq 0$. Choose $f \in I(X)$ from the vanishing ideal. We have $f|_X = 0$. Write $f = \prod_{j=1}^l f_j^{m_j}$. So $X \subseteq Z(f) = \bigcup_{j=1}^l Z(f_j)$. Since X is irreducible, $X \subseteq Z(f_j)$ for some j . Note that both X and $Z(f_j)$ are irreducible of dimension $n - 1$.

Claim. Let $Y \subseteq X$ both be irreducible and finite dimensional. Y is closed. Then $Y = X$ if and only if $\dim(Y) = \dim(X)$.

Proof: ($\Rightarrow$) Obvious. ($\Leftarrow$) Let $Y_0 \subsetneq Y_1 \subsetneq \dots \subsetneq Y_d = Y$ be a chain of length $d = \dim(Y)$. $d = \dim(X)$ implies this chain is maximal in X . It follows that $Y_d = X$.

By the claim above, $X = Z(f_j)$. This proves (3). □

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Q. Wait...I thought we already did $\dim(Z(f))$ in Proposition 25 and $\dim(\text{component})$ in Theorem 20?

A. Well actually...We chose f from $K[x_1, \dots, x_n] \setminus K = K[\mathbb{A}_K^n] \setminus K$ in both Proposition 25 and Theorem 20. And (3.) of Theorem 20 has some strong assumptions that we usually do not have. If $f \in K[X \neq \mathbb{A}_K^n] \setminus K$, things can get a little complicated.

Lemma (reduction). Let X be an irreducible affine variety. If for all $f \in K[X] \setminus K$, $\dim(Z(f)) = \dim(X) - 1$, then every component of $Z(f)$ has dimension $\dim(X) - 1$.

Proof. We proceed by contradiction. Suppose X has an $f \in K[X] \setminus K$ such that $Z(f) = \bigcup_{i=1}^r Z(f)_i$ has a component of dimension $\dim(Z(f)_i) \leq \dim(X) - 2$. Without loss of generality, we assume this is the first component in the union. Choose $p \in Z(f)_1 \setminus \bigcup_{i=2}^r Z(f)_i$. Since the union of $Z(f)_i$ is closed, we can choose some basic open $U = D(g)$ such that $p \in U$ and $U \subseteq X \setminus \bigcup_{i=2}^r Z(f)_i$. As a subset of common zeros, U is an affine variety. X being irreducible implies U is irreducible. By Proposition 10, $K[U] = K[X][g^{-1}]$. We claim that $\dim(U) = \dim(X)$. Indeed, $\text{Frac}(K[U]) = \text{Frac}(K[X])$ implies $\text{Trdeg}(\text{Frac}(K[U])) = \text{Trdeg}(\text{Frac}(K[X]))$. By Proposition 26, $\dim(U) = \dim(X)$. Consider $f|_U \in K[X] \setminus K$ on U . Notice that $Z(f|_U) = Z(f) \cap U = Z(f)_1 \cap U \subseteq U = D(g)$. This set is affine and open in side $Z(f)_1$. In particular, it is open where $g \neq 0$. But

$$\dim(Z(f|_U)) \leq \dim(Z(f)_1) \leq \dim(U) - 2 = \dim(X) - 2$$

Clearly, $f|_U$ violates the assumption of this lemma. This is a contradiction. □

Corollary. Let X be an irreducible affine variety, and let $U \subseteq X$ be an open set. Then $\dim(U) = \dim(X)$. $\square$

Theorem 21 (global principal ideal theorem). Let X be an irreducible affine variety of dimension n . Let $f \in K[X] \setminus K$. Then every component of $Z(f)$ has dimension $\dim(X) - 1$.

Proof. This theorem is proved in several steps. But before we start, let us clear up some easier cases. When $\dim(X) = 0$, this is vacuously true. When $\dim(X) = 1$, this is trivial. When $X = \mathbb{A}_K^n$, this is given by Theorem 20. Now we start the real deal:

Step 1: We use the above lemma to reduce the theorem. By Theorem 6, there exists a finite surjective map $\varphi : X \rightarrow \mathbb{A}_K^n$. This is because $n = \dim(X)$ is the number we will get if we run noether normalization on $K[X]$. In other words, $K[x_1, \dots, x_n] \simeq K[\mathbb{A}_K^n] \hookrightarrow K[X]$ is a finite injective map. By 2/12, the dual map $\varphi : X \rightarrow \mathbb{A}_K^n$ is dominant. Since $X \subseteq X$ is closed, Proposition 22 tells us that $\varphi(X)$ is closed. It follows that $\varphi(X) = \overline{\varphi(X)} = \mathbb{A}_K^n$. In particular, φ is integral. We want to show for any $f \in K[X] \setminus K$, $\dim(Z(f)) = \dim(X) - 1$.

Step 2: Let $\varphi^\# : K[x_1, \dots, x_n] \hookrightarrow K[X]$ be the associated ring map. Since f is integral over $K[x_1, \dots, x_n]$, there exists some monic $H \in K[x_1, \dots, x_n][T]$ such that $H(f) = 0$ in $K[X]$ (i.e. view it as $H(T) \in K[X][T]$). Since $K[x_1, \dots, x_n][T]$ is a UFD, we can factor $H = H_1 H_2 \dots H_r$ into irreducibles. All of which are monic in T . By Proposition 11, $K[X]$ is a domain. So $0 = H(f) = H_1(f) H_2(f) \dots H_r(f)$ implies $H_i(f) = 0$ for some i . We now assume H is irreducible.

Step 3: Set $X' = Z(H) \subseteq \mathbb{A}_K^{n+1}$. Since H is irreducible, X' is an irreducible hypersurface in $\mathbb{A}_K^{n+1}$. Otherwise, $X' = Z(H_1) \cup Z(H_2) = Z(H_1 H_2)$ would imply H has a nonconstant factorization. By Theorem 20, $\dim(X') = n = \dim(X)$. Define a projection map $\pi : X' \rightarrow \mathbb{A}_K^n$ given by $(x_1, \dots, x_n, T) \mapsto (x_1, \dots, x_n)$. This map is clearly surjective. It is also finite because

$$K[\mathbb{A}_K^n] = K[x_1, \dots, x_n] \hookrightarrow K[x_1, \dots, x_n][T] / \langle H(T) \rangle = K[X']$$

is finite due to reasons stated in *Step 1*. We can write $K[X']$ like this because $X' \subseteq \mathbb{A}_K^{n+1}$. However, X may not be affine. For this reason, we cannot say much about $K[X]$.

Step 4: Recall that $H(f) = 0$ in $K[X][T]$. Define an evaluation map $\psi^\# : K[X'] \rightarrow K[X]$ by $x_i \mapsto \varphi^\#(x_i)$ and $T \mapsto f$. This is well-defined because $H(f) = 0$ in $K[X]$. Here is the correspondence between rings and varieties

$$\begin{array}{ccc} K[\mathbb{A}_K^n] & \xleftarrow{\varphi^\#} & K[X] \\ & \searrow \pi^\# & \nearrow \psi^\# \\ & K[x_1, \dots, x_n][T] / \langle H \rangle & \end{array} \quad \begin{array}{ccc} \mathbb{A}_K^n & \xleftarrow{\varphi} & X \\ & \searrow \pi & \nearrow \psi \\ & X' & \end{array}$$

Define a map $\chi : K[x_1, \dots, x_n][T] \rightarrow K[x_1, \dots, x_n] \langle f \rangle$ by $q = \sum a_k(x_1, \dots, x_n) T^k \mapsto \sum a_k(x_1, \dots, x_n) f^k = q(f)$. This map is surjective with $\ker(\chi) = \langle H \rangle$. Thus, we get an isomorphism $K[x_1, \dots, x_n][T] / \langle H \rangle \simeq K[x_1, \dots, x_n] \langle f \rangle = K[X']$. We claim that $\psi : X \rightarrow X'$ is also a finite surjective map. Indeed, $K[X]$ is already finite over $K[x_1, \dots, x_n]$. This implies⁵ that $K[X]$ is finite over $K[x_1, \dots, x_n] \langle f \rangle$.

Step 5: Recall that the map $\psi^\# : K[x_1, \dots, x_n] / \langle H \rangle \rightarrow K[X]$ sends T to f . Also recall that Theorem 10 says for any regular map $\psi : X \rightarrow X'$, its dual map is given by $\psi^\#(g) = g \circ \psi$, where $g \in K[X']$. Note that $T : X' \rightarrow \mathbb{A}_K^1$ given by $(x_1, \dots, x_n, T) \mapsto T$ is a regular function on $K[X']$. Thus, $T \in K[X']$. It follows that $\psi^\#(T) = T \circ \psi = f$ since $\psi^\#$ sends T to f . Now consider $Z(f) \subseteq X$ and $Z(T) \subseteq X'$. Since $f = T \circ \psi$, one has

$$x \in Z(f) \iff f(x) = 0 \iff T(\psi(x)) = 0 \iff \psi(x) \in Z(T)$$

Hence $Z(f) = \psi^{-1}(Z(T))$. We have shown $\psi : X \rightarrow X'$ is a finite surjection. By Proposition 23, $\psi|_{\psi^{-1}(Z(T))} : \psi^{-1}(Z(T)) \rightarrow Z(T)$ is finite. $\psi|_{\psi^{-1}(Z(T))=Z(f)}$ is surjective because if $y \in Z(T)$, surjectivity of ψ gives some $x \in X$ with $\psi(x) = y$, and then $T(y) = 0$ implies $f(x) = 0$, so $x \in Z(f)$. Therefore, $\psi|_{Z(f)} : Z(f) \rightarrow Z(T)$ is a finite surjection as well. By Theorem 19, $\dim(Z(f)) = \dim(Z(T))$. Now we need to show $\dim(Z(T)) = n - 1$.

Step 6: Consider $K[x_1, \dots, x_n] \subseteq K[x_1, \dots, x_n, T] / \langle H \rangle$. We claim that $\langle T \rangle \cap K[x_1, \dots, x_n] = \langle a_0 \rangle$, where $H(T) = \sum_{i=0}^m a_i T^i$. a_i are in $K[x_1, \dots, x_n]$. Since H is monic, $a_m = 1$. Observe that $a_0 = -\sum_{i=1}^m a_i T^i \in \langle T \rangle$. So $a_0 \in \langle T \rangle \cap K[x_1, \dots, x_n]$. Now we want to show the other direction. For any arbitrary $b = \sum_{j=0}^{m-1} b_j T^j \in K[x_1, \dots, x_n, T] / \langle H \rangle$,

⁵Recall that $K[x_1, \dots, x_n] \langle f \rangle$ is the smallest ring containing $K[x_1, \dots, x_n]$ and f . So $K[x_1, \dots, x_n] \subseteq K[x_1, \dots, x_n] \langle f \rangle$.

a typical element of $\langle T \rangle$ would be

$$T \cdot b = \sum_{j=0}^{m-2} b_j T^{j+1} - b_{m-1} \left(\sum_{i=0}^{m-1} a_i T^i \right) = -a_0 b_{m-1} + c_1 T + c_2 T^2 + \dots + c_{m-1} T^{m-1}$$

Note that $T \cdot b \in K[x_1, \dots, x_n]$ (and hence $T \cdot b \in \langle T \rangle \cap K[x_1, \dots, x_n]$) if and only if $c_1 = c_2 = \dots = c_{m-1} = 0$. It follows that $T \cdot b = -a_0 b_{m-1} \in \langle a_0 \rangle$. Now we can use this claim to finish up the proof.

Step 7: Let $R = K[x_1, \dots, x_n]$, $A = K[x_1, \dots, x_n, T]/\langle H \rangle$, and $I = \langle T \rangle \cap K[x_1, \dots, x_n]$. Note that $IA \subseteq \langle T \rangle$. Then $A/\langle T \rangle \simeq (A/IA)/(\langle T \rangle/IA)$ is a quotient of the R/I -module A/IA . We know that A is finite over R . It follows⁶ that A/IA is finite over R/I . Any quotient of a finite R/I -module is still finite. So

$$K[x_1, \dots, x_n]/\langle T \rangle \cap K[x_1, \dots, x_n] = K[x_1, \dots, x_n]/\langle a_0 \rangle \hookrightarrow A/\langle T \rangle = K[x_1, \dots, x_n, T]/\langle H, T \rangle$$

is finite. By Proposition 24, $\dim(K[x_1, \dots, x_n]/\langle a_0 \rangle) = \dim(K[x_1, \dots, x_n, T]/\langle H, T \rangle)$. Suppose the dimension of $K[x_1, \dots, x_n]/\langle a_0 \rangle$ is less or equal to $n - 2$. By Proposition 25, $\langle a_0 \rangle$ is a unit. This is because for $a_0 \in K[x_1, \dots, x_n]$, $\dim(Z(a_0)) = \dim(K[x_1, \dots, x_n]/\langle a_0 \rangle) = n - 1$. And $\dim(Z(a_0)) < n - 1$ if and only if $\langle a_0 \rangle = K[x_1, \dots, x_n]$. So T is also a unit⁷. This is a contradiction. $\square$

Week 8

Did more corollaries of the global ideal theorem including the Catenary property. Local data like germ and stalk. Identified stalks with local rings. Local dimension theory and prime avoidance.

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Let $\varphi : X \rightarrow Y$ be surjective or dense. We want to relate $\dim(X)$, $\dim(Y)$, and $\dim(\varphi^{-1}(y))$ for any $y \in Y$. The global principal ideal theorem was a big deal. This allows us to talk about a few more things:

- immediate corollaries
- local dimensions and local rings
- "local principal ideal theorem"
- dimension of base and fiber

Corollary. Let X be an affine variety. Let $Y \subseteq Z$ be irreducible closed subsets of X such that $\dim(Y) \leq \dim(Z) - 2$. Then there exists an irreducible closed subset such that $Y \subsetneq W \subsetneq Z$.

Proof. Since $\dim(Y) = Z - 2$, we have $Y \subsetneq Z$. Now consider $K[Z]$. Because $Y \subsetneq Z$, the ideal $I(Z) \subseteq I(Y)$ are strict inclusions of prime ideals (irreducible implies domain...) in $K[X]$. So $I(Y)/I(Z) \neq 0$. In $K[Z]$, the ideal $I(Y)$ becomes $I(Y)/I(Z)$. And since $Y \neq \emptyset$, $I(Y)/I(Z)$ is not the unit ideal. $I(Y)$ being prime implies $I(Y)/I(Z)$ is a nonzero proper prime ideal of $K[Z]$. So

$$0 \subsetneq I(Y)/I(Z) \subsetneq K[X]/I(Z) = K[Z]$$

Choose a nonzero $f \in I(Y)$ so that $Y \subseteq Z(f)$. Since Y is irreducible, $Y \subseteq W$, where W is some component of $Z(f)$. By Theorem 21, $\dim(W) = \dim(Z) - 1 > \dim(Y)$. $\square$

Theorem 22 (Catenary property). Let X be an affine variety. Suppose $Z \subsetneq W$ are two irreducible subsets of X . Write $\dim(Z) = i$ and $\dim(W) = i + j$ for some nonnegative integers i and j with $j > 0$. Then there exists a chain $Z = Z_0 \subsetneq \dots \subsetneq Z_j = W$ of affine irreducibles with $\dim(Z_l) = i + l$ for all l .

Proof. Perform induction on the Corollary above. $\square$

Corollary. Let X be an irreducible affine variety. Every maximal chain of irreducible subsets of X has a length equal to the dimension of X .

⁶<https://math.stackexchange.com/questions/2807783/>

⁷Remember that $a_0 \in \langle T \rangle \cap K[x_1, \dots, x_n] \subseteq \langle T \rangle$. If a_0 is a unit of K , $\langle T \rangle = K[x_1, \dots, x_n, T]/\langle H \rangle$.

Corollary. Let A be a finitely generated K -algebra. Let $\mathfrak{p} \subsetneq \mathfrak{q}$ be primes. Then every chain of primes starting in $\mathfrak{p}$, ending in $\mathfrak{q}$, is of length $\dim(A/\mathfrak{p}) - \dim(A/\mathfrak{q})$.

Corollary. Let A be a finitely generated K -algebra (a domain), then for all primes $\mathfrak{p} \subsetneq A$,

$$\dim(A) = \dim(A/\mathfrak{p}) + \text{ht}(\mathfrak{p})$$

Corollary. Let X be irreducible and $\dim(X) = n$. Let $\{f_1, \dots, f_r\} \subseteq K[X]$. Suppose $Z(f_1, \dots, f_r) \neq \emptyset$, then every component of $Z(f_1, \dots, f_r)$ has dimension greater or equal to $n - r$.

Proof. Induct on r . If $r = 1$, use Theorem 21. Suppose the corollary is true for $Z := Z(f_1, \dots, f_{r-1})$. Let $\{Z_i\}_{i=1}^t$ be the components of Z . So, by induction, Z is a variety and $\dim(Z_i) \geq n - (r - 1) = n - r + 1$ for all i . We want to show each component of $Z(f_r|_Z)$, that is also inside Z , has dimension greater or equal to $n - r$. Suppose W is a component of $Z(f_r|_Z)$ such that $\dim(W) \leq n - r - 1$. Since W is irreducible and closed, $W \subseteq Z_i$ for some Z_i . So W is a component of $Z(f_r|_{Z_i}) \subseteq Z(f_r|_Z)$. Assume $\dim(W) = n - r - 1$. Then

$$\dim(Z_i) - \dim(W) \geq 2$$

But Theorem 21 says for any $f_r|_{Z_i} \in K[Z_i] \setminus K$, every component of $Z(f_r|_{Z_i})$ has dimension $\dim(Z_i) - 1$. In particular, $\dim(Z_i) - \dim(W) = 1$. This is a contradiction. Thus, $\dim(W) \geq n - r$ as required. $\square$

Q. What about the converse? Say $X \subseteq \mathbb{A}_K^n$ and $\dim(X) = n - r$. Is $X = Z(f_1, \dots, f_r)$?

A. No. But if we choose $p \in X$ and work "near" p , it can be true.

Vocabulary 8.1 (germ). Let X be a variety and $p \in X$. A germ of a function on X near p is a pair (U_p, f) . $p \in U_p$ is open. $f : U \rightarrow \mathbb{A}_K^1$ is regular. Two germs (U_p, f) and (W_p, g) are equivalent if there exists an open $V_p \subseteq U_p \cap W_p$ such that $f|_{V_p} = g|_{V_p}$. If p is well-known, we sometimes write U , V , and W instead.

Vocabulary 8.2 (stalk). The stalk at p , denoted as $\mathcal{O}_{X,p}$, is the set of equivalence classes of germs near p .

Lemma. $\mathcal{O}_{X,p}$ is a ring.

Proof. $(X, 0)$ and $(X, 1)$ are the 0 and 1. $(U, f) + (W, g) = (U \cap W, f + g)$. $(U, f) \cdot (W, g) = (U \cap W, fg)$. $\square$

Vocabulary 8.3 (local ring). A local ring is a ring with a single maximal ideal.

Lemma. Let $\mathfrak{m}_p \subseteq \mathcal{O}_{X,p}$ be an ideal generated by $[(U, f)]$ with $f(p) = 0$. Then $\mathfrak{m}_p$ is a maximal ideal and $\mathcal{O}_{X,p}$ is a local ring.

Proof. Consider the evaluation map $\text{ev}_p : \mathcal{O}_{X,p} \rightarrow K$ given by $[(X, \lambda)] \mapsto \lambda(p)$. This map is surjective because we can just set λ to a constant map. Moreover, $\ker(\text{ev}_p)$ corresponds to precisely functions such that $f(p) = 0$. In other words, $\ker(\text{ev}_p) = \mathfrak{m}_p$. Now we want to show $\mathcal{O}_{X,p} \setminus \mathfrak{m}_p$ is a set of units in $\mathcal{O}_{X,p}$. Let $[(U, f)] \in \mathcal{O}_{X,p}$ such that $f(p) \neq 0$. Since f is locally rational, $f = (f_1/f_2)|_X$ for some f_1, f_2 defined on V_p . Due to continuity, f_1 is nonzero near p . It follows that f_2/f_1 is well-defined on some W_p . So $(W, 1/f)$ is a germ. $f \in \mathcal{O}_{X,p}$ a unit. $\square$

Theorem 23. Let X be an affine variety. Let $p \in X$. Let $\mathfrak{m} \subseteq K[X]$ be a maximal ideal centered at p . Then

$$(K[X] \setminus \mathfrak{m}_p)^{-1} K[X] = K[X]_{\mathfrak{m}_p} \simeq \mathcal{O}_{X,p}$$

Proof. Let $\varphi : K[X] \rightarrow \mathcal{O}_{X,p}$ be a map given by $f \mapsto (X, f)$. If $f \notin \mathfrak{m}_p$, then $f(p) \neq 0$. So $f = f_1/f_2$ is invertible near p . It follows that $\varphi(f) \in \mathcal{O}_{X,p}$ is a unit as well. The induced map $\chi : (K[X] \setminus \mathfrak{m}_p)^{-1} K[X] \rightarrow \mathcal{O}_{X,p}$ exists by Theorem A2. For simplicity, write $S = (K[X] \setminus \mathfrak{m}_p)^{-1}$. Now we want to show χ is surjective and injective.

Surjective: Let (U, f) be a germ. Since basic opens generate the Zariski topology, it suffices to prove the theorem on basic opens. Without loss of generality, assume $U = D(g)$ for some $g \in K[X]$. Suppose $(U, f) = (D(g), f_1/f_2)$, where $f_2 \neq 0$ on $D(g)$. So $(U, f) = (D(g), f_1 g / f_2 g) = (D(f_2 g), f_1 g / f_2 g)$. If $p \in D(f_2 g)$, then $f_2(p) \neq 0$ and $g(p) \neq 0$. This implies $f_2 g \in S$ is invertible. So $1/f_2 g \cdot f_1 g \in S^{-1} K[X]$.

Injective: Suppose $\chi(f/g) = (f/g, D(g)) = 0$ in $\mathcal{O}_{X,p}$ so that $f/g \in \ker(\chi)$. Since $g(p) \neq 0$, $p \in D(g)$. Note that g is invertible on basic open $D(g)$. This makes f/g regular. With the identification from Proposition 10,

$$\frac{f}{g} \in K[D(g)] \simeq K[X][g^{-1}]$$

Now we claim $f/g = 0$. By the definition of the equivalence of germs, there exists⁸ some neighborhood $U_p \subseteq D(g)$ such that $f/g|_{U_p} = 0$. Without loss of generality, assume $U = D(h)$ with $h(p) \neq 0$. Since f/g is regular on $D(h)$ and it is 0 on $D(h)$, $f/g = 0$ inside $K[D(h)] \simeq K[X][h^{-1}]$. Finally, $f/g = 0$ in $S^{-1}K[X]$ because there exists a map $K[X][h^{-1}] \rightarrow S^{-1}K[X]$ (since $h' \in S$), and it sends the element f/g to 0 in $S^{-1}K[X]$. $\square$

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Vocabulary 8.4 (local dimension). $\dim_p(X) = \max\{\dim(X_i) | X_i \text{ is a component, } p \in X_i\}$.

Theorem 24. Let X be an affine variety. $\dim(\mathcal{O}_{X,p}) = \dim_p(X)$.

Proof. Let X_r be a component such that $\dim_p(X) = \dim(X_r)$. By Theorem 23, we have $\mathcal{O}_{X,p} \simeq K[X]_{\mathfrak{m}_p} = (K[X] \setminus \mathfrak{m}_p)^{-1}K[X]$. Theorem A3 gives us a correspondence between prime ideals. So

$$\begin{aligned} \dim(\mathcal{O}_{X,p}) = \max\{r | \mathfrak{p}_0 \supsetneq \dots \supsetneq \mathfrak{p}_r, \mathfrak{p}_i \subseteq K[X]_{\mathfrak{m}_p}\} &\iff \max\{r | \mathfrak{q}_0 \supsetneq \dots \supsetneq \mathfrak{q}_r, \mathfrak{q}_i \subseteq \mathfrak{m}_p \subseteq K[X]\} \\ &\iff \max\{r | p \in Z_0 \subsetneq \dots \subsetneq Z_r, Z_i \subseteq \mathbb{A}_K^n \text{ clsd and irred}\} \end{aligned}$$

Note that $\dim(\mathcal{O}_{X,p})$ is at most $\dim_p(X)$. Since every chain of irreducibles must be contained in some component of X , $\dim(\mathcal{O}_{X,p}) \leq \dim_p(X)$. On the other hand, a chain with length $\dim(X_r)$ exists according to Theorem 22. Here, the chain starts with p which corresponds to $\mathfrak{m}_p$. By the Nutz, $p \in \mathbb{A}_K^n$ is closed because $\mathfrak{m}_p$ is radical. Since $X_r \subseteq X$, $\dim(\mathcal{O}_{X,p}) \geq \dim_p(X)$. $\square$

Theorem 25 (prime avoidance). Let $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ be prime ideals of a commutative ring R . Then for any ideal $I \subseteq R$, $I \subseteq \bigcup_{i=1}^r \mathfrak{p}_i$ implies $I \subseteq \mathfrak{p}_s$ for some s .

Proof. We show the contrapositive statement: If $I \not\subseteq \mathfrak{p}_i$ for prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$, then $I \not\subseteq \bigcup_{i=1}^n \mathfrak{p}_i$. Now perform induction on n . Assume this result holds for $n-1$.

$$I \not\subseteq \mathfrak{p}_{1, \dots, \hat{i}, \dots, n} \iff \exists x_i \notin \mathfrak{p}_{j \neq i} \quad (a)$$

$$I \not\subseteq \bigcup_{j=1, j \neq i}^n \mathfrak{p}_j \iff \exists x_i \notin \bigcup_{j \neq i} \mathfrak{p}_j \quad (b)$$

The induction hypothesis assumes $(a) \Rightarrow (b)$. If $x_i \notin \mathfrak{p}_i$ in all these cases as well, then we can conclude that $x_i \notin \bigcup_{i=1}^n \mathfrak{p}_i$. We are done. Otherwise, we assume $x_i \in \mathfrak{p}_i$ in every case. Define $y = \sum_{k=1}^n \prod_{j \neq k} x_j$. We claim that $y \notin \mathfrak{p}_i$ for all i and $y \in I$. It suffices to check the $i=1$ case. Since $x_1 \in \mathfrak{p}_1$,

$$y = x_2 x_3 \cdots x_n + \sum_{k=2}^n \prod_{j \neq k} x_j$$

Note that when $k \geq 2$, each summand contains the factor $x_1 \in \mathfrak{p}_1$. So $y \in \mathfrak{p}_1$ if and only if $x_2 x_3 \cdots x_n \in \mathfrak{p}_1$. Since $\mathfrak{p}_1$ is prime, $x_j \in \mathfrak{p}_1$ for some $j \in \{2, \dots, n\}$. This contradicts the defining property of x_j above, because $j \neq 1$ and we chose $x_j \notin \bigcup_{l \neq j} \mathfrak{p}_l$, in particular $x_j \notin \mathfrak{p}_1$. On the other hand, $y \in I$ because every summand has a factor in I . Conclude that $y \notin \bigcup_{i=1}^n \mathfrak{p}_i$. $\square$

Theorem 26. Let X be a variety and $p \in X$. Then $\dim_p(X)$ equals the least integer n such that there exists an open U_p and n functions $f_1, \dots, f_n \in K[U]$ such that $Z(f_1, \dots, f_n) = \{p\}$.

Proof. We perform induction on $n = \dim_p(X)$. If $n = 0$, $\{p\}$ itself is a component of X . The ideal not generated by any function gives the whole component. So it is $\{p\}$. Now suppose $n > 0$. Let $X_1, \dots, X_r$ be components containing $\{p\}$. Let $X' := \bigcup_i X_i$. Due to reverse inclusions, irreducible components correspond to minimal prime ideals. Each X_i corresponds to a prime ideal $\mathfrak{p}_i \subseteq K[X']$. For each $X_i \supsetneq \{p\}$, we have $\mathfrak{p}_i \subsetneq \mathfrak{m}_p$. By Theorem 25 (prime avoidance), we must have

$$\mathfrak{m}_p \not\subseteq \bigcup_{i=1}^r \mathfrak{p}_i$$

⁸The map $\psi : A[h^{-1}] \rightarrow S^{-1}A$ is given by $a/h^n \mapsto a/h^n$. So every denominator allowed in $A[h^{-1}]$ is also allowed in $S^{-1}A$.

Now choose $f_1 \in \mathfrak{m}_p$ such that f_1 is not contained in any $\mathfrak{p}_i$. That is, $f_1(0) = 0$ but $f_1|_{X_i} \neq 0$ for all i . Theorem 21 implies $\dim(Z(f_1)) = n - 1$ and $p \in Z(f_1)$. By the induction hypothesis applied to the affine variety $Z(f_1)$ at p , there is a basic open neighborhood $W = D(g) \cap Z(f_1)$ of p (with some $g \in A$ and $g(p) \neq 0$) and functions $\bar{f}_2, \dots, \bar{f}_n \in K[W] = A_g / \langle f_1 \rangle$ such that $Z_{Z(f_1)}(\bar{f}_2, \dots, \bar{f}_n) = \{p\}$ inside W . Can find some basic open U such that $Z_U(f_2, \dots, f_n) = Z_W(f_2, \dots, \bar{f}_n) = \{p\}$. On the same U , we have that $f_1(p) = 0$. So $Z_U(f_1, \dots, f_n) = Z_W(f_2, \dots, \bar{f}_n) = \{p\}$. $\square$

Week 9

spring break.

Week 10

Shown results related to the dimension of local rings. Calculated the dimension of fibers of dominant maps. Introduced the projective space. Defined a few more things related to $\mathbb{P}_K^n$. Proved the homogeneous Nutz.

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Proposition 27. Let $\mathfrak{m}_p$ be the unique maximal ideal of $\mathcal{O}_{X,p}$. The Krull dimension $\dim(\mathcal{O}_{X,p})$ is the least integer n such that there exist $\{f_1, \dots, f_n\} \subseteq \mathfrak{m}_p$ satisfying $\sqrt{\langle f_1, \dots, f_n \rangle} = \mathfrak{m}_p$. In particular, $\dim(\mathcal{O}_{X,p})$ is less or equal to the number of generators of $\mathfrak{m}_p$.

Proof. Our goal is to show there exist some $\{f_1, \dots, f_n\} \subseteq \mathcal{O}_{X,p}$ such that $\sqrt{\langle f_1, \dots, f_n \rangle} = \mathfrak{m}_p$ if and only if there exists $U \subseteq X$, $p \in U$, $\{f'_1, \dots, f'_n\} \subseteq K[U]$ with $Z(f'_1, \dots, f'_n) = \{p\}$. Theorem 26 will give the result from here.

($\Rightarrow$) There exists an open $U \ni p$ such that $f_1, \dots, f_n$ are defined on U because each $f_i \in \mathcal{O}_{X,p} \simeq K[U]_{\mathfrak{m}_p}$ is a germ. Shrink U if necessary. Choose $f'_i \in K[U]$ whose germs are f_i at p . Since we are working with polynomial rings, we get noetherian coordinate rings and their noetherian localizations. As a result, we can find g_i defined on U (again, shrink U if necessary) such that $\mathfrak{m}_p = \langle g_1, \dots, g_n \rangle$ in $\mathcal{O}_{X,p}$. By assumption, each $g_i \in \sqrt{\langle f_1, \dots, f_n \rangle} = \mathfrak{m}_p$ has some N_i such that $g_i^{N_i} \in \langle f_1, \dots, f_n \rangle$. We may write $g_i^{N_i} = \sum_{j=1}^n a_{ij} f_j$, where $a_{ij} \in \mathcal{O}_{X,p}$. Without loss of generality, assume $a_{ij} \in K[U]$. If not, we can multiply the common denominator and shrink U to where the denominator is invertible. So $g_i^{N_i} = \sum_{j=1}^n a_{ij} f'_j$ and $\mathfrak{m}_p = \langle g_1, \dots, g_n \rangle \subseteq \sqrt{\langle f'_1, \dots, f'_n \rangle}$ in $K[U]$. Since germs of g_i generate the maximal ideal $\mathfrak{m}_p$ in $K[U]$, $\sqrt{\langle f'_1, \dots, f'_n \rangle} = \mathfrak{m}_p$. It follows that $Z(f'_1, \dots, f'_n) = \{p\}$ in U .

($\Leftarrow$) $Z(f'_1, \dots, f'_n) = \{p\}$ implies $\sqrt{\langle f'_1, \dots, f'_n \rangle} = \mathfrak{m}_p$ in $K[U]$. Since localization commutes with taking radicals, $\sqrt{\langle f_1, \dots, f_n \rangle} = \mathfrak{m}_p$ in $\mathcal{O}_{X,p}$. $\square$

Example. The $f_1, \dots, f_n$ may not be generators. Take $K[X] = K[x, y] / \langle x^2 - y^3 \rangle$ and $p = 0$. The local ring is $\mathcal{O}_{X,0} = (K[x, y] \setminus \mathfrak{m}_0)^{-1} K[t^2, t^3]$ with the unique maximal ideal $\mathfrak{m}_0 = \langle t^2, t^3 \rangle$. Despite $\mathfrak{m}_0$ being non-principal, it can be the radical of a principal ideal: $\mathfrak{m}_0 = \sqrt{\langle t^2 \rangle}$. Indeed, t^2 may not be the generator. By Proposition 20, $\dim(X) = 1$. This number matches $n = 1$ from Proposition 27.

Theorem 27. Let X, Y be affine irreducible varieties. Let $\varphi : X \rightarrow Y$ be dominant. These things hold

1. $\dim(X) \geq \dim(Y)$.
2. If φ is surjective, for all $p \in Y$, $\dim(\varphi^{-1}(p)) \geq \dim(X) - \dim(Y) \geq 0$.

Proof. (1.) By Proposition 9, φ is dominant if and only if $\varphi^\sharp : K[Y] \rightarrow K[X]$ is injective. So when it comes to the fraction fields, we have $K(Y) \subseteq K(X)$. It follows that $\text{Trdeg}(K(Y)/K) \leq \text{Trdeg}(K(X)/K)$. By Proposition 26, $\dim(Y) \leq \dim(X)$. (2.) Let $p \in Y$. Let $U \subseteq X$ be open. Suppose $\dim(Y) = n$. By Theorem 26, there exists $\{f_1, \dots, f_n\}$ such that $Z(f_1, \dots, f_n) = \{p\}$. In particular, $Z(f_1, \dots, f_n) \cap U = \{p\} \cap U = \{p\}$. So

$$x \in \varphi^{-1}(p) \iff \varphi(x) \in Z(f_1, \dots, f_n) \cap U \iff f_i(\varphi(x)) = 0 \iff x \in Z(f_1 \circ \varphi, \dots, f_n \circ \varphi)$$

It follows that $\varphi^{-1}(p) \cap \varphi^{-1}(U) = Z(f_1 \circ \varphi, \dots, f_n \circ \varphi) \cap \varphi^{-1}(U)$. Note that $\varphi^{-1}(U)$ is open and $\varphi^{-1}(U) \cap \varphi^{-1}(p)$ is open in $\varphi^{-1}(p)$. Since $\varphi^{-1}(p)$ is subset of an irreducible variety, it is irreducible. By the Corollary after the reduction

lemma in 3/1, $\dim(\varphi^{-1}(U) \cap \varphi^{-1}(p)) = \dim(\varphi^{-1}(p))$. Now choose an affine open neighborhood $x \in W \subseteq \varphi^{-1}(U)$. So $\varphi^{-1}(p) \cap W = Z_W((f_1 \circ \varphi)|_W, \dots, (f_n \circ \varphi)|_W)$. By 3/1 again, $\dim(W) = \dim(X)$. Combining all this:

$$\begin{aligned} \dim(\varphi^{-1}(p)) &= \dim(\varphi^{-1}(U) \cap \varphi^{-1}(p)) \geq \dim(W \cap \varphi^{-1}(p)) \\ &= \dim(Z_W((f_1 \circ \varphi)|_W, \dots, (f_n \circ \varphi)|_W)) \\ &\geq \dim(W) - n = \dim(X) - \dim(Y) \end{aligned} \quad (3/4)$$

□

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Theorem 28. Let X, Y be affine irreducible varieties. Let $\varphi : X \rightarrow Y$ be dominant. There exists open nonempty $U \subseteq Y$ such that for all $p \in U$, $\dim(\varphi^{-1}(p)) = \dim(X) - \dim(Y) \geq 0$. In particular, $\varphi^{-1}(p) \neq \emptyset$, i.e. $U \subseteq \varphi(X)$.

Proof. We have $\varphi^\sharp : A := K[Y] \hookrightarrow B := K[X]$. A, B are finite over K . So we can write $B = K[\alpha_1, \dots, \alpha_r]$. Since $K \subseteq A$, $B = K[\alpha_1, \dots, \alpha_r] \subseteq A[\alpha_1, \dots, \alpha_r]$. On the other hand, $A[\alpha_1, \dots, \alpha_r] \subseteq B$. Therefore, it is safe to say $\{\alpha_1, \dots, \alpha_r\}$ are algebra generators of B over A . Let $S = K[Y] \setminus \{0\}$ be a multiplicative closed set of $K[Y]$. There is an inclusion $K(Y) := S^{-1}K[Y] \hookrightarrow S^{-1}K[X]$. The set $\{\alpha_1/1, \dots, \alpha_r/1\}$ generates $S^{-1}K[X]$ over $K(Y)$. Indeed, if $\beta \in S^{-1}K[X]$, then $\beta = f/s$, where $f = \sum_n c_{n_1, \dots, n_r} \alpha_1^{n_1} \cdots \alpha_r^{n_r}$ and $c_{n_1, \dots, n_r}$ are constants from $K[Y]$. It follows that $f = \sum_n (c_{n_1, \dots, n_r}/s) \alpha_1^{n_1} \cdots \alpha_r^{n_r}$. Applying Theorem 6 to $S^{-1}K[X]$, we know there exist $\{t_1, \dots, t_d\} \subseteq S^{-1}K[X]$ such that $K(Y) \langle t_1, \dots, t_d \rangle = K(Y)[t_1, \dots, t_d]$. Moreover, $S^{-1}K[X]$ is finitely generated as a $K(Y)[t_1, \dots, t_d]$ -module. Now we have an important claim:

Claim. There exists $0 \neq f \in K[Y] = A$ such that $\{t_1, \dots, t_d\} \subseteq K[X][f^{-1}] = B[f^{-1}]$ and

- ① $A[f^{-1}] \langle t_1, \dots, t_d \rangle = A[f^{-1}][t_1, \dots, t_d]$.
- ② $A[f^{-1}][t_1, \dots, t_d] \subseteq B[f^{-1}]$ is given by module finiteness (i.e. $B[f^{-1}]$ is finite as a module over $A[f^{-1}][t_1, \dots, t_d]$). Note that $A[f^{-1}] \subseteq A[f^{-1}][t_1, \dots, t_d]$.

Proof: We know that $\{t_1, \dots, t_d\} \subseteq S^{-1}B$. So write $t_i = b_i/s_i$. Setting $f = s_1 \cdots s_d$, we get $t_i \in B[f^{-1}]$. Now prove each claim separately

- ① Suppose there exists a $p \in A[f^{-1}][y_1, \dots, y_d]$ such that $p(t_1, \dots, t_d) = 0$ in $B[f^{-1}]$. Recall that localizing at x and y is the same as localizing at xy . Consequentially, $p(t_1, \dots, t_d) = 0$ in $S^{-1}B$. Earlier in the proof, we showed that $S^{-1}B$ is a finite $K(Y)$ -algebra. Since $t_i \in S^{-1}B$ are algebraically independent over $K(Y)$, all coefficients of p are 0 in $K(Y)$. But coefficients of p lie in $A[f^{-1}] \subseteq S^{-1}A = K(Y)$. They must have been 0 in $A[f^{-1}]$. This shows that each t_i is algebraically independent over $A[f^{-1}]$. Therefore, they generate a polynomial ring.

Recall that B is a finite A -algebra. Following the reasoning in the beginning of this proof, we conclude that $B[f^{-1}]$ is a finitely generated $A[f^{-1}]$ -algebra. But $A[f^{-1}] \subseteq A[f^{-1}][t_1, \dots, t_d] \subseteq B[f^{-1}]$, we say that there exists some $\{x_1, \dots, x_n\}$ such that $B[f^{-1}]$ is generated as an $A[f^{-1}][t_1, \dots, t_d]$ -algebra. We need to show each $x_i \in B[f^{-1}] \subseteq S^{-1}B$ is integral over $A[f^{-1}] \langle t_1, \dots, t_d \rangle$. From Theorem 6, we saw that $S^{-1}B$ is a finitely generated $S^{-1}A \langle t_1, \dots, t_d \rangle$ -module. Since finiteness implies integrality, each x_i is integral over $S^{-1}A \langle t_1, \dots, t_d \rangle$. So x_i satisfies some monic $p_i(x_i) = 0$ with $p_i \in S^{-1}A \langle t_1, \dots, t_d \rangle [z]$.

- ② Let a be a coefficient of p_i . We may write it as a finite sum $a = \sum_{\mathbf{I}} (a_{\mathbf{I}}/s_{\mathbf{I}}) t^{\mathbf{I}}$. Define

$$\tilde{f} := f \cdot \left(\prod_{\{\mathbf{I} | s_{\mathbf{I}} \neq 0\}} s_{\mathbf{I}} \right)$$

Now update our f to $\tilde{f}$ so that $a \in A[f^{-1}] \langle t_1, \dots, t_d \rangle$. We may further update f by multiplying other elements of S if necessary. So assume that $p_i \in A[f^{-1}] \langle t_1, \dots, t_d \rangle [z]$ and $p_i(x_i) = 0$. Conclude that x_i is integral over $A[f^{-1}] \langle t_1, \dots, t_d \rangle$. By Theorem 4, $B[f^{-1}]$ is finite over $A[f^{-1}] \langle t_1, \dots, t_d \rangle$.

Set $U = Y \setminus Z(f) = D(f)$. If $\varphi : \varphi^{-1}(U) \rightarrow U$, then the dual map is $\varphi^\sharp : K[U] \rightarrow K[\varphi^{-1}(D(f)) = X \setminus Z(f \circ \varphi)]$. In basic opens, $\varphi^\sharp : K[D(f)] \rightarrow K[D(f \circ \varphi) = D(\varphi^\sharp(f))]$. By Proposition 10, $\varphi^\sharp : A[f^{-1}] \rightarrow B[(\varphi^\sharp f)^{-1}]$. Recall

that $f \in K[Y]$ and that $K[Y] \subseteq K[X]$ is an inclusion. We get the correspondence

$$\begin{array}{ccc} \varphi : \varphi^{-1}(U) & \longrightarrow & U \\ \uparrow \simeq & & \\ X \setminus Z(f \circ \varphi) & & \end{array} \quad \begin{array}{ccc} \varphi^\# : A[f^{-1}] & \longrightarrow & B[f^{-1}] \\ \downarrow & \nearrow & \\ A[f^{-1}][t_1, \dots, t_d] & & \end{array} \quad (\clubsuit)$$

Now we need to show $\varphi : \varphi^{-1}(U) \rightarrow U$ is surjective. Let $\mathfrak{m} \subseteq A[f^{-1}]$ be maximal. It follows that $\mathfrak{n} = \mathfrak{m} + \langle t_1, \dots \rangle$ is a maximal ideal of $A[f^{-1}][t_1, \dots, t_d]$ such that $\mathfrak{n} \cap A[f^{-1}] = \mathfrak{m}$ because the quotient $A[f^{-1}][t_1, \dots, t_d]/\mathfrak{n} \simeq A[f^{-1}]/\mathfrak{m}$. By the important claim above, $A[f^{-1}][t_1, \dots, t_d] \subseteq B[f^{-1}]$ is integral. By Theorem 5, there exists a maximal ideal $\mathfrak{n}' \subseteq B[f^{-1}]$ such that $\mathfrak{n}' \cap A[f^{-1}][t_1, \dots, t_d] = \mathfrak{n}$. This proves surjectivity because every point of U , denoted as $\mathfrak{m}$, has a point of $\varphi^{-1}(U)$, denoted as $\mathfrak{n}'$, that maps to it. Now compute the dimension of the fiber. The factorization in $(\clubsuit)$ induces

$$\varphi^{-1}(U) \xrightarrow{\psi} U \times \mathbb{A}^d \xrightarrow{\pi} U$$

where ψ is the finite map and π is the projection. Since $\varphi|_{\varphi^{-1}(U)} = \pi \circ \psi$, $\varphi^{-1}(p) = \psi^{-1}(\pi^{-1}(p))$. The preimage of a point in U is affine in $U \times \mathbb{A}^d$ because for $\mathfrak{m} \subseteq A[f^{-1}]$ maximal, $A[f^{-1}][t_1, \dots, t_d]/\mathfrak{m} \simeq K[t_1, \dots, t_d] = K[\mathbb{A}^d]$. This means $\pi^{-1}(p) = \mathbb{A}^d$. Note that ψ is finite and surjective. By Theorem 19, $\dim(\varphi^{-1}(\pi^{-1}(p))) = d$. Finally, d was the number such that $S^{-1}B$ is finite over $S^{-1}A[t_1, \dots, t_d] = K(Y)[t_1, \dots, t_d]$ as a module. So

$$\dim(\varphi^{-1}(p)) = d = \text{Trdeg}(K(X)/K) - \text{Trdeg}(K(Y)/K) \stackrel{\text{prop 26}}{=} \dim(X) - \dim(Y)$$

□

Remark. There exists a noetherian topological space $X = \{p_1, p_2\}$ such that $\{p_1\}$ is closed, $\{p_2\}$ is open but not closed, and $\overline{\{p_2\}} = X$. Let $Y = \{p\}$. Let the map $Y \rightarrow X$ given by $p \mapsto p_2$ be dominant. But $\dim(Y) = 0 < \dim(X) = 1$. Aside, this is $\mathbb{Z}_p \hookrightarrow \mathbb{Q}_p$ with a dual map $\text{spec}(\mathbb{Q}_p) \rightarrow \text{spec}(\mathbb{Z}_p)$.

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Vocabulary 10.1 (projective space $\mathbb{P}_K^n$). Let K be a field and let $n \geq 0$ be an integer. The projective space is

$$\mathbb{P}_K^n = (\mathbb{A}_K^{n+1} \setminus \{0\})/K^\times$$

$K^\times$ acts by scaling. We may view $\mathbb{P}_K^n$ as lines in $\mathbb{A}_K^{n+1}$ that pass through 0. When $n = 0$, $\mathbb{P}_K^0 = \{\text{pt}\}$. When $n = 1$, $\mathbb{P}_K^1 = (\mathbb{A}_K^2 \setminus \{0\})/K^\times$. Label points of $\mathbb{P}_K^1$ as $[x : y]$ and $[\lambda x : \lambda y] = [x : y]$ for all $\lambda \in K^\times$. The equivalence relation on $\mathbb{A}_K^2$ is $(a_0, a_1) \sim (b_0, b_1)$ if and only if $(a_0, a_1) = (\lambda b_0, \lambda b_1)$. Suppose $(x, y) \in \mathbb{A}_K^2$. If either x or y becomes 0, then $\mathbb{P}_K^1$ would collapse into a single point under this equivalence. So we require $xy \neq 0$. This gives $(x, y) \simeq (1, y' := y/x)$. With this requirement in mind, clearly $\mathbb{P}_K^1 = U_0 \cup U_1$, where $U_0 = \{[x : y] | x \neq 0\}$ and $U_1 = \{[x : y] | y \neq 0\}$. If $[x : y] \in U_0$, then $[x : y] = [1 : y/x]$. Similarly, if $[x : y] \in U_1$, then $[x : y] = [x/y : 1]$. There are bijections $U_i \rightarrow \mathbb{A}_K^1$. In this bijective correspondence, $[1, 0/x]$ corresponds to 0 in $\mathbb{A}_K^1$ while $[1, y/0]$ corresponds to ∞ . Also observe that $U_0 \cap U_1 = \mathbb{A}_K^1 \setminus \{0\}$. So

$$\mathbb{P}_K^1 = U_0 \cup (U_1 \setminus U_0) = U_0 \cup \{[0 : 1]\} = \mathbb{A}_K^1 \cup \{\infty\}$$

Example. Let $K = \mathbb{C}$. The Riemann sphere is defined as

$$\mathbb{C}^2 \setminus \{0\} / \mathbb{C}^\times = \{(z, w) \in \mathbb{C}^2 | \|(z, w)\| = 1\} / \mathbb{S}^1 = \mathbb{S}^3 / \mathbb{S}^1 \simeq \mathbb{S}^2$$

Remark. Following the construction in 10.1, we can write out the coordinates on $\mathbb{P}_K^n$ as $[x_0 : x_1 : \dots : x_n]$. And

$$\mathbb{P}_K^n = \bigcup_{i=0}^n (U_i = \{[x_0 : \dots : x_n] | x_i \neq 0\})$$

with bijections

$$\begin{aligned}\mathbb{A}_K^n &\longrightarrow U_i \\ (t_1, \dots, t_n) &\longmapsto [t_1 : \dots : t_{i-1} : 1 : t_{i+1} : \dots : t_n] \\ \left(\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i}\right) &\longleftarrow [x_0 : \dots : x_n]\end{aligned}$$

If $K = \mathbb{C}$, then $\mathbb{P}_{\mathbb{C}}^n = \{v \in \mathbb{C}^{n+1} \mid \|v\| = 1\} / \mathbb{S}^1$ is a compact Hausdorff space, in which each $U_i \simeq \mathbb{C}^n$ is open.

Problem. We want to define closed sets of $\mathbb{P}_K^n$ via $Z = Z(f_1, \dots, f_r)$, where each $f_i \in K[x_0, \dots, x_n]$. But $Z(f_i) \subseteq \mathbb{A}_K^{n+1}$ may not be $K^\times$ -invariant.

Solution. Restrict $f_i \in K[x_0, \dots, x_n]$ to homogeneous polynomials with respect to the total degree. Indeed, f is homogeneous if and only if for all $\lambda \in K^\times$, $f(\lambda x_0, \dots, \lambda x_n) = \lambda^{\deg(f)} f(x_0, \dots, x_n)$.

Corollary. If f is homogeneous, then $(a_0, \dots, a_n) \in Z(f)$ if and only if $(\lambda a_0, \dots, \lambda a_n) \in Z(f)$ for all $\lambda \in K^\times$. $\square$

Vocabulary 10.2 (zero set in $\mathbb{P}_K^n$). Let f be homogeneous. $Z(f) := \{[a_0 : \dots : a_n] \mid f(a_0, \dots, a_n) = 0\} \subseteq \mathbb{P}_K^n$.

Vocabulary 10.3 (homogeneous ideal). Let $A = \bigoplus_{i=0}^{\infty} A_i$ be a graded ring. An ideal $I \subseteq A$ is homogeneous if

$$I = \bigoplus_{i=0}^{\infty} (I \cap A_i)$$

Example. Suppose $K[a, b] = \bigoplus_{i=0}^{\infty} A_i$ is graded by total degrees. Each A_i consists of monomials like $x^\alpha y^\beta$, where $\alpha + \beta = i$. Let $r = x^3 + x^2 y + y^2 + x$ be an arbitrary element. We can write $r = a_3 + a_2 + a_1$, where $a_3 = x^3 + x^2 y \in A_3$, $a_2 = y^2 \in A_2$, and $a_1 \in A_1$. In other words, r is a sum of homogeneous elements.

Proposition 28. Let $A = \bigoplus_{i=0}^{\infty} A_i$ be a graded ring. An ideal $I \subseteq A$ is homogeneous if and only if $I = \langle f_i \rangle$, where each f_i is a homogeneous element.

Proof. ($\Rightarrow$) If $I = \bigoplus_{i=0}^{\infty} (I \cap A_i)$, then I is generated by $\{I \cap A_i\}_{i=0}^{\infty}$. Moreover, each $I \cap A_i$ consists of homogeneous elements. ($\Leftarrow$) Suppose each f_i is homogeneous of degree d_i . If $f \in I$, then $f = \sum_{i=1}^n a_i f_i$ for $a_i \in A$. Let $a_i = \sum_{j=0}^{N_i} a_{ij}$ for homogeneous a_{ij} . Write

$$f = \sum_{i=1}^n \sum_{j=0}^{N_i} (a_{ij} f_i), \quad a_{ij} f_i \in I \cap A_{d_i + \deg(a_{ij})}$$

So $f = \sum_j h_j$, where each h_j is homogeneous and in I . $\square$

Vocabulary 10.4 (zero set of a homogeneous ideal). Let $I \subseteq K[x_0, \dots, x_n]$ be a homogeneous ideal. $Z(I) := \bigcap_{f \in I} Z(f)$. If $I = \langle f_1, \dots, f_r \rangle$ with homogeneous f_i , then $Z(I) := \bigcap_{i=1}^r Z(f_i)$. Each $Z(f_i)$ is in the sense of 10.2.

Remark. Since $K[x_0, \dots, x_n]$ is noetherian, I is generated by a finite set of homogeneous elements.

Vocabulary 10.5 (Zariski topology on $\mathbb{P}_K^n$). The Zariski topology on $\mathbb{P}_K^n$ is the one whose closed sets are

$$\{Z(I) \mid I \subseteq K[x_0, \dots, x_n] \text{ homogeneous ideal}\}$$

Lemma. If $Z_i = Z(I_i)$, we immediately get these properties

1. $\bigcup_{i=1}^n Z_i = Z(\bigcap_{i=1}^n I_i)$.
2. $\bigcap_{i \in J} Z_i = Z(\sum_{i \in J} I_i)$.

$\square$

Vocabulary 10.6 (radical homogeneous ideal). Let $I \subseteq K[x_0, \dots, x_n]$ be a homogeneous ideal.

$$\sqrt{I}^* = \{f \in K[x_0, \dots, x_n] \mid f \text{ homogeneous and } f^{N \geq 1} \in I\}$$

Vocabulary 10.7 (ideal of $\mathbb{P}_K^n$). Let $Z \subseteq \mathbb{P}_K^n$ be closed. Define

$$I(Z) = \{f \in K[x_0, \dots, x_n] \mid f \text{ homogeneous and } f(Z) = 0\}$$

Vocabulary 10.8 (coordinate ring in $\mathbb{P}_K^n$). Let $Y \subseteq \mathbb{P}_K^n$ be closed. The homogeneous coordinate ring of Y embedded in is

$$K[x_0, \dots, x_n]/I(Y)$$

Theorem 29 (homogeneous Nutz). If $\mathfrak{a} \subseteq S := K[x_0, \dots, x_n]$ is a homogeneous ideal, and if $f \in S$ is a homogeneous polynomial with $\deg(f) > 0$ such that $f(P) = 0$ for all $P \in Z(\mathfrak{a}) \subseteq \mathbb{P}_K^n$, then $f^q \in \mathfrak{a}$ for some $q > 0$. In particular,

$$I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}^*$$

Proof. Assume Let $\mathfrak{a} \subseteq S$ be a homogeneous ideal and $f \in S$ a homogeneous polynomial with $\deg f > 0$, such that $f(P) = 0$ for all $P \in Z(\mathfrak{a})$ in $\mathbb{P}_K^n$. Then f vanishes on all representatives for points in $Z(\mathfrak{a})$, so in particular, $f^r \in \mathfrak{a}$ when thinking of $Z(\mathfrak{a})$ as being in $\mathbb{A}_K^{n+1}$. $\square$

11 Week 11

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Vocabulary 11.1 (hypersurface in $\mathbb{P}_K^n$). A hypersurface $H \subseteq \mathbb{P}_K^n$ is a subset $Z(f)$ for some homogeneous $f \in K[x_0, \dots, x_n]$ such that $\deg(f) > 0$.

Remark. Each $U_i = \{[x_0 : \dots : x_n] | x_i \neq 0\}$ has $\mathbb{P}_K^n \setminus U_i = Z(x_i)$.

Vocabulary 11.2 (cone). Let $Y \subseteq \mathbb{P}_K^n$ be a closed subvariety. Let $C(Y) \subseteq \mathbb{A}_K^{n+1}$ be

$$C(Y) = \{p \in \mathbb{A}_K^{n+1} \setminus \{0\} | \pi(p) \in Y\} \cup \{0\} = Z_{\mathbb{A}_K^{n+1}}(f_i)$$

where $f_i \in I(Y)$ (and therefore, $f_i \in K[x_0, \dots, x_n]$) and $\pi : \mathbb{A}_K^{n+1} \rightarrow \mathbb{P}_K^n$ is the projection map.

Vocabulary 11.3 (projective dimension). Let $Y \subseteq \mathbb{P}_K^n$ be a closed subvariety. $\dim(Y) = \dim(C(Y)) - 1$.

Proposition 29. Let $H \subseteq \mathbb{P}_K^n$ be a hypersurface. Let $Y \subseteq \mathbb{P}_K^n$ be such that $\dim(Y) \geq 1$. We have that $H \cap Y \neq \emptyset$. In $n = 2$, this implies the fact that any two curves in $\mathbb{P}_K^2$ always intersect.

Proof. We know $C(H) = Z_{\mathbb{A}_K^{n+1}}(f)$ for some homogeneous f . And $C(Y)$ is a closed set such that $\dim(C(Y)) \geq 2$. Note that $C(H) \cap C(Y) \neq \emptyset$ as they both contain 0. Here $C(Y)$ need not to be irreducible. Let W be an irreducible component of $C(Y)$ with maximal dimension. If $C(Y) = \bigcup_i W_i$, then $\dim(C(Y)) = \max_i \dim(W_i)$. Choose W to be some W_j such that $\dim(W_j) = \dim(C(Y))$. Since $W \subseteq C(Y)$, $\dim(Z_{\mathbb{A}_K^{n+1}}(f) \cap C(Y)) \geq \dim(Z_{\mathbb{A}_K^{n+1}}(f) \cap W)$. If $W \subseteq Z_{\mathbb{A}_K^{n+1}}(f)$, then $Z_{\mathbb{A}_K^{n+1}}(f) \cap W = W$ and

$$\dim(C(H) \cap C(Y)) = \dim(Z_{\mathbb{A}_K^{n+1}}(f) \cap W) = \dim(W) \geq \dim(W) - 1 = \dim(C(Y)) - 1 \geq 1$$

If $W \not\subseteq Z_{\mathbb{A}_K^{n+1}}(f)$, then $f|_W$ is not constant 0 in $K[W]$. Moreover, $f|_W \notin K$ because f is homogeneous⁹ of degree at least 1. So Theorem 21 applies to the irreducible affine variety W with the function $f|_W$. In this case, every irreducible component of $Z_{\mathbb{A}_K^{n+1}}(f) \cap W$ has dimension $\dim(W) - 1$. Since both contain 0, $Z_{\mathbb{A}_K^{n+1}}(f) \cap W \neq \emptyset$ and $\dim(Z_{\mathbb{A}_K^{n+1}}(f) \cap W) = \dim(W) - 1$. This again gives

$$\dim(C(H) \cap C(Y)) = \dim(Z_{\mathbb{A}_K^{n+1}}(f) \cap W) = \dim(W) - 1 = \dim(C(Y)) - 1 \geq 1$$

If $C(H) \cap C(Y)$ contains only $\{0\}$, its dimension would be 0. We have shown that it contains more than just $\{0\}$. So $\{0\} \subsetneq C(H) \cap C(Y)$. It follows that $(C(H) \cap C(Y)) \setminus \{0\} \neq \emptyset$ and $\pi^{-1}(H \cap Y) \neq \emptyset$. Thus, $H \cap Y \neq \emptyset$. $\square$

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Proposition 30. Projections like

$$\pi_1 : \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^n \quad \pi_2 : \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^m$$

⁹Homogeneous implies $f(0) = 0$. If f restricts to a constant $c \in W$, then $c = f|_W(0) = f(0) = 0$. This retracts to the previous case where $W \subseteq Z_{\mathbb{A}_K^{n+1}}(f)$.

are morphisms.

Proof. Let $X = \mathbb{P}^n$, $Y = \mathbb{P}^m$ and consider the first projection p_1 . Let $([a_0, \dots, a_n], [b_0, \dots, b_m]) \in \mathbb{P}^n \times \mathbb{P}^m$ and assume $a_0 \neq 0$ and $b_0 \neq 0$, then $([a_0, \dots, a_n], [b_0, \dots, b_m])$, as a point in the image of Segre embedding, has homogeneous coordinates $[w_{00}, \dots, w_{mn}]$, while $w_{00} \neq 0$. Then $p_1([a_0, \dots, a_n], [b_0, \dots, b_m]) = [w_{00}, w_{10}, \dots, w_{n0}]$. This is a locally rational map, hence a morphism. $\square$

Now let us recall a corollary that we proved before introducing rational maps. We may use this later on.

Corollary. Let X be irreducible and let $\varphi, \psi : X \rightarrow Y$. If $\varphi = \psi$ on some $\emptyset \neq U \subseteq X$, then $\varphi = \psi$ on X .

Lemma (rational map). Let X, Y be irreducible varieties. A **rational map** $X \dashrightarrow Y$ is an equivalence class of pairs (U, φ) , where $\varphi \neq \emptyset \subseteq X$ open, $\varphi : U \rightarrow Y$.

Proof. $(U, \varphi) \sim (V, \psi)$ if and only if there exists $W \subseteq U \cap V$ nonempty such that $\varphi = \psi$ on W . This is an equivalence relation. $\square$

Remark. In fact, there exists a unique morphism $\varphi : U \cup V \rightarrow Y$ such that $\varphi|_U = \varphi$ and $\varphi|_V = \psi$ (do not ask me why). This is also known as the **complex continuation**.

Example. If X is irreducible, $Y = \mathbb{A}^1$, the set of rational maps $X \dashrightarrow \mathbb{A}^1$ is denoted $K(X)$. This is the function field of X . If X is affine, $K(X) = \text{Frac}(K[X])$. Element of $\text{Frac}(K[X]) \ni f$ is $f = f_1/f_2 \in K[D(f_2)]$.

Appendix

Here, we discuss results that we should have proved. And also additional readings that you might find interesting.

Complementary

Theorem A1. Let A be a commutative ring with 1, and let $S \subseteq A$ be a subset such that $1 \in S$ and $s_1, s_2 \in S$ implies $s_1 s_2 \in S$. We call such an S a multiplicative subset. Then

1. the following defines an equivalence relation on the set $A \times S$: $(a_1, s_1) \sim (a_2, s_2)$ if and only if there exists some $s_3 \in S$ such that $s_3(s_2 a_1 - a_2 s_1) = 0$.
2. the quotient $(A \times S)/\sim$ has the structure of a ring under the operations

$$(a_1, s_1) + (a_2, s_2) = (s_2 a_1 + s_1 a_2, s_1 s_2) \quad (a_1, s_1) \cdot (a_2, s_2) = (a_1 a_2, s_1 s_2).$$

3. Let the ring above be denoted A_S (also often denoted $S^{-1}A$). The map $\varphi : A \rightarrow A_S$ given by $a \mapsto (a, 1)$ is a ring homomorphism.
4. $A_S = 0$ if and only if $0 \in S$. In particular, if $S = \{t^i\}_{i \geq 0}$ then $A_S = 0$ precisely when t is nilpotent.
5. Suppose $S = \{t^i\}_{i \geq 0}$ for some $t \in A$. There is an isomorphism $A_S \simeq A[x]/(xt - 1)$.

Theorem A2 (universal property of localization). If B is any commutative ring and $\psi : A \rightarrow B$ is a ring homomorphism such that $\psi(S) \subseteq B^\times$, then there exists a unique ring homomorphism $\chi : A_S \rightarrow B$ given by $\chi \circ \varphi = \psi$. Moreover, if $\varphi' : A \rightarrow C$ is another ring map satisfying the same universal property, then there is a unique isomorphism $A_S \simeq C$ making the triangles commute.

Theorem A3. Let $S \subseteq A$ be a multiplicative subset. There is a bijection between

$$\{\text{prime ideals } \mathfrak{p} \subset A \mid \mathfrak{p} \cap S = \emptyset\} \leftrightarrow \{\text{prime ideals of } A_S\}$$

given by $\mathfrak{p} \mapsto A_S \cdot \varphi(\mathfrak{p})$; the inverse sends a prime $\mathfrak{q} \subset A_S$ to its preimage $\varphi^{-1}(\mathfrak{q})$.

Theorem A4. Let $\mathfrak{p} \subset A$ be a prime ideal. Then $S = A \setminus \mathfrak{p}$ is a multiplicative subset and that A_S is a ring with a unique maximal ideal.

Theorem A5. Let $\varphi : A \rightarrow B$ be a ring homomorphism. Then

1. suppose $S \subseteq A$ is a multiplicative subset. $\varphi(S) \subseteq B$ is a multiplicative, and there is an induced map

$$\varphi_S : A_S \longrightarrow B_{\varphi(S)}.$$

2. suppose φ is integral. Then φ_S is as well.

Lemma (Zorn). Let X be a non-empty poset. If every non-empty chain in X has an upper bound, then X has a maximal element. $\square$

Theorem A6. Any proper ideal $I \subseteq R$ is contained in a maximal ideal.

Proof. Choose any proper ideal $I_0 \subsetneq R$. Let X be the set of all proper ideals of R which contain I_0 . If I_0 is maximal, we are done. Otherwise, $I_0 \subsetneq I_1 \subsetneq R$. We either find the maximal ideal or get a possibly infinite chain ordered by inclusion

$$I_0 \subsetneq I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \dots \subsetneq R$$

Set $J_0 = \bigcup_{k=0}^{\infty} I_k$.

J_0 is an ideal: It is not empty because the chain is not empty (at least I_0 itself is in the chain). If $a, b \in J_0$, there exist A, B in the chain such that $a \in A$ and $b \in B$. Since A, B are members of the chain, either $A \subseteq B$ or $A \supseteq B$, so $a \pm b \in J_0$. If $a \in J_0$ and $r \in R$, then there exists A in a chain such that $a \in A$, and thus $ra, ar \in A \subseteq J_0$.

J_0 is a proper ideal: Since all ideals are proper, J_0 is not the whole ring (If $J_0 = R$ then $1 \in J_0$ and $1 \in I_k$ for some $k \Rightarrow I_k = R$ is not a proper ideal).

Now we conclude that such $J_0 \in X$ and they are upper bounds of the chain. Applying Zorn's lemma to X , we see that X has a maximal element M . $\square$

Proposition A1. Some analogous results for modules:

1. Let M be an R -module. Every submodule is finitely generated if and only if every chain stabilizes.
2. Any finitely generated module over a noetherian ring is noetherian.

Vocabulary A1 (noetherian module). A module satisfying Proposition 1A is called noetherian.

Additional Readings

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